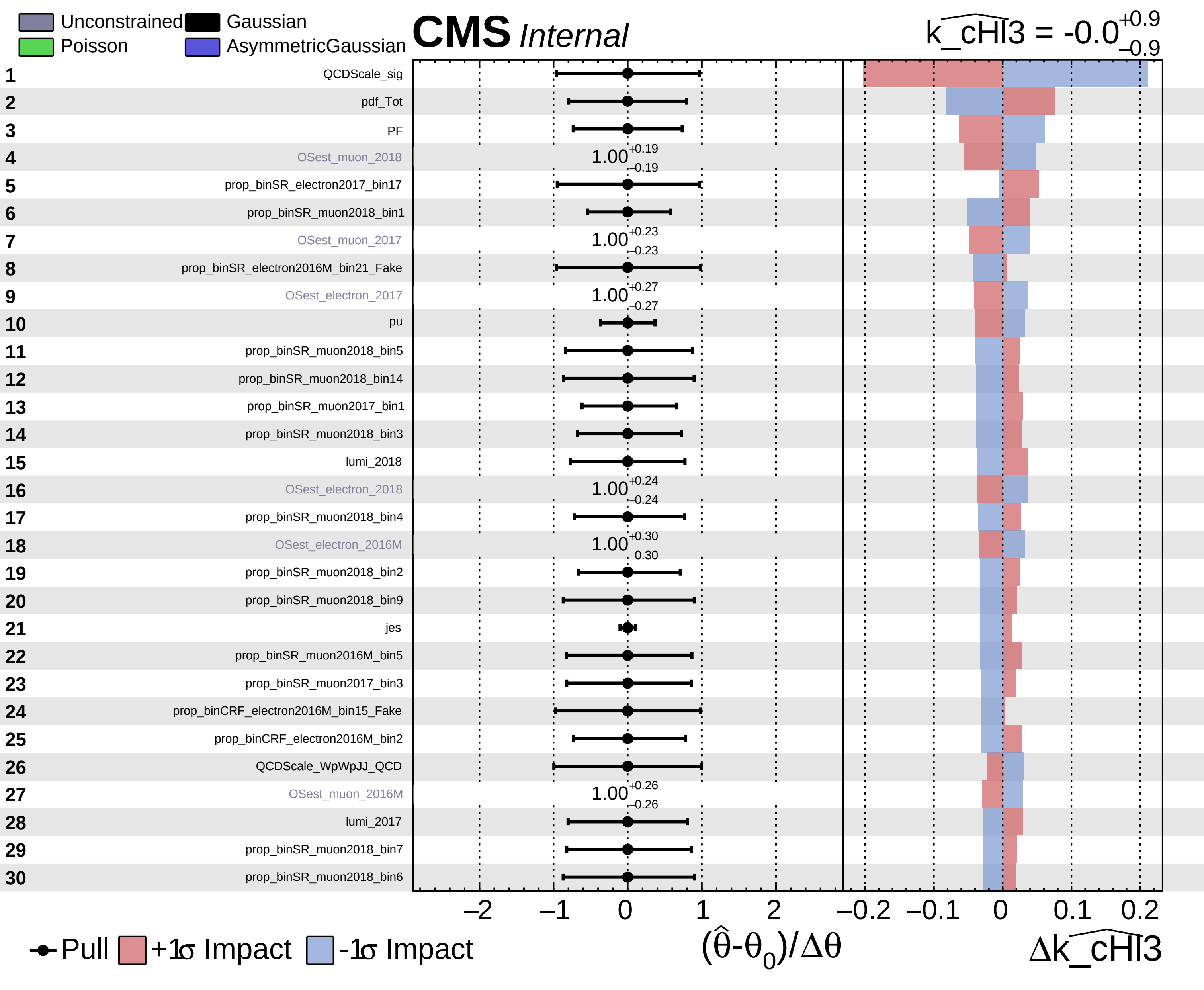
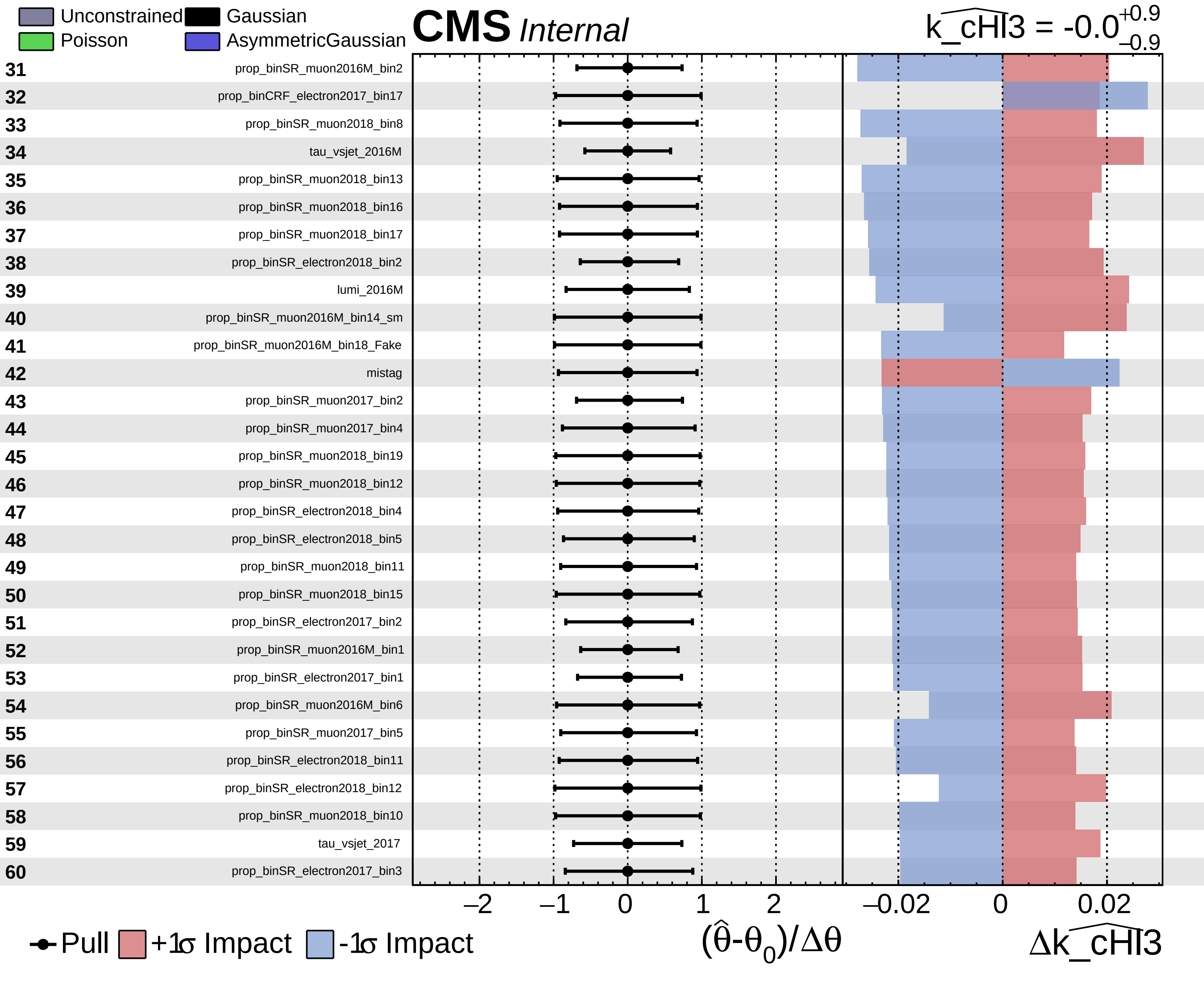
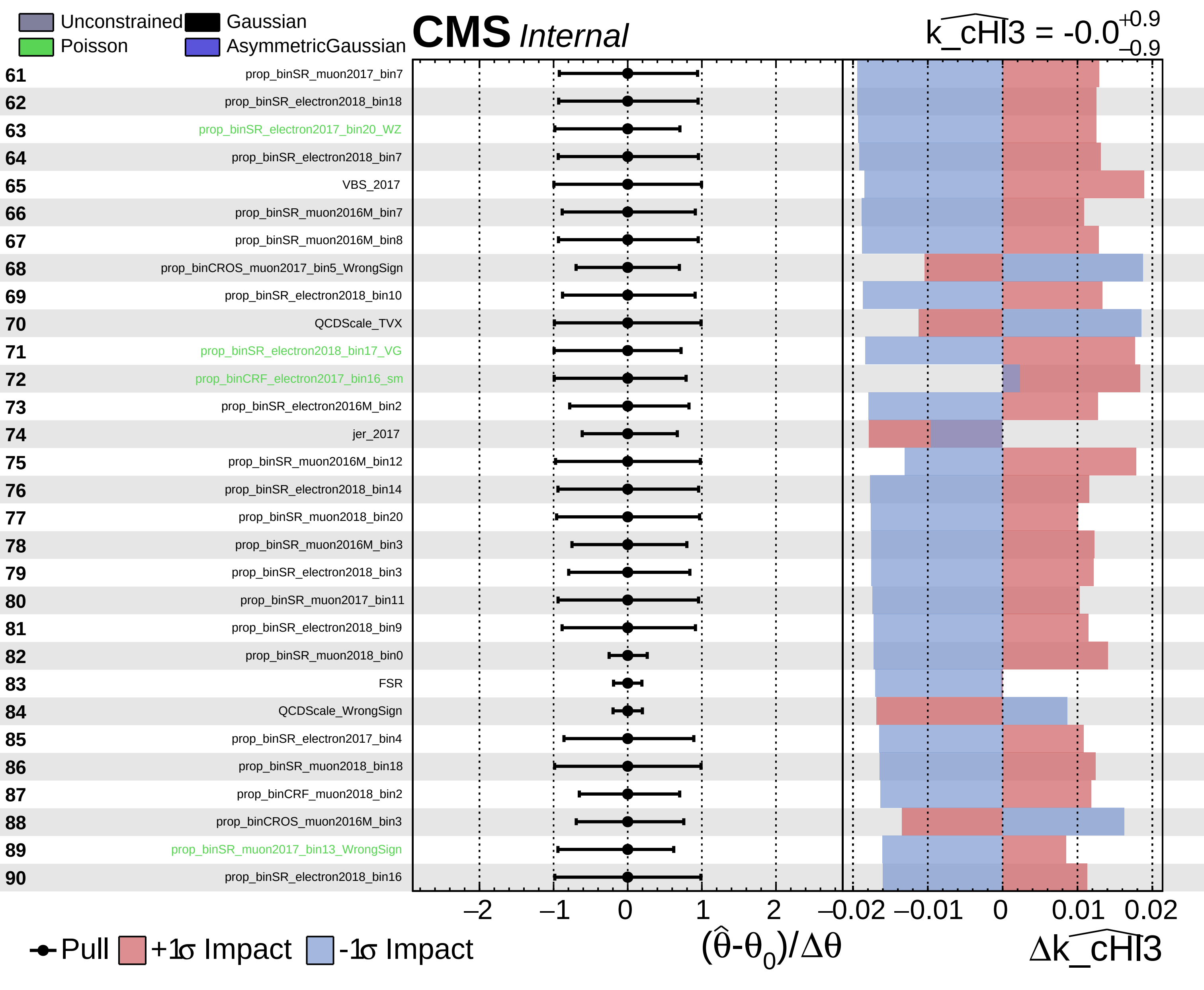


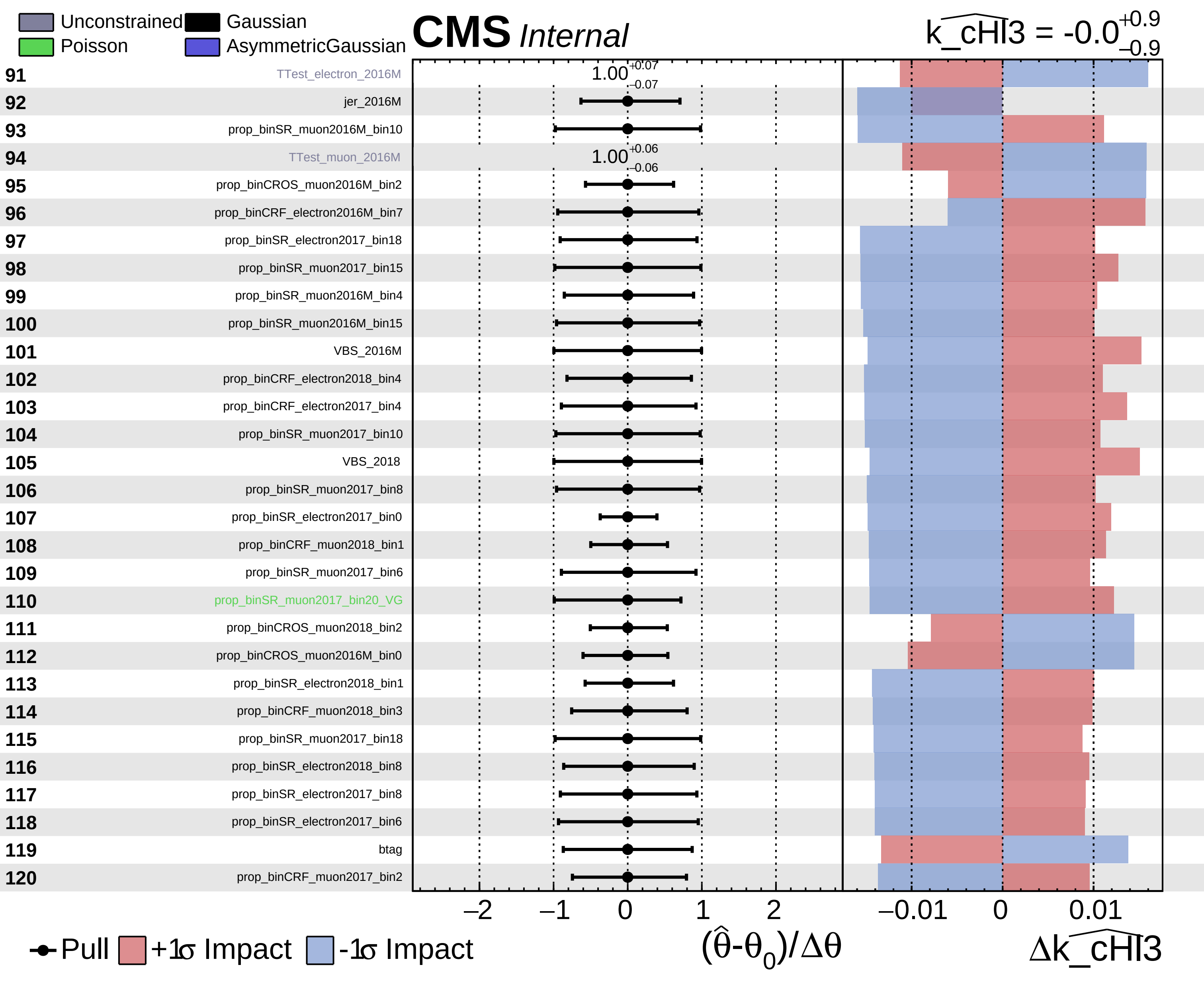
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1 QCDScale_sig
2 pdf_Tot
3 PF
4 OSest_muon_2018 1.00 +0.19 −0.19
5 prop_binSR_electron2017_bin17
6 prop_binSR_muon2018_bin1
7 OSest_muon_2017 1.00 +0.23 −0.23
8 prop_binSR_electron2016M_bin21_Fake
9 OSest_electron_2017 1.00 +0.27 −0.27
10 pu
11 prop_binSR_muon2018_bin5
12 prop_binSR_muon2018_bin14
13 prop_binSR_muon2017_bin1
14 prop_binSR_muon2018_bin3
15 lumi_2018
16 OSest_electron_2018 1.00 +0.24 −0.24
17 prop_binSR_muon2018_bin4
18 OSest_electron_2016M 1.00 +0.30 −0.30
19 prop_binSR_muon2018_bin2
20 prop_binSR_muon2018_bin9
21 jes
22 prop_binSR_muon2016M_bin5
23 prop_binSR_muon2017_bin3
24 prop_binCRF_electron2016M_bin15_Fake
25 prop_binCRF_electron2016M_bin2
26 QCDScale_WpWpJJ_QCD
27 OSest_muon_2016M 1.00 +0.26 −0.26
28 lumi_2017
29 prop_binSR_muon2018_bin7
30 prop_binSR_muon2018_bin6
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

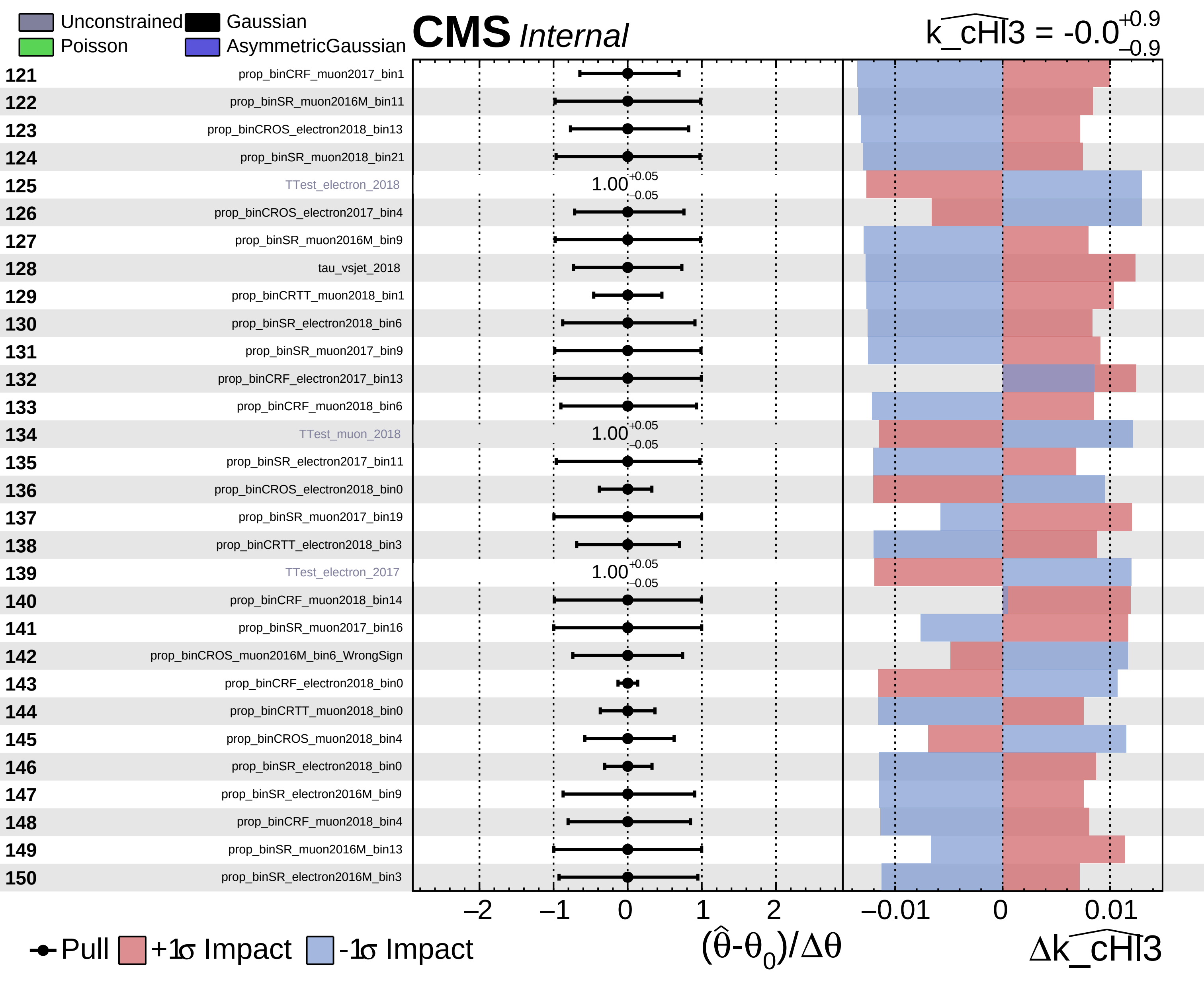


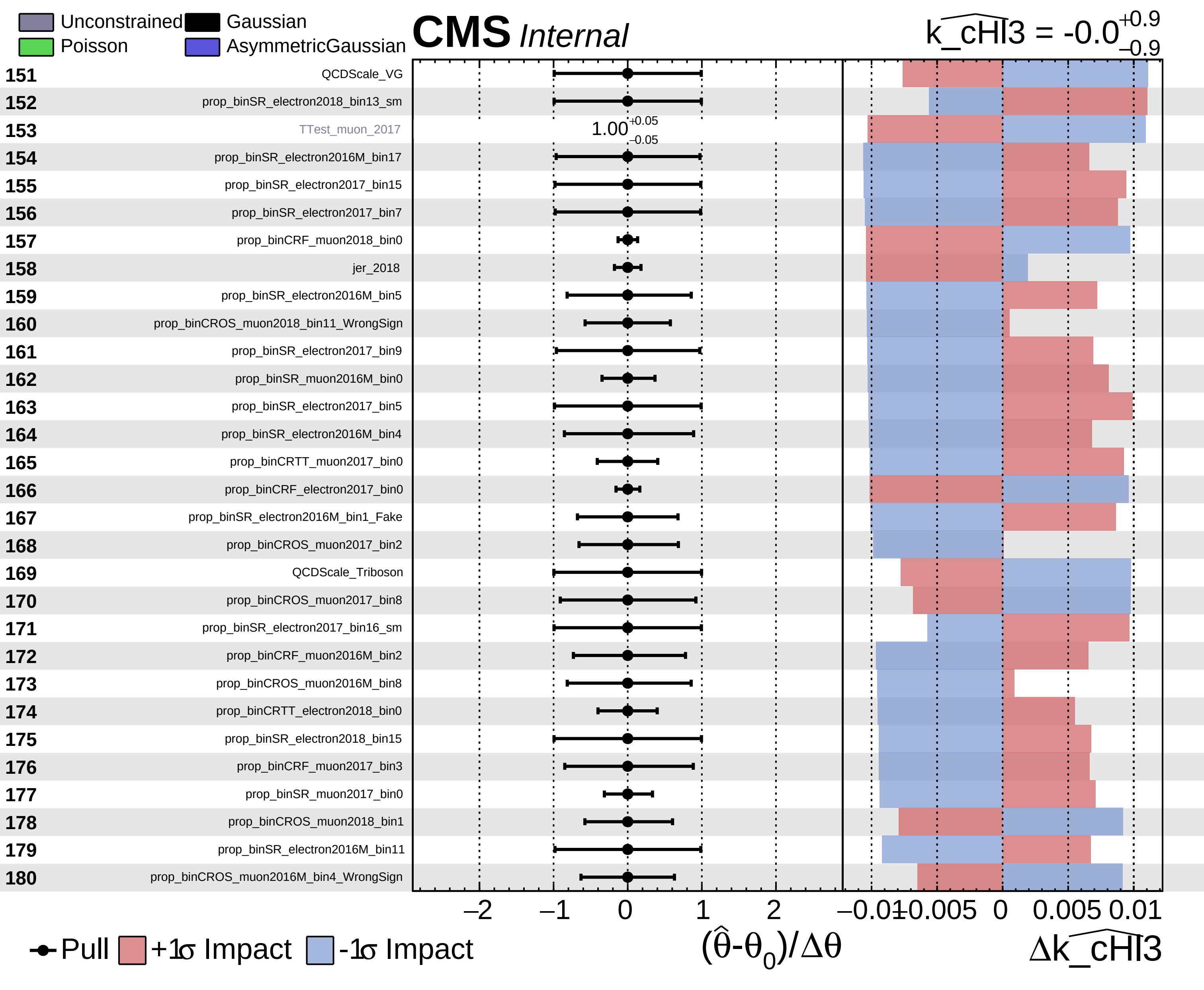
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{\text{k\_cHl3}} = -0.0^{+0.9}_{-0.9}$
31 prop_binSR_muon2016M_bin2
32 prop_binCRF_electron2017_bin17
33 prop_binSR_muon2018_bin8
34 tau_vsjet_2016M
35 prop_binSR_muon2018_bin13
36 prop_binSR_muon2018_bin16
37 prop_binSR_muon2018_bin17
38 prop_binSR_electron2018_bin2
39 lumi_2016M
40 prop_binSR_muon2016M_bin14_sm
41 prop_binSR_muon2016M_bin18_Fake
42 mistag
43 prop_binSR_muon2017_bin2
44 prop_binSR_muon2017_bin4
45 prop_binSR_muon2018_bin19
46 prop_binSR_muon2018_bin12
47 prop_binSR_electron2018_bin4
48 prop_binSR_electron2018_bin5
49 prop_binSR_muon2018_bin11
50 prop_binSR_muon2018_bin15
51 prop_binSR_electron2017_bin2
52 prop_binSR_muon2016M_bin1
53 prop_binSR_electron2017_bin1
54 prop_binSR_muon2016M_bin6
55 prop_binSR_muon2017_bin5
56 prop_binSR_electron2018_bin11
57 prop_binSR_electron2018_bin12
58 prop_binSR_muon2018_bin10
59 tau_vsjet_2017
60 prop_binSR_electron2017_bin3
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{\text{k\_cHl3}}$



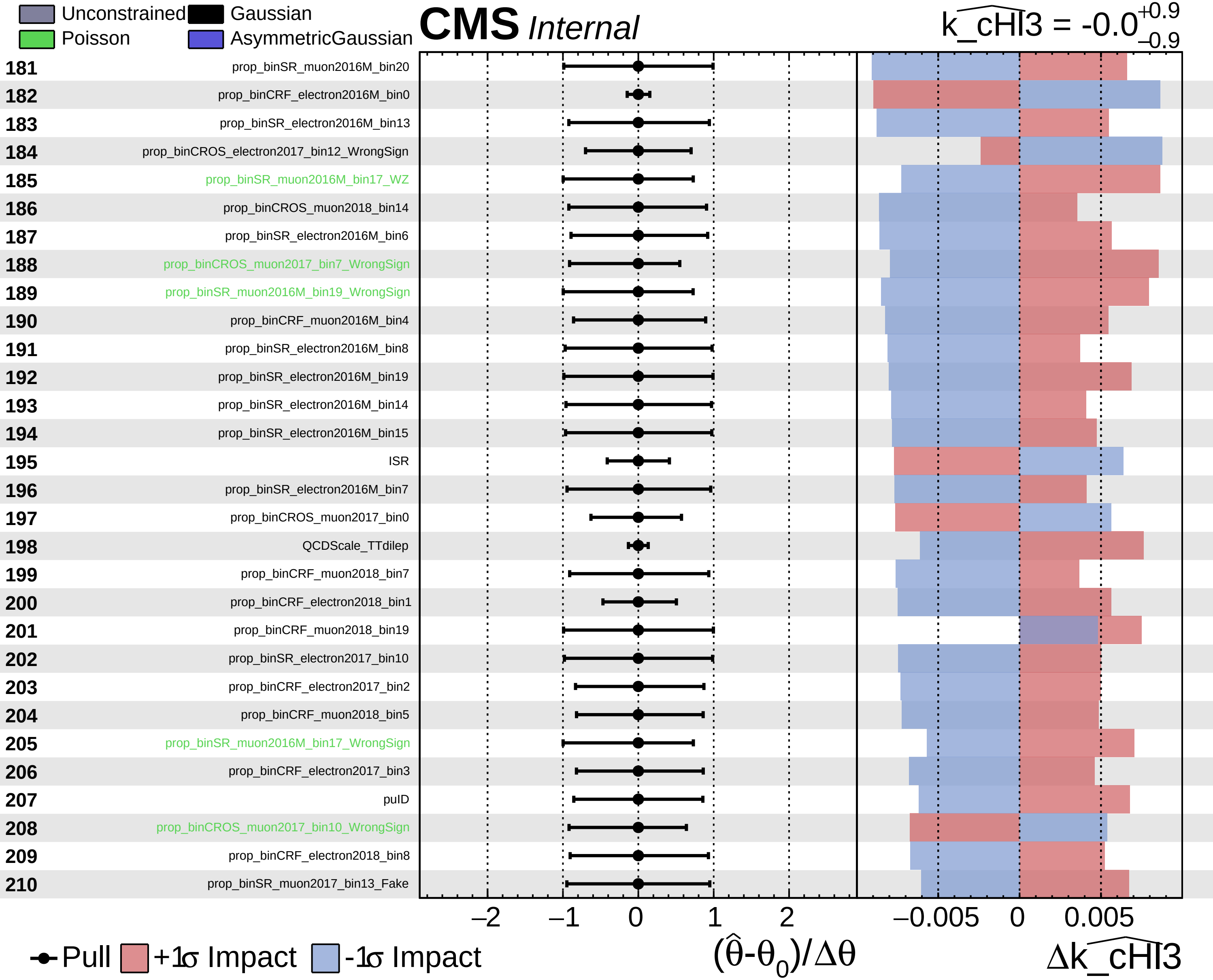
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
61 prop_binSR_muon2017_bin7
62 prop_binSR_electron2018_bin18
63 prop_binSR_electron2017_bin20_WZ
64 prop_binSR_electron2018_bin7
65 VBS_2017
66 prop_binSR_muon2016M_bin7
67 prop_binSR_muon2016M_bin8
68 prop_binCROS_muon2017_bin5_WrongSign
69 prop_binSR_electron2018_bin10
70 QCDScale_TVX
71 prop_binSR_electron2018_bin17_VG
72 prop_binCRF_electron2017_bin16_sm
73 prop_binSR_electron2016M_bin2
74 jer_2017
75 prop_binSR_muon2016M_bin12
76 prop_binSR_electron2018_bin14
77 prop_binSR_muon2018_bin20
78 prop_binSR_muon2016M_bin3
79 prop_binSR_electron2018_bin3
80 prop_binSR_muon2017_bin11
81 prop_binSR_electron2018_bin9
82 prop_binSR_muon2018_bin0
83 FSR
84 QCDScale_WrongSign
85 prop_binSR_electron2017_bin4
86 prop_binSR_muon2018_bin18
87 prop_binCRF_muon2018_bin2
88 prop_binCROS_muon2016M_bin3
89 prop_binSR_muon2017_bin13_WrongSign
90 prop_binSR_electron2018_bin16
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
91 TTest_electron_2016M
92 jer_2016M
93 prop_binSR_muon2016M_bin10
94 TTest_muon_2016M
95 prop_binCROS_muon2016M_bin2
96 prop_binCRF_electron2016M_bin7
97 prop_binSR_electron2017_bin18
98 prop_binSR_muon2017_bin15
99 prop_binSR_muon2016M_bin4
100 prop_binSR_muon2016M_bin15
101 VBS_2016M
102 prop_binCRF_electron2018_bin4
103 prop_binCRF_electron2017_bin4
104 prop_binSR_muon2017_bin10
105 VBS_2018
106 prop_binSR_muon2017_bin8
107 prop_binSR_electron2017_bin0
108 prop_binCRF_muon2018_bin1
109 prop_binSR_muon2017_bin6
110 prop_binSR_muon2017_bin20_VG
111 prop_binCROS_muon2018_bin2
112 prop_binCROS_muon2016M_bin0
113 prop_binSR_electron2018_bin1
114 prop_binCRF_muon2018_bin3
115 prop_binSR_muon2017_bin18
116 prop_binSR_electron2018_bin8
117 prop_binSR_electron2017_bin8
118 prop_binSR_electron2017_bin6
119 btag
120 prop_binCRF_muon2017_bin2
1.00 +0.07 −0.07
1.00 +0.06 −0.06
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

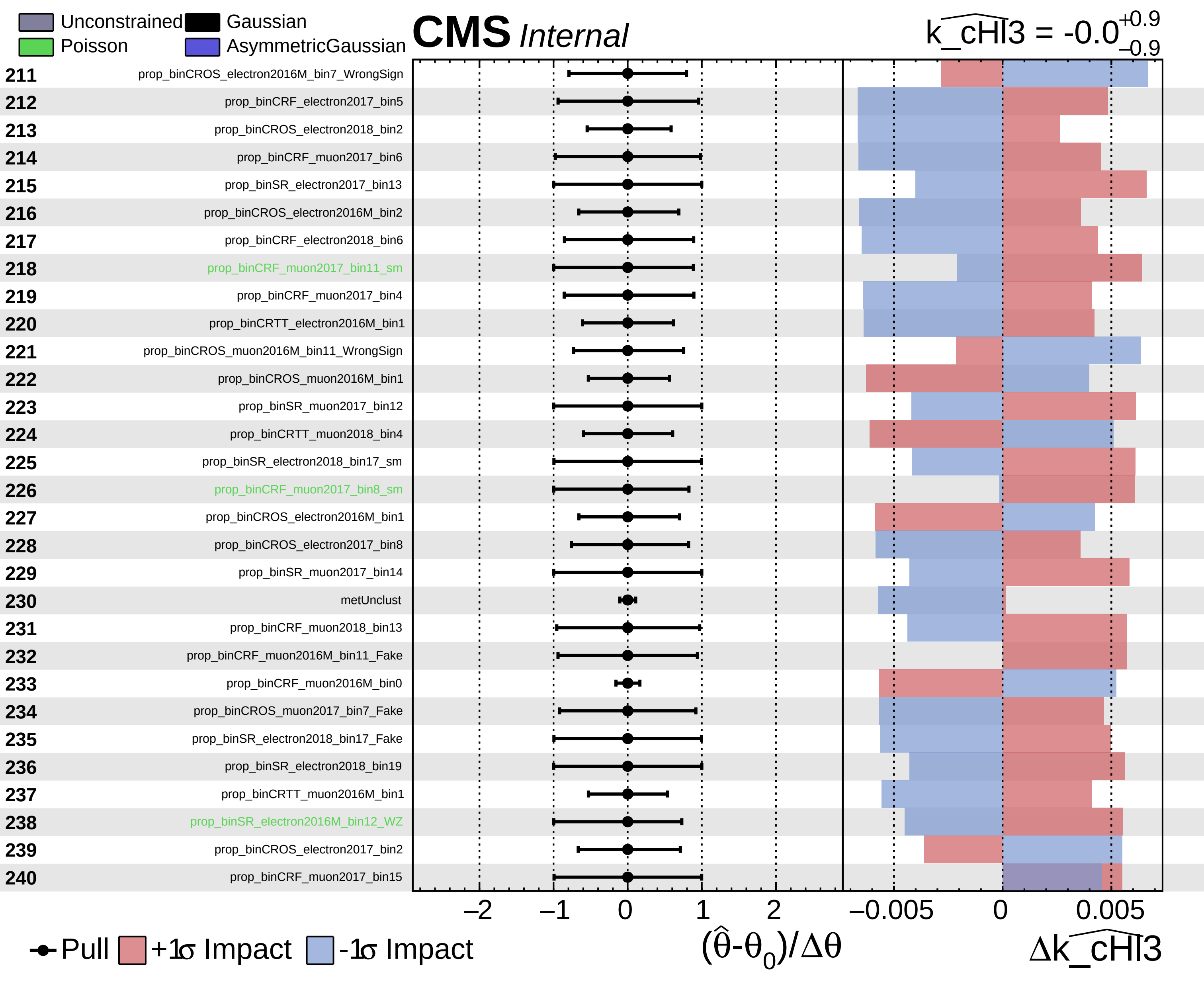

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
121 prop_binCRF_muon2017_bin1
122 prop_binSR_muon2016M_bin11
123 prop_binCROS_electron2018_bin13
124 prop_binSR_muon2018_bin21
125 TTest_electron_2018
1.00 +0.05 −0.05
126 prop_binCROS_electron2017_bin4
127 prop_binSR_muon2016M_bin9
128 tau_vsjet_2018
129 prop_binCRTT_muon2018_bin1
130 prop_binSR_electron2018_bin6
131 prop_binSR_muon2017_bin9
132 prop_binCRF_electron2017_bin13
133 prop_binCRF_muon2018_bin6
134 TTest_muon_2018
1.00 +0.05 −0.05
135 prop_binSR_electron2017_bin11
136 prop_binCROS_electron2018_bin0
137 prop_binSR_muon2017_bin19
138 prop_binCRTT_electron2018_bin3
139 TTest_electron_2017
1.00 +0.05 −0.05
140 prop_binCRF_muon2018_bin14
141 prop_binSR_muon2017_bin16
142 prop_binCROS_muon2016M_bin6_WrongSign
143 prop_binCRF_electron2018_bin0
144 prop_binCRTT_muon2018_bin0
145 prop_binCROS_muon2018_bin4
146 prop_binSR_electron2018_bin0
147 prop_binSR_electron2016M_bin9
148 prop_binCRF_muon2018_bin4
149 prop_binSR_muon2016M_bin13
150 prop_binSR_electron2016M_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



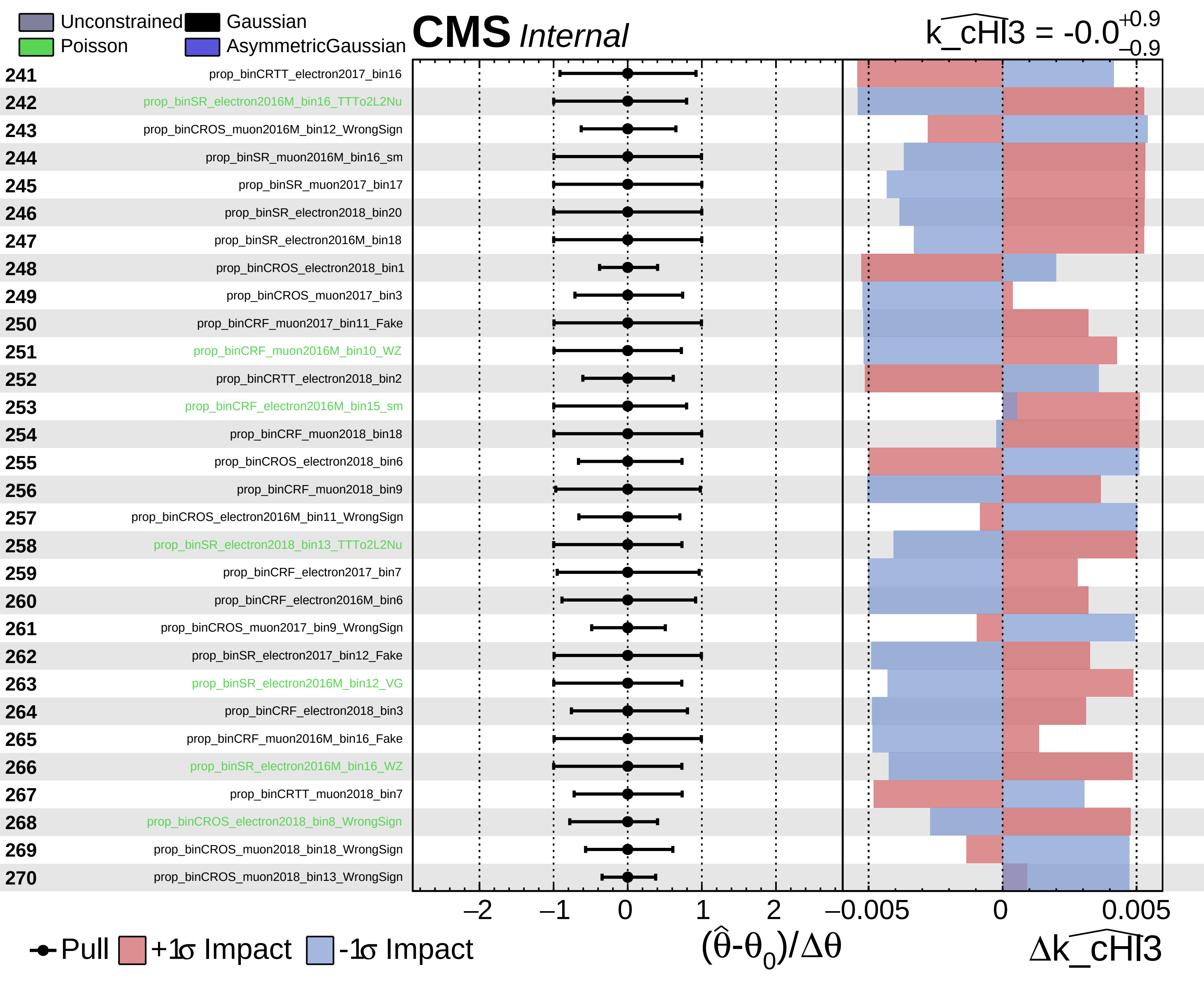
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
151 QCDScale_VG
152 prop_binSR_electron2018_bin13_sm
153 TTest_muon_2017
1.00 +0.05 −0.05
154 prop_binSR_electron2016M_bin17
155 prop_binSR_electron2017_bin15
156 prop_binSR_electron2017_bin7
157 prop_binCRF_muon2018_bin0
158 jer_2018
159 prop_binSR_electron2016M_bin5
160 prop_binCROS_muon2018_bin11_WrongSign
161 prop_binSR_electron2017_bin9
162 prop_binSR_muon2016M_bin0
163 prop_binSR_electron2017_bin5
164 prop_binSR_electron2016M_bin4
165 prop_binCRTT_muon2017_bin0
166 prop_binCRF_electron2017_bin0
167 prop_binSR_electron2016M_bin1_Fake
168 prop_binCROS_muon2017_bin2
169 QCDScale_Triboson
170 prop_binCROS_muon2017_bin8
171 prop_binSR_electron2017_bin16_sm
172 prop_binCRF_muon2016M_bin2
173 prop_binCROS_muon2016M_bin8
174 prop_binCRTT_electron2018_bin0
175 prop_binSR_electron2018_bin15
176 prop_binCRF_muon2017_bin3
177 prop_binSR_muon2017_bin0
178 prop_binCROS_muon2018_bin1
179 prop_binSR_electron2016M_bin11
180 prop_binCROS_muon2016M_bin4_WrongSign
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

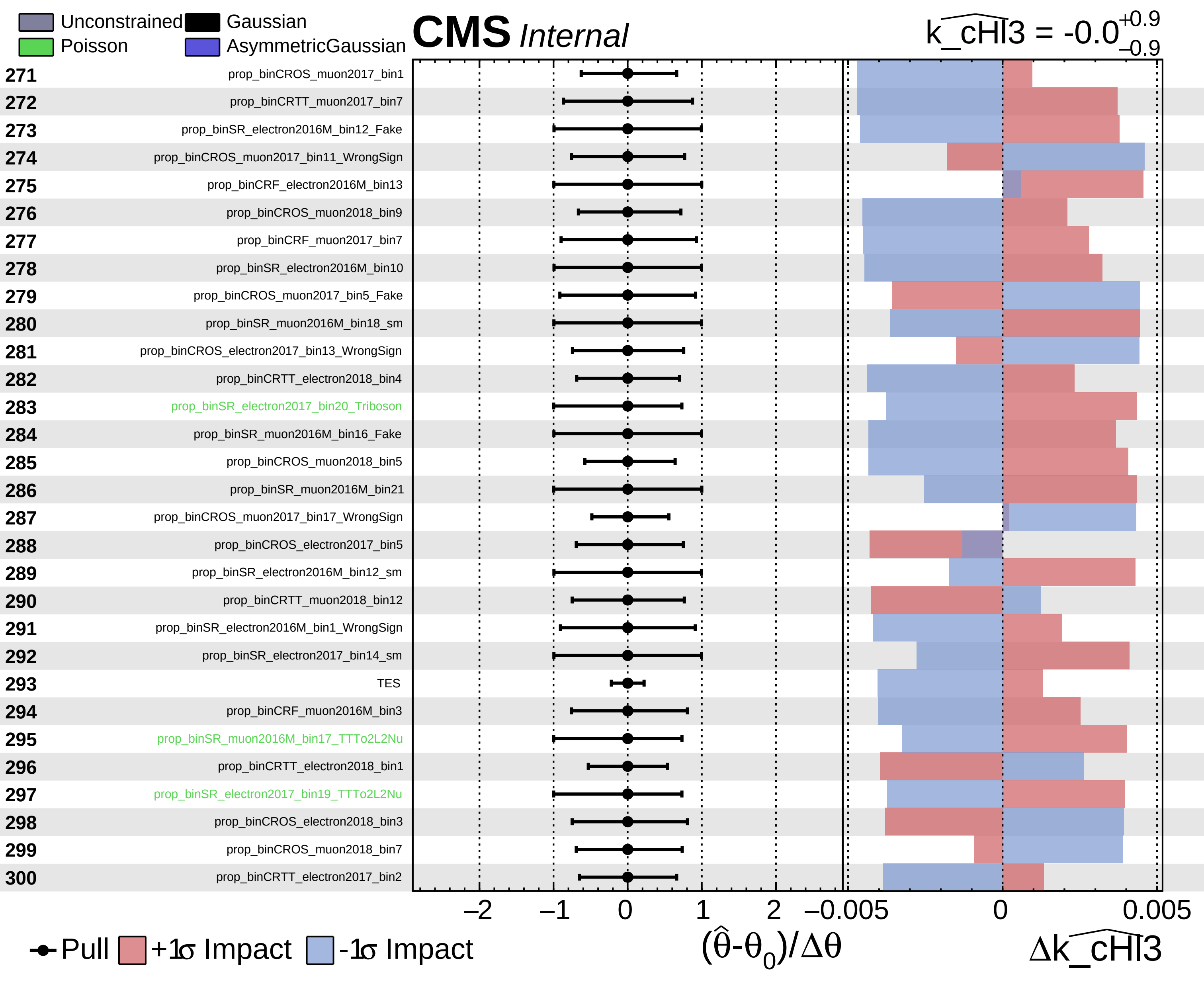


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
181 prop_binSR_muon2016M_bin20
182 prop_binCRF_electron2016M_bin0
183 prop_binSR_electron2016M_bin13
184 prop_binCROS_electron2017_bin12_WrongSign
185 prop_binSR_muon2016M_bin17_WZ
186 prop_binCROS_muon2018_bin14
187 prop_binSR_electron2016M_bin6
188 prop_binCROS_muon2017_bin7_WrongSign
189 prop_binSR_muon2016M_bin19_WrongSign
190 prop_binCRF_muon2016M_bin4
191 prop_binSR_electron2016M_bin8
192 prop_binSR_electron2016M_bin19
193 prop_binSR_electron2016M_bin14
194 prop_binSR_electron2016M_bin15
195 ISR
196 prop_binSR_electron2016M_bin7
197 prop_binCROS_muon2017_bin0
198 QCDScale_TTdilep
199 prop_binCRF_muon2018_bin7
200 prop_binCRF_electron2018_bin1
201 prop_binCRF_muon2018_bin19
202 prop_binSR_electron2017_bin10
203 prop_binCRF_electron2017_bin2
204 prop_binCRF_muon2018_bin5
205 prop_binSR_muon2016M_bin17_WrongSign
206 prop_binCRF_electron2017_bin3
207 puID
208 prop_binCROS_muon2017_bin10_WrongSign
209 prop_binCRF_electron2018_bin8
210 prop_binSR_muon2017_bin13_Fake
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

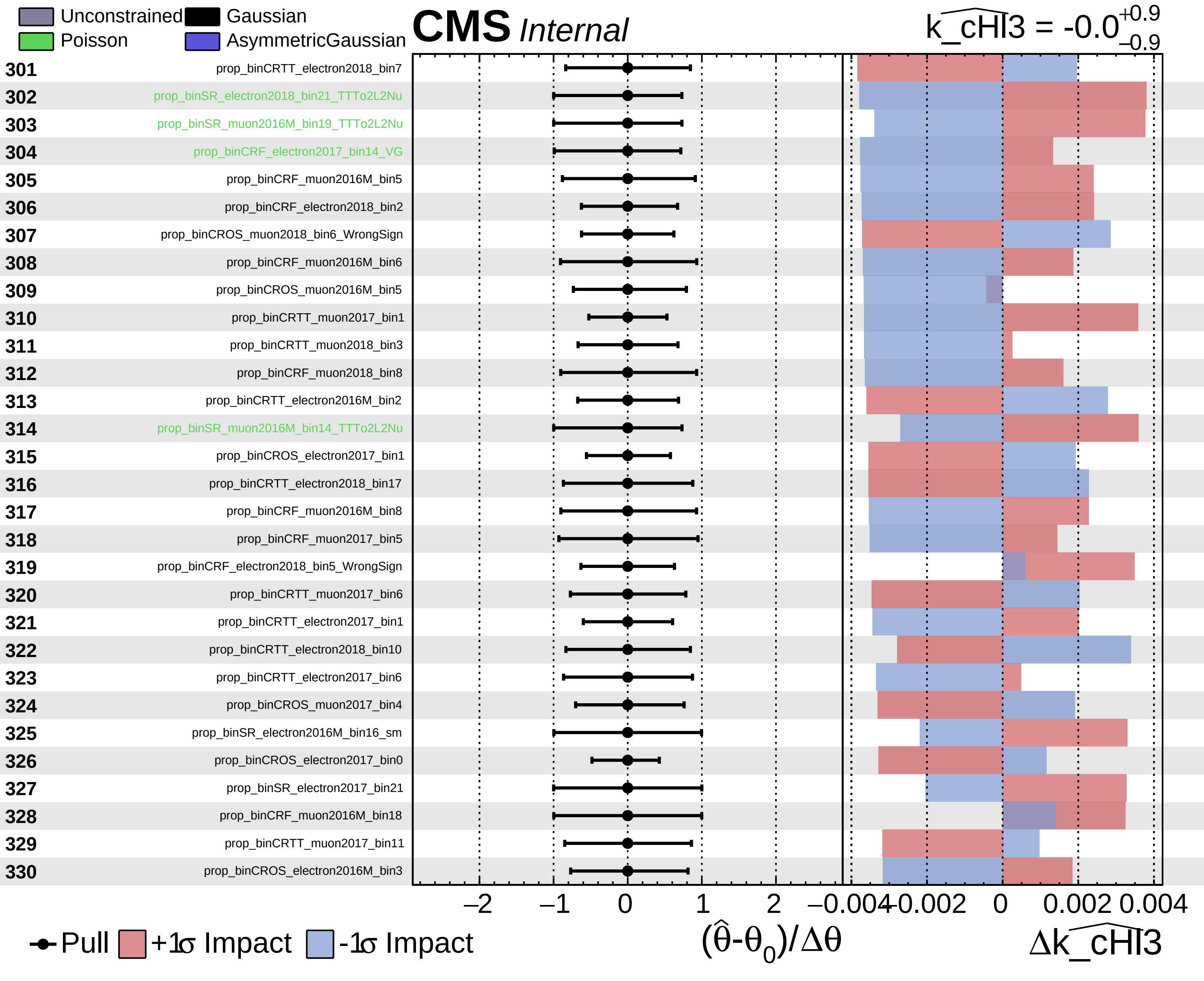


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
211 prop_binCROS_electron2016M_bin7_WrongSign
212 prop_binCRF_electron2017_bin5
213 prop_binCROS_electron2018_bin2
214 prop_binCRF_muon2017_bin6
215 prop_binSR_electron2017_bin13
216 prop_binCROS_electron2016M_bin2
217 prop_binCRF_electron2018_bin6
218 prop_binCRF_muon2017_bin11_sm
219 prop_binCRF_muon2017_bin4
220 prop_binCRTT_electron2016M_bin1
221 prop_binCROS_muon2016M_bin11_WrongSign
222 prop_binCROS_muon2016M_bin1
223 prop_binSR_muon2017_bin12
224 prop_binCRTT_muon2018_bin4
225 prop_binSR_electron2018_bin17_sm
226 prop_binCRF_muon2017_bin8_sm
227 prop_binCROS_electron2016M_bin1
228 prop_binCROS_electron2017_bin8
229 prop_binSR_muon2017_bin14
230 metUnclust
231 prop_binCRF_muon2018_bin13
232 prop_binCRF_muon2016M_bin11_Fake
233 prop_binCRF_muon2016M_bin0
234 prop_binCROS_muon2017_bin7_Fake
235 prop_binSR_electron2018_bin17_Fake
236 prop_binSR_electron2018_bin19
237 prop_binCRTT_muon2016M_bin1
238 prop_binSR_electron2016M_bin12_WZ
239 prop_binCROS_electron2017_bin2
240 prop_binCRF_muon2017_bin15
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

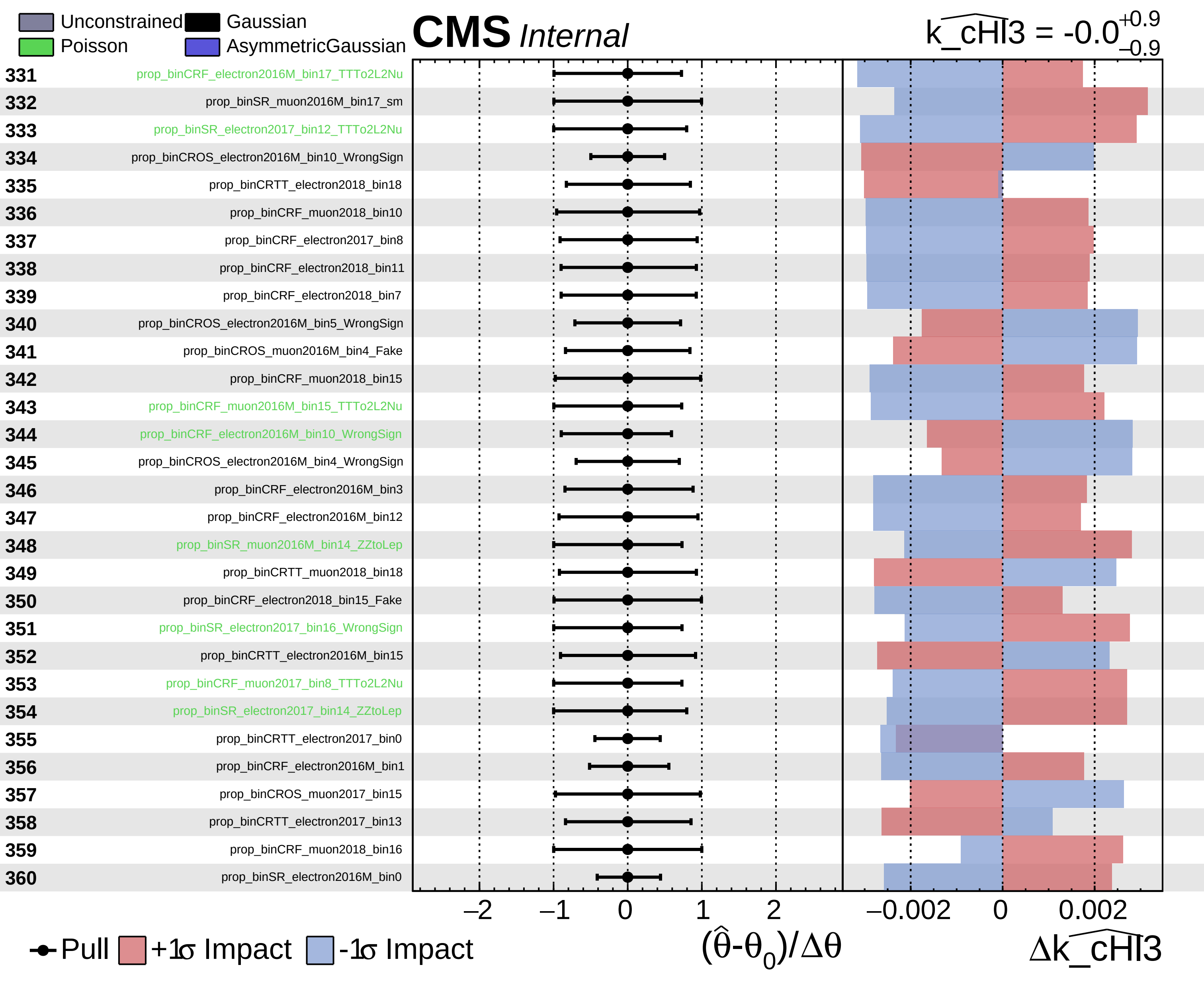

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
241 prop_binCRTT_electron2017_bin16
242 prop_binSR_electron2016M_bin16_TTTo2L2Nu
243 prop_binCROS_muon2016M_bin12_WrongSign
244 prop_binSR_muon2016M_bin16_sm
245 prop_binSR_muon2017_bin17
246 prop_binSR_electron2018_bin20
247 prop_binSR_electron2016M_bin18
248 prop_binCROS_electron2018_bin1
249 prop_binCROS_muon2017_bin3
250 prop_binCRF_muon2017_bin11_Fake
251 prop_binCRF_muon2016M_bin10_WZ
252 prop_binCRTT_electron2018_bin2
253 prop_binCRF_electron2016M_bin15_sm
254 prop_binCRF_muon2018_bin18
255 prop_binCROS_electron2018_bin6
256 prop_binCRF_muon2018_bin9
257 prop_binCROS_electron2016M_bin11_WrongSign
258 prop_binSR_electron2018_bin13_TTTo2L2Nu
259 prop_binCRF_electron2017_bin7
260 prop_binCRF_electron2016M_bin6
261 prop_binCROS_muon2017_bin9_WrongSign
262 prop_binSR_electron2017_bin12_Fake
263 prop_binSR_electron2016M_bin12_VG
264 prop_binCRF_electron2018_bin3
265 prop_binCRF_muon2016M_bin16_Fake
266 prop_binSR_electron2016M_bin16_WZ
267 prop_binCRTT_muon2018_bin7
268 prop_binCROS_electron2018_bin8_WrongSign
269 prop_binCROS_muon2018_bin18_WrongSign
270 prop_binCROS_muon2018_bin13_WrongSign
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



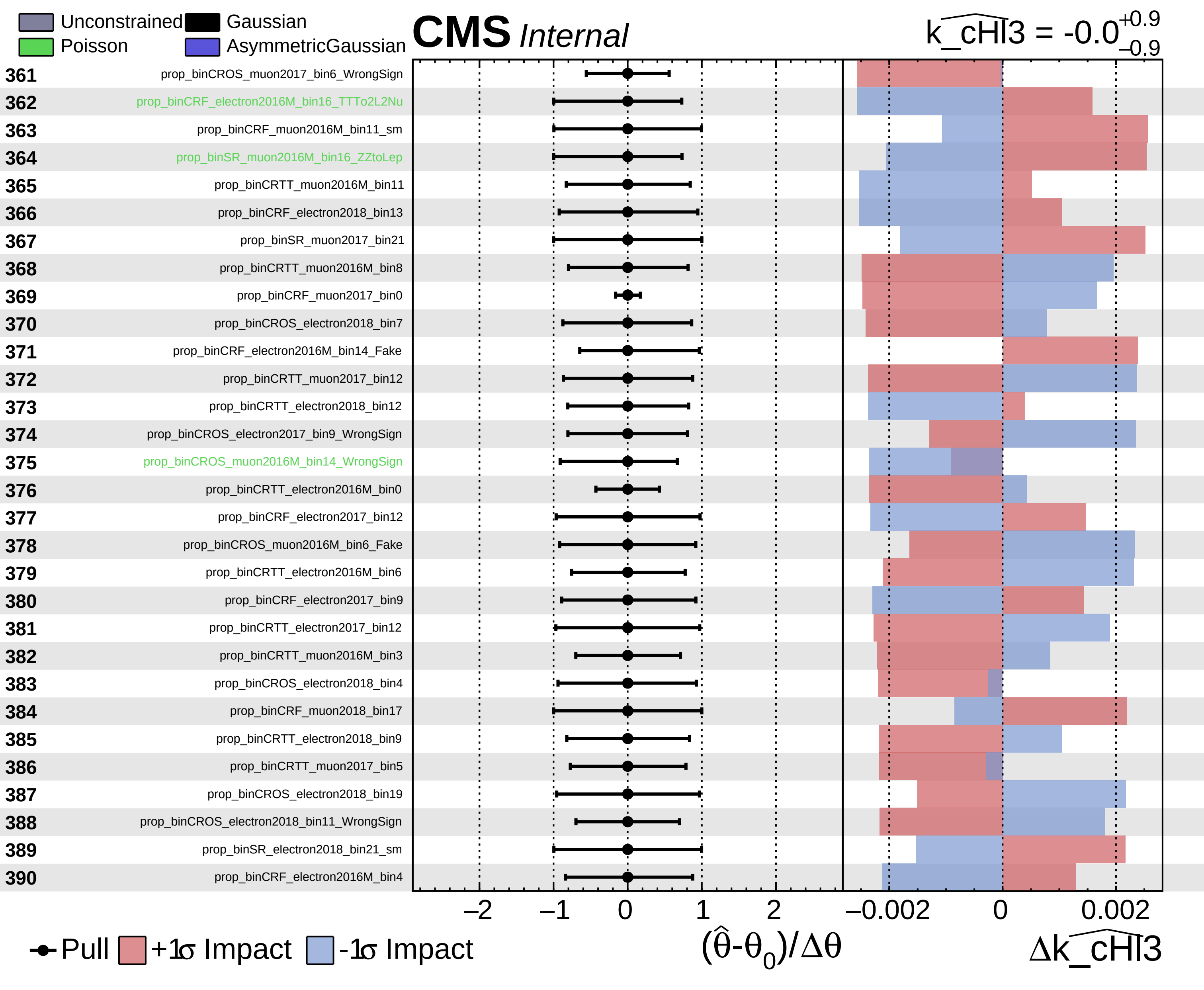
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
271 prop_binCROS_muon2017_bin1
272 prop_binCRTT_muon2017_bin7
273 prop_binSR_electron2016M_bin12_Fake
274 prop_binCROS_muon2017_bin11_WrongSign
275 prop_binCRF_electron2016M_bin13
276 prop_binCROS_muon2018_bin9
277 prop_binCRF_muon2017_bin7
278 prop_binSR_electron2016M_bin10
279 prop_binCROS_muon2017_bin5_Fake
280 prop_binSR_muon2016M_bin18_sm
281 prop_binCROS_electron2017_bin13_WrongSign
282 prop_binCRTT_electron2018_bin4
283 prop_binSR_electron2017_bin20_Triboson
284 prop_binSR_muon2016M_bin16_Fake
285 prop_binCROS_muon2018_bin5
286 prop_binSR_muon2016M_bin21
287 prop_binCROS_muon2017_bin17_WrongSign
288 prop_binCROS_electron2017_bin5
289 prop_binSR_electron2016M_bin12_sm
290 prop_binCRTT_muon2018_bin12
291 prop_binSR_electron2016M_bin1_WrongSign
292 prop_binSR_electron2017_bin14_sm
293 TES
294 prop_binCRF_muon2016M_bin3
295 prop_binSR_muon2016M_bin17_TTTo2L2Nu
296 prop_binCRTT_electron2018_bin1
297 prop_binSR_electron2017_bin19_TTTo2L2Nu
298 prop_binCROS_electron2018_bin3
299 prop_binCROS_muon2018_bin7
300 prop_binCRTT_electron2017_bin2
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



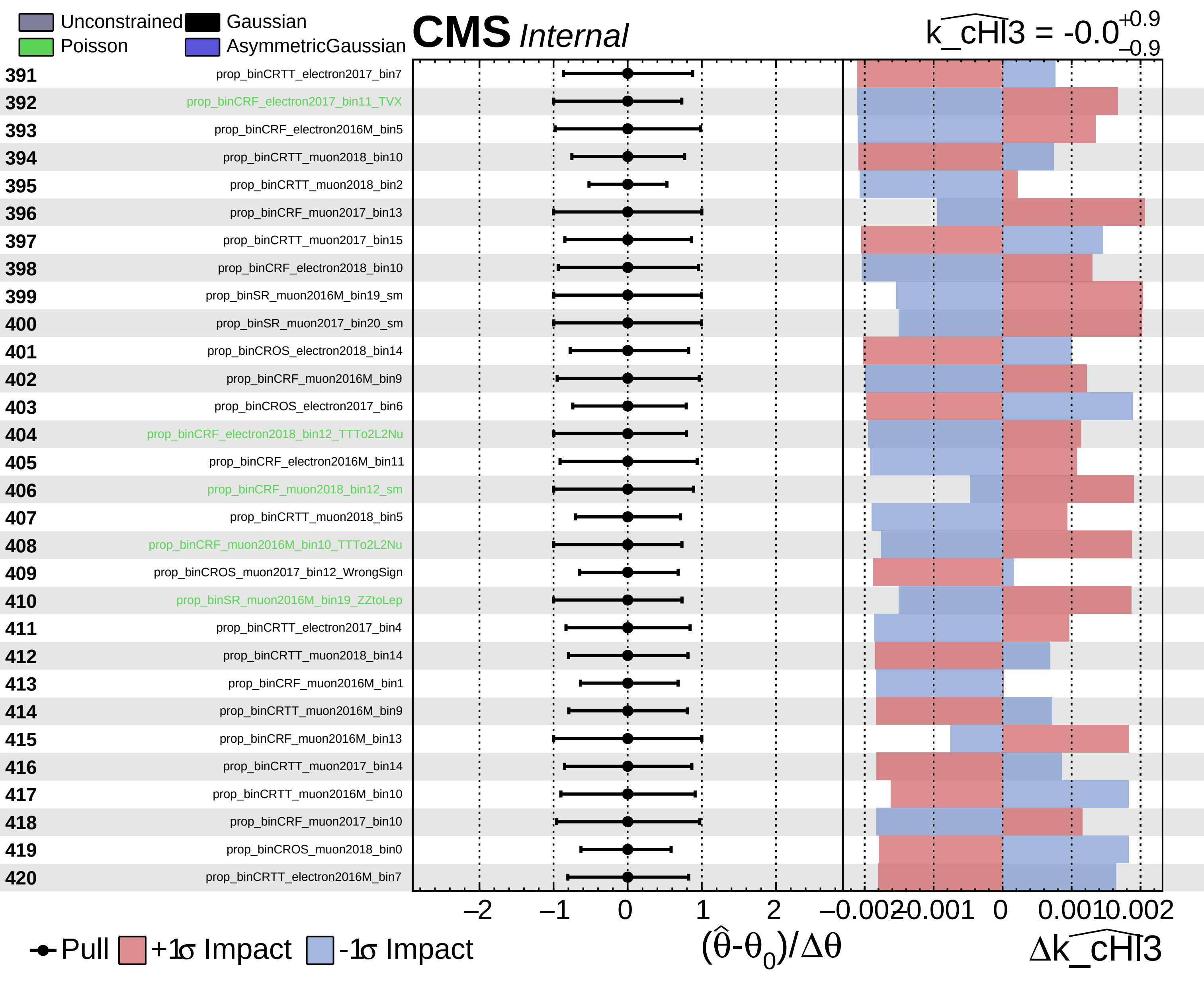
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
301 prop_binCRTT_electron2018_bin7
302 prop_binSR_electron2018_bin21_TTTo2L2Nu
303 prop_binSR_muon2016M_bin19_TTTo2L2Nu
304 prop_binCRF_electron2017_bin14_VG
305 prop_binCRF_muon2016M_bin5
306 prop_binCRF_electron2018_bin2
307 prop_binCROS_muon2018_bin6_WrongSign
308 prop_binCRF_muon2016M_bin6
309 prop_binCROS_muon2016M_bin5
310 prop_binCRTT_muon2017_bin1
311 prop_binCRTT_muon2018_bin3
312 prop_binCRF_muon2018_bin8
313 prop_binCRTT_electron2016M_bin2
314 prop_binSR_muon2016M_bin14_TTTo2L2Nu
315 prop_binCROS_electron2017_bin1
316 prop_binCRTT_electron2018_bin17
317 prop_binCRF_muon2016M_bin8
318 prop_binCRF_muon2017_bin5
319 prop_binCRF_electron2018_bin5_WrongSign
320 prop_binCRTT_muon2017_bin6
321 prop_binCRTT_electron2017_bin1
322 prop_binCRTT_electron2018_bin10
323 prop_binCRTT_electron2017_bin6
324 prop_binCROS_muon2017_bin4
325 prop_binSR_electron2016M_bin16_sm
326 prop_binCROS_electron2017_bin0
327 prop_binSR_electron2017_bin21
328 prop_binCRF_muon2016M_bin18
329 prop_binCRTT_muon2017_bin11
330 prop_binCROS_electron2016M_bin3
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3

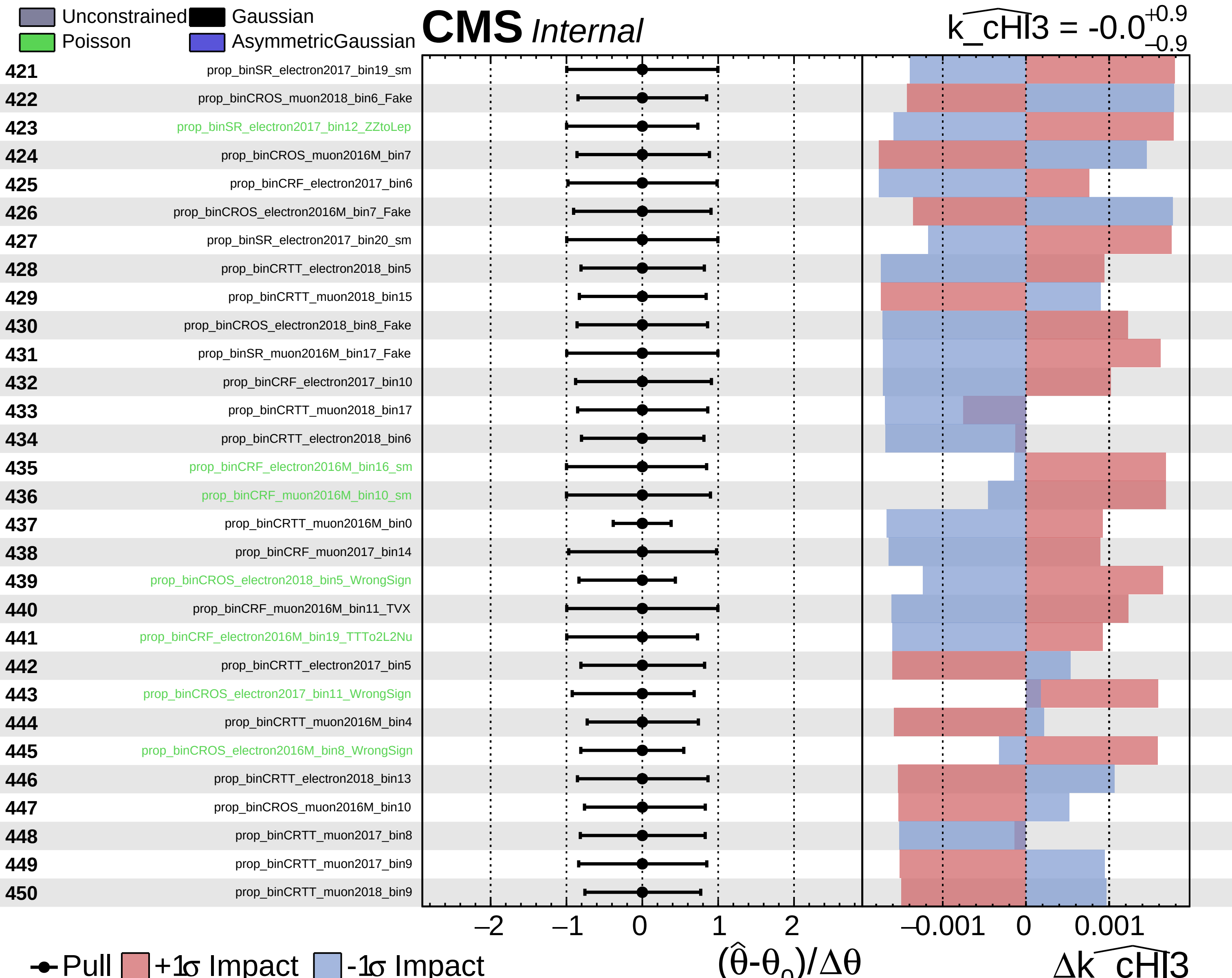


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
331 prop_binCRF_electron2016M_bin17_TTTo2L2Nu
332 prop_binSR_muon2016M_bin17_sm
333 prop_binSR_electron2017_bin12_TTTo2L2Nu
334 prop_binCROS_electron2016M_bin10_WrongSign
335 prop_binCRTT_electron2018_bin18
336 prop_binCRF_muon2018_bin10
337 prop_binCRF_electron2017_bin8
338 prop_binCRF_electron2018_bin11
339 prop_binCRF_electron2018_bin7
340 prop_binCROS_electron2016M_bin5_WrongSign
341 prop_binCROS_muon2016M_bin4_Fake
342 prop_binCRF_muon2018_bin15
343 prop_binCRF_muon2016M_bin15_TTTo2L2Nu
344 prop_binCRF_electron2016M_bin10_WrongSign
345 prop_binCROS_electron2016M_bin4_WrongSign
346 prop_binCRF_electron2016M_bin3
347 prop_binCRF_electron2016M_bin12
348 prop_binSR_muon2016M_bin14_ZZtoLep
349 prop_binCRTT_muon2018_bin18
350 prop_binCRF_electron2018_bin15_Fake
351 prop_binSR_electron2017_bin16_WrongSign
352 prop_binCRTT_electron2016M_bin15
353 prop_binCRF_muon2017_bin8_TTTo2L2Nu
354 prop_binSR_electron2017_bin14_ZZtoLep
355 prop_binCRTT_electron2017_bin0
356 prop_binCRF_electron2016M_bin1
357 prop_binCROS_muon2017_bin15
358 prop_binCRTT_electron2017_bin13
359 prop_binCRF_muon2018_bin16
360 prop_binSR_electron2016M_bin0
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

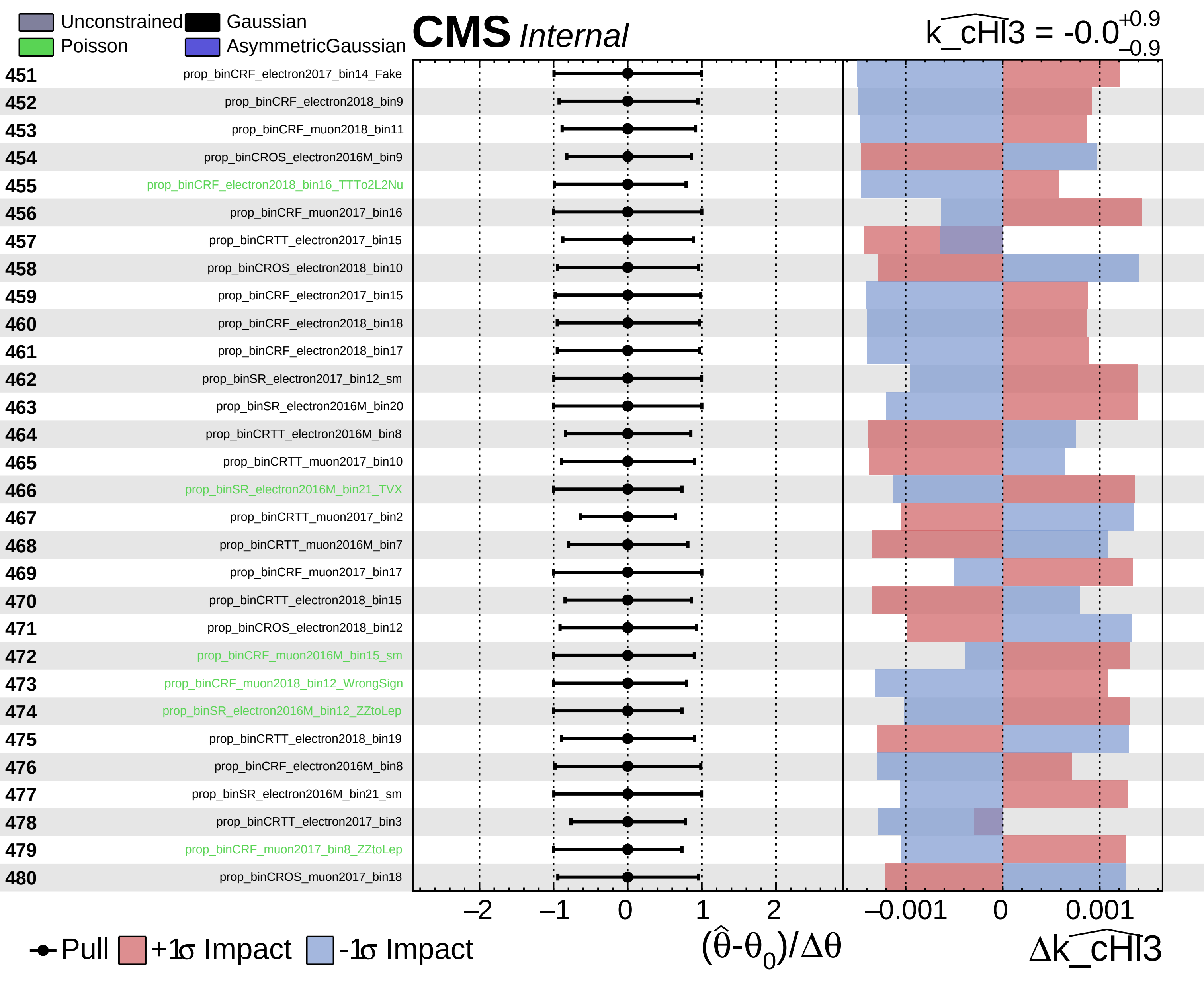


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
361 prop_binCROS_muon2017_bin6_WrongSign
362 prop_binCRF_electron2016M_bin16_TTTo2L2Nu
363 prop_binCRF_muon2016M_bin11_sm
364 prop_binSR_muon2016M_bin16_ZZtoLep
365 prop_binCRTT_muon2016M_bin11
366 prop_binCRF_electron2018_bin13
367 prop_binSR_muon2017_bin21
368 prop_binCRTT_muon2016M_bin8
369 prop_binCRF_muon2017_bin0
370 prop_binCROS_electron2018_bin7
371 prop_binCRF_electron2016M_bin14_Fake
372 prop_binCRTT_muon2017_bin12
373 prop_binCRTT_electron2018_bin12
374 prop_binCROS_electron2017_bin9_WrongSign
375 prop_binCROS_muon2016M_bin14_WrongSign
376 prop_binCRTT_electron2016M_bin0
377 prop_binCRF_electron2017_bin12
378 prop_binCROS_muon2016M_bin6_Fake
379 prop_binCRTT_electron2016M_bin6
380 prop_binCRF_electron2017_bin9
381 prop_binCRTT_electron2017_bin12
382 prop_binCRTT_muon2016M_bin3
383 prop_binCROS_electron2018_bin4
384 prop_binCRF_muon2018_bin17
385 prop_binCRTT_electron2018_bin9
386 prop_binCRTT_muon2017_bin5
387 prop_binCROS_electron2018_bin19
388 prop_binCROS_electron2018_bin11_WrongSign
389 prop_binSR_electron2018_bin21_sm
390 prop_binCRF_electron2016M_bin4
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3

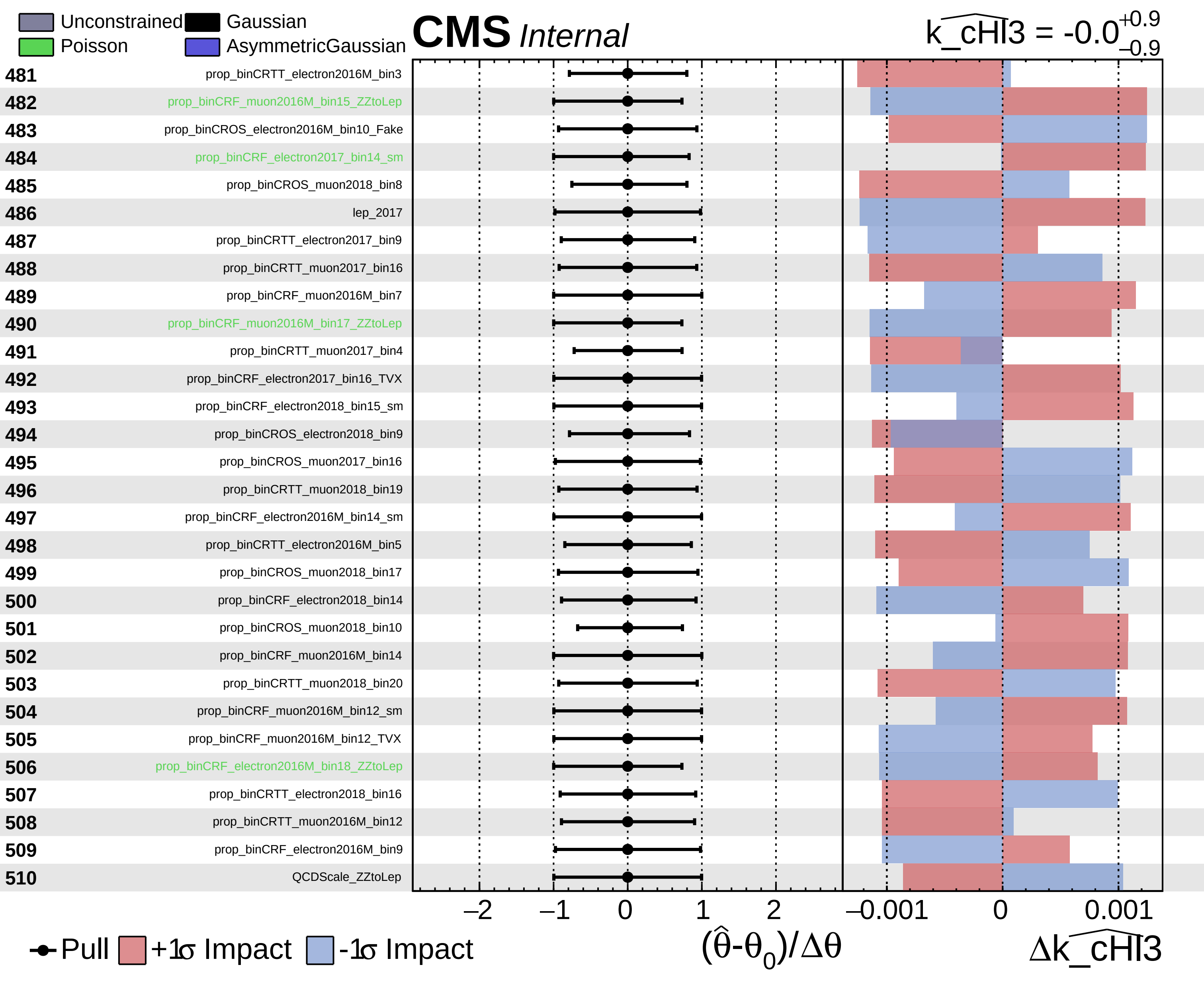

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
391 prop_binCRTT_electron2017_bin7
392 prop_binCRF_electron2017_bin11_TVX
393 prop_binCRF_electron2016M_bin5
394 prop_binCRTT_muon2018_bin10
395 prop_binCRTT_muon2018_bin2
396 prop_binCRF_muon2017_bin13
397 prop_binCRTT_muon2017_bin15
398 prop_binCRF_electron2018_bin10
399 prop_binSR_muon2016M_bin19_sm
400 prop_binSR_muon2017_bin20_sm
401 prop_binCROS_electron2018_bin14
402 prop_binCRF_muon2016M_bin9
403 prop_binCROS_electron2017_bin6
404 prop_binCRF_electron2018_bin12_TTTo2L2Nu
405 prop_binCRF_electron2016M_bin11
406 prop_binCRF_muon2018_bin12_sm
407 prop_binCRTT_muon2018_bin5
408 prop_binCRF_muon2016M_bin10_TTTo2L2Nu
409 prop_binCROS_muon2017_bin12_WrongSign
410 prop_binSR_muon2016M_bin19_ZZtoLep
411 prop_binCRTT_electron2017_bin4
412 prop_binCRTT_muon2018_bin14
413 prop_binCRF_muon2016M_bin1
414 prop_binCRTT_muon2016M_bin9
415 prop_binCRF_muon2016M_bin13
416 prop_binCRTT_muon2017_bin14
417 prop_binCRTT_muon2016M_bin10
418 prop_binCRF_muon2017_bin10
419 prop_binCROS_muon2018_bin0
420 prop_binCRTT_electron2016M_bin7
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



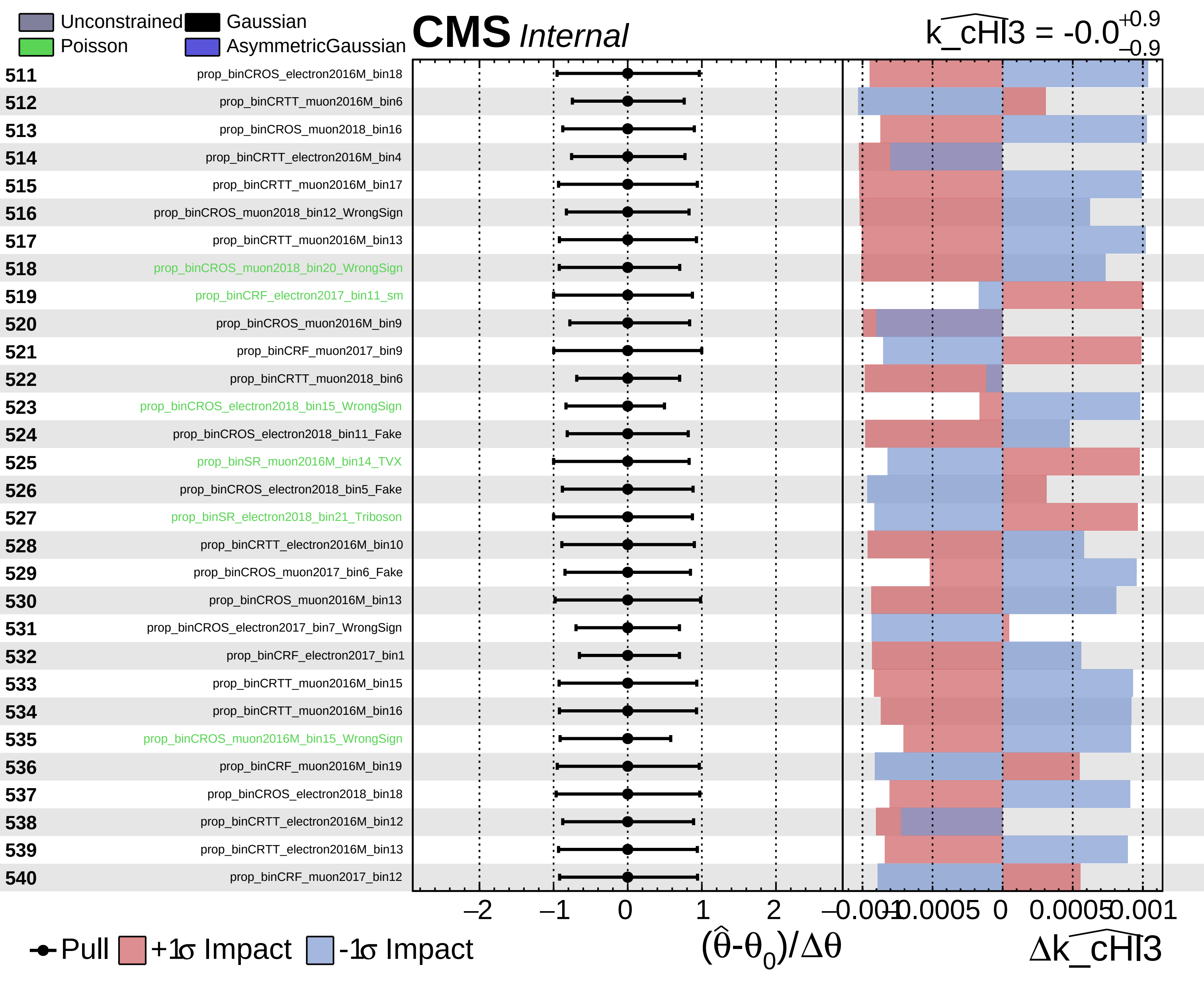
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
421 prop_binSR_electron2017_bin19_sm
422 prop_binCROS_muon2018_bin6_Fake
423 prop_binSR_electron2017_bin12_ZZtoLep
424 prop_binCROS_muon2016M_bin7
425 prop_binCRF_electron2017_bin6
426 prop_binCROS_electron2016M_bin7_Fake
427 prop_binSR_electron2017_bin20_sm
428 prop_binCRTT_electron2018_bin5
429 prop_binCRTT_muon2018_bin15
430 prop_binCROS_electron2018_bin8_Fake
431 prop_binSR_muon2016M_bin17_Fake
432 prop_binCRF_electron2017_bin10
433 prop_binCRTT_muon2018_bin17
434 prop_binCRTT_electron2018_bin6
435 prop_binCRF_electron2016M_bin16_sm
436 prop_binCRF_muon2016M_bin10_sm
437 prop_binCRTT_muon2016M_bin0
438 prop_binCRF_muon2017_bin14
439 prop_binCROS_electron2018_bin5_WrongSign
440 prop_binCRF_muon2016M_bin11_TVX
441 prop_binCRF_electron2016M_bin19_TTTo2L2Nu
442 prop_binCRTT_electron2017_bin5
443 prop_binCROS_electron2017_bin11_WrongSign
444 prop_binCRTT_muon2016M_bin4
445 prop_binCROS_electron2016M_bin8_WrongSign
446 prop_binCRTT_electron2018_bin13
447 prop_binCROS_muon2016M_bin10
448 prop_binCRTT_muon2017_bin8
449 prop_binCRTT_muon2017_bin9
450 prop_binCRTT_muon2018_bin9
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



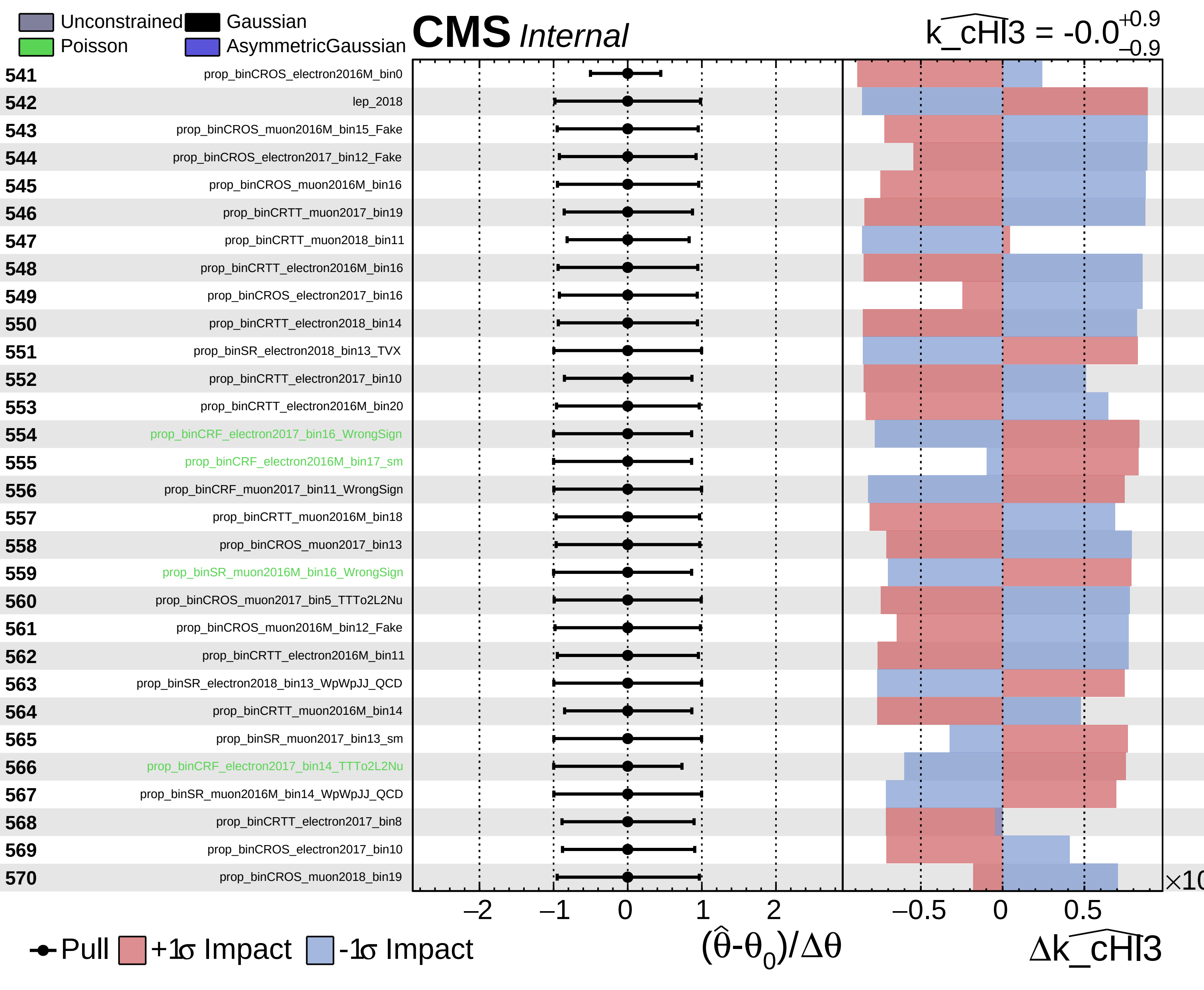
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0+0.9−0.9
451 prop_binCRF_electron2017_bin14_Fake
452 prop_binCRF_electron2018_bin9
453 prop_binCRF_muon2018_bin11
454 prop_binCROS_electron2016M_bin9
455 prop_binCRF_electron2018_bin16_TTTo2L2Nu
456 prop_binCRF_muon2017_bin16
457 prop_binCRTT_electron2017_bin15
458 prop_binCROS_electron2018_bin10
459 prop_binCRF_electron2017_bin15
460 prop_binCRF_electron2018_bin18
461 prop_binCRF_electron2018_bin17
462 prop_binSR_electron2017_bin12_sm
463 prop_binSR_electron2016M_bin20
464 prop_binCRTT_electron2016M_bin8
465 prop_binCRTT_muon2017_bin10
466 prop_binSR_electron2016M_bin21_TVX
467 prop_binCRTT_muon2017_bin2
468 prop_binCRTT_muon2016M_bin7
469 prop_binCRF_muon2017_bin17
470 prop_binCRTT_electron2018_bin15
471 prop_binCROS_electron2018_bin12
472 prop_binCRF_muon2016M_bin15_sm
473 prop_binCRF_muon2018_bin12_WrongSign
474 prop_binSR_electron2016M_bin12_ZZtoLep
475 prop_binCRTT_electron2018_bin19
476 prop_binCRF_electron2016M_bin8
477 prop_binSR_electron2016M_bin21_sm
478 prop_binCRTT_electron2017_bin3
479 prop_binCRF_muon2017_bin8_ZZtoLep
480 prop_binCROS_muon2017_bin18
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



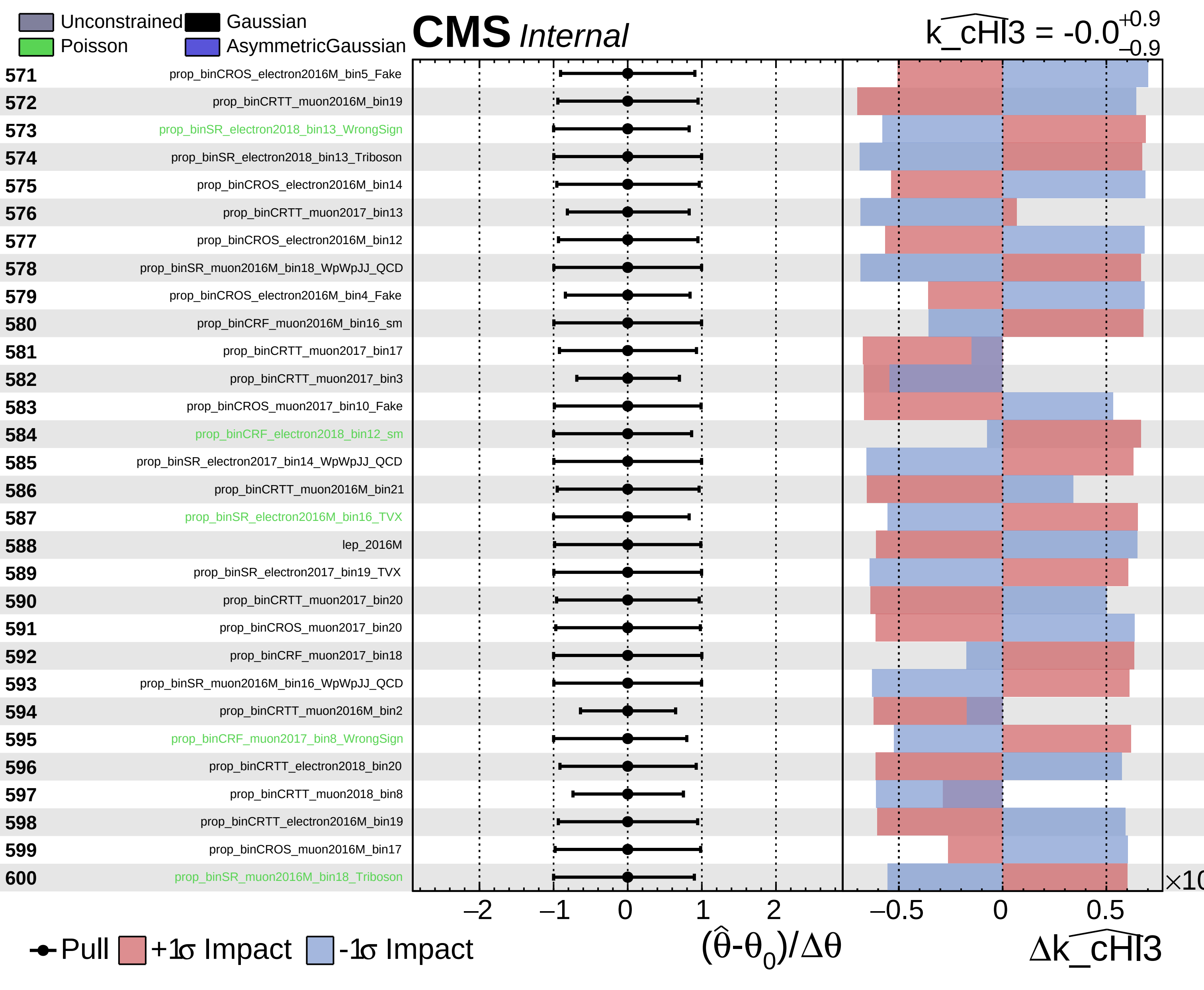
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
481 prop_binCRTT_electron2016M_bin3
482 prop_binCRF_muon2016M_bin15_ZZtoLep
483 prop_binCROS_electron2016M_bin10_Fake
484 prop_binCRF_electron2017_bin14_sm
485 prop_binCROS_muon2018_bin8
486 lep_2017
487 prop_binCRTT_electron2017_bin9
488 prop_binCRTT_muon2017_bin16
489 prop_binCRF_muon2016M_bin7
490 prop_binCRF_muon2016M_bin17_ZZtoLep
491 prop_binCRTT_muon2017_bin4
492 prop_binCRF_electron2017_bin16_TVX
493 prop_binCRF_electron2018_bin15_sm
494 prop_binCROS_electron2018_bin9
495 prop_binCROS_muon2017_bin16
496 prop_binCRTT_muon2018_bin19
497 prop_binCRF_electron2016M_bin14_sm
498 prop_binCRTT_electron2016M_bin5
499 prop_binCROS_muon2018_bin17
500 prop_binCRF_electron2018_bin14
501 prop_binCROS_muon2018_bin10
502 prop_binCRF_muon2016M_bin14
503 prop_binCRTT_muon2018_bin20
504 prop_binCRF_muon2016M_bin12_sm
505 prop_binCRF_muon2016M_bin12_TVX
506 prop_binCRF_electron2016M_bin18_ZZtoLep
507 prop_binCRTT_electron2018_bin16
508 prop_binCRTT_muon2016M_bin12
509 prop_binCRF_electron2016M_bin9
510 QCDScale_ZZtoLep
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



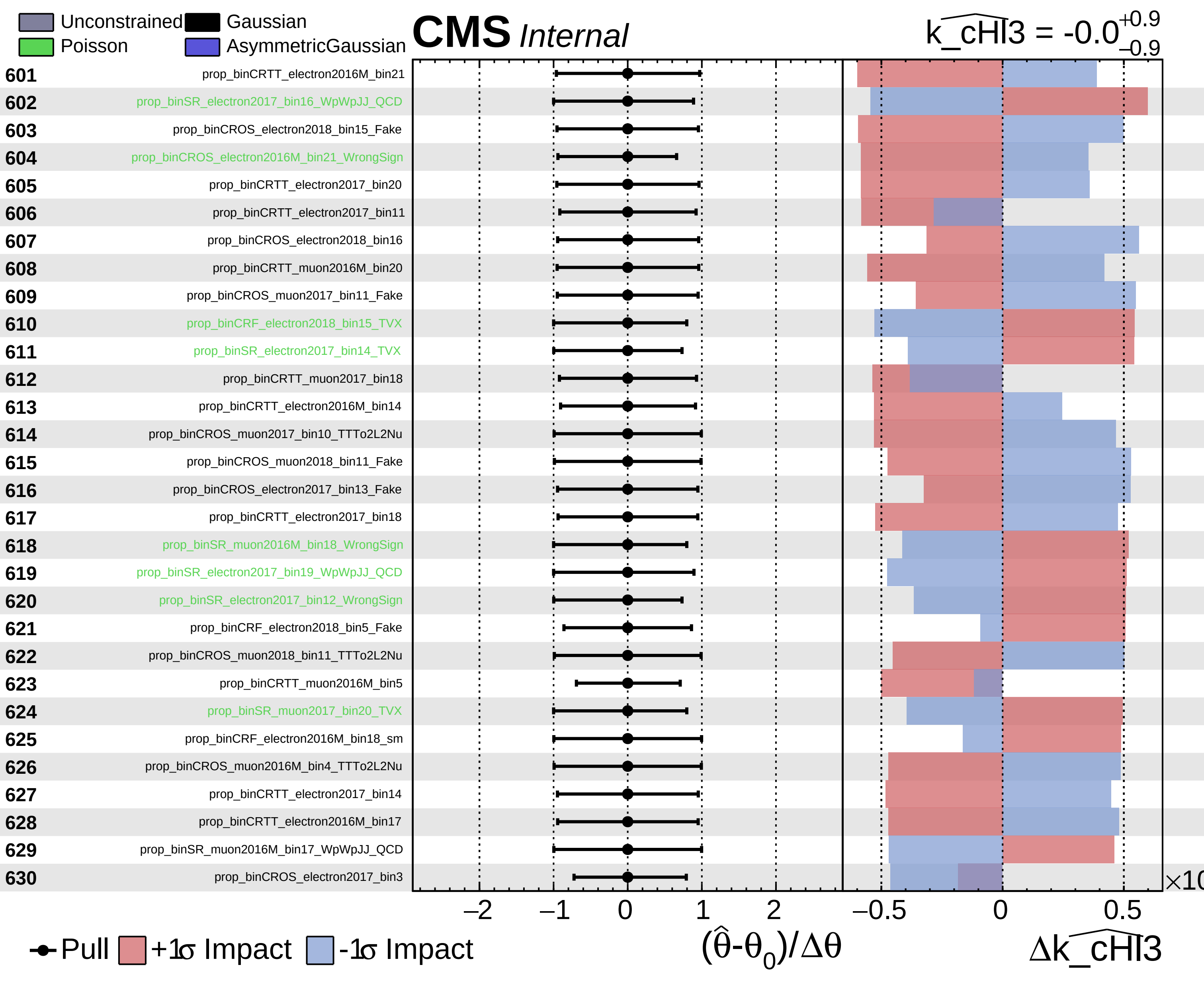
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
511 prop_binCROS_electron2016M_bin18
512 prop_binCRTT_muon2016M_bin6
513 prop_binCROS_muon2018_bin16
514 prop_binCRTT_electron2016M_bin4
515 prop_binCRTT_muon2016M_bin17
516 prop_binCROS_muon2018_bin12_WrongSign
517 prop_binCRTT_muon2016M_bin13
518 prop_binCROS_muon2018_bin20_WrongSign
519 prop_binCRF_electron2017_bin11_sm
520 prop_binCROS_muon2016M_bin9
521 prop_binCRF_muon2017_bin9
522 prop_binCRTT_muon2018_bin6
523 prop_binCROS_electron2018_bin15_WrongSign
524 prop_binCROS_electron2018_bin11_Fake
525 prop_binSR_muon2016M_bin14_TVX
526 prop_binCROS_electron2018_bin5_Fake
527 prop_binSR_electron2018_bin21_Triboson
528 prop_binCRTT_electron2016M_bin10
529 prop_binCROS_muon2017_bin6_Fake
530 prop_binCROS_muon2016M_bin13
531 prop_binCROS_electron2017_bin7_WrongSign
532 prop_binCRF_electron2017_bin1
533 prop_binCRTT_muon2016M_bin15
534 prop_binCRTT_muon2016M_bin16
535 prop_binCROS_muon2016M_bin15_WrongSign
536 prop_binCRF_muon2016M_bin19
537 prop_binCROS_electron2018_bin18
538 prop_binCRTT_electron2016M_bin12
539 prop_binCRTT_electron2016M_bin13
540 prop_binCRF_muon2017_bin12
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



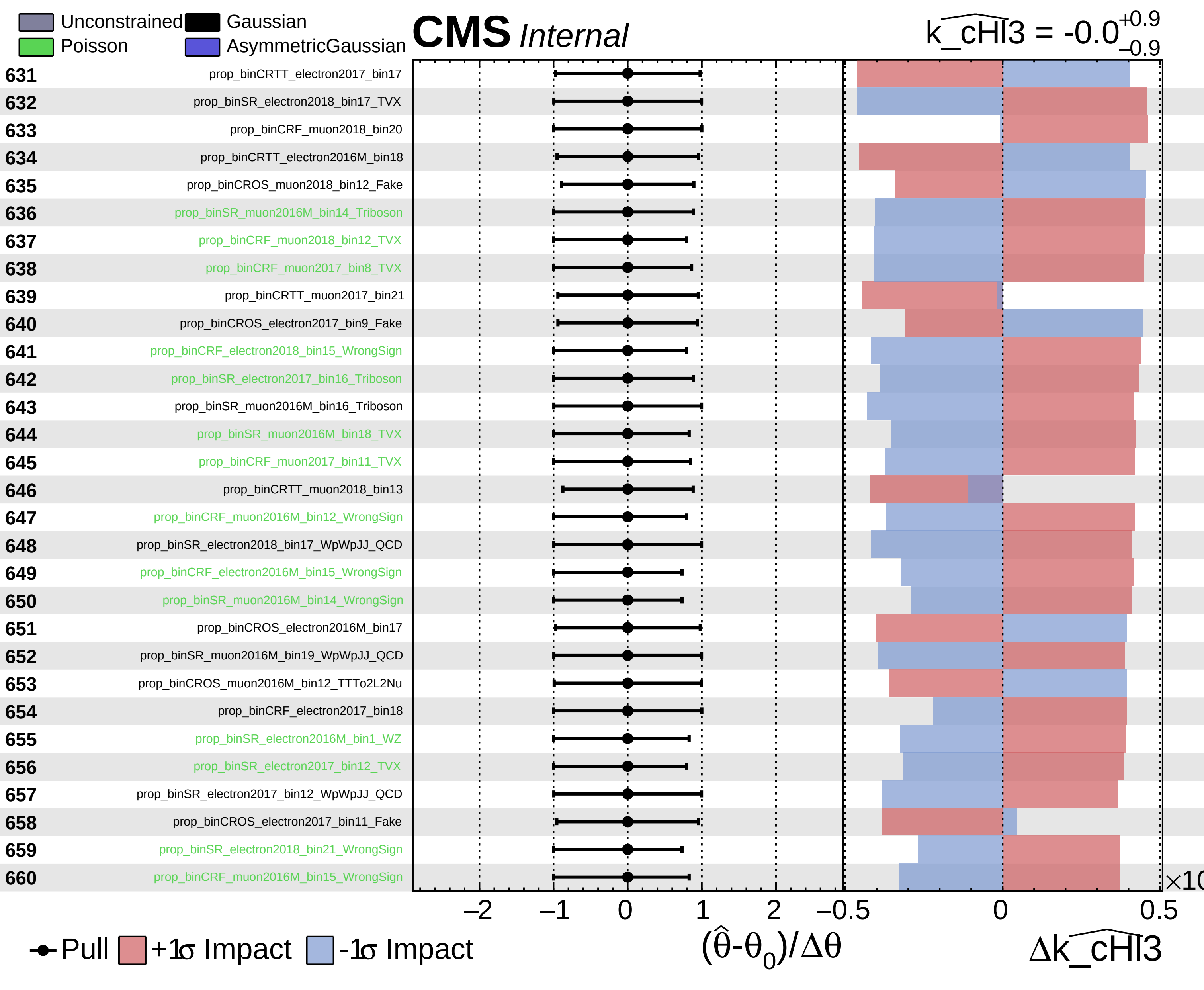
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHl3} = -0.0^{+0.9}_{-0.9}$
541 prop_binCROS_electron2016M_bin0
542 lep_2018
543 prop_binCROS_muon2016M_bin15_Fake
544 prop_binCROS_electron2017_bin12_Fake
545 prop_binCROS_muon2016M_bin16
546 prop_binCRTT_muon2017_bin19
547 prop_binCRTT_muon2018_bin11
548 prop_binCRTT_electron2016M_bin16
549 prop_binCROS_electron2017_bin16
550 prop_binCRTT_electron2018_bin14
551 prop_binSR_electron2018_bin13_TVX
552 prop_binCRTT_electron2017_bin10
553 prop_binCRTT_electron2016M_bin20
554 prop_binCRF_electron2017_bin16_WrongSign
555 prop_binCRF_electron2016M_bin17_sm
556 prop_binCRF_muon2017_bin11_WrongSign
557 prop_binCRTT_muon2016M_bin18
558 prop_binCROS_muon2017_bin13
559 prop_binSR_muon2016M_bin16_WrongSign
560 prop_binCROS_muon2017_bin5_TTTo2L2Nu
561 prop_binCROS_muon2016M_bin12_Fake
562 prop_binCRTT_electron2016M_bin11
563 prop_binSR_electron2018_bin13_WpWpJJ_QCD
564 prop_binCRTT_muon2016M_bin14
565 prop_binSR_muon2017_bin13_sm
566 prop_binCRF_electron2017_bin14_TTTo2L2Nu
567 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
568 prop_binCRTT_electron2017_bin8
569 prop_binCROS_electron2017_bin10
570 prop_binCROS_muon2018_bin19
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHl3}$



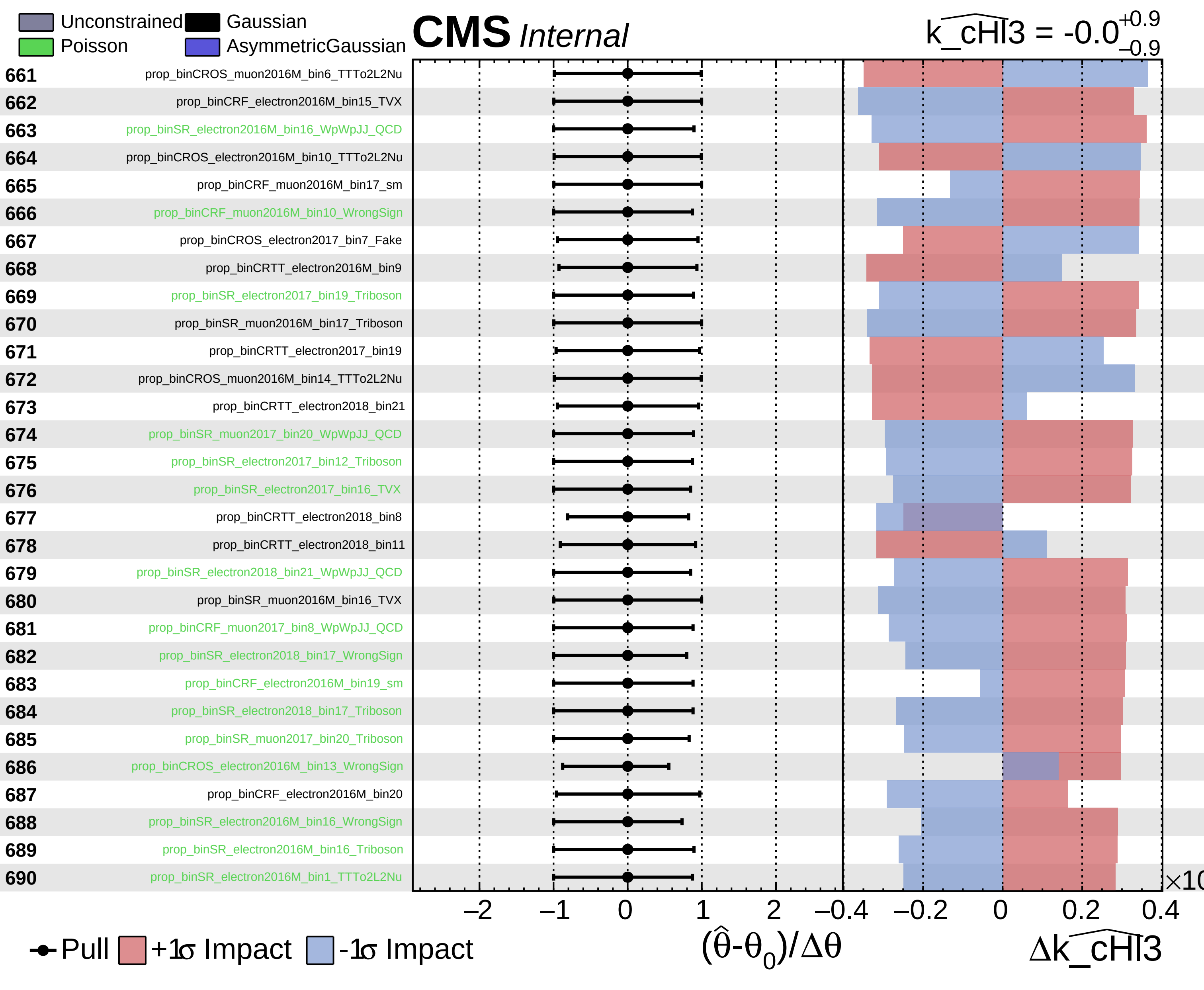
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
571 prop_binCROS_electron2016M_bin5_Fake
572 prop_binCRTT_muon2016M_bin19
573 prop_binSR_electron2018_bin13_WrongSign
574 prop_binSR_electron2018_bin13_Triboson
575 prop_binCROS_electron2016M_bin14
576 prop_binCRTT_muon2017_bin13
577 prop_binCROS_electron2016M_bin12
578 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
579 prop_binCROS_electron2016M_bin4_Fake
580 prop_binCRF_muon2016M_bin16_sm
581 prop_binCRTT_muon2017_bin17
582 prop_binCRTT_muon2017_bin3
583 prop_binCROS_muon2017_bin10_Fake
584 prop_binCRF_electron2018_bin12_sm
585 prop_binSR_electron2017_bin14_WpWpJJ_QCD
586 prop_binCRTT_muon2016M_bin21
587 prop_binSR_electron2016M_bin16_TVX
588 lep_2016M
589 prop_binSR_electron2017_bin19_TVX
590 prop_binCRTT_muon2017_bin20
591 prop_binCROS_muon2017_bin20
592 prop_binCRF_muon2017_bin18
593 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
594 prop_binCRTT_muon2016M_bin2
595 prop_binCRF_muon2017_bin8_WrongSign
596 prop_binCRTT_electron2018_bin20
597 prop_binCRTT_muon2018_bin8
598 prop_binCRTT_electron2016M_bin19
599 prop_binCROS_muon2016M_bin17
600 prop_binSR_muon2016M_bin18_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



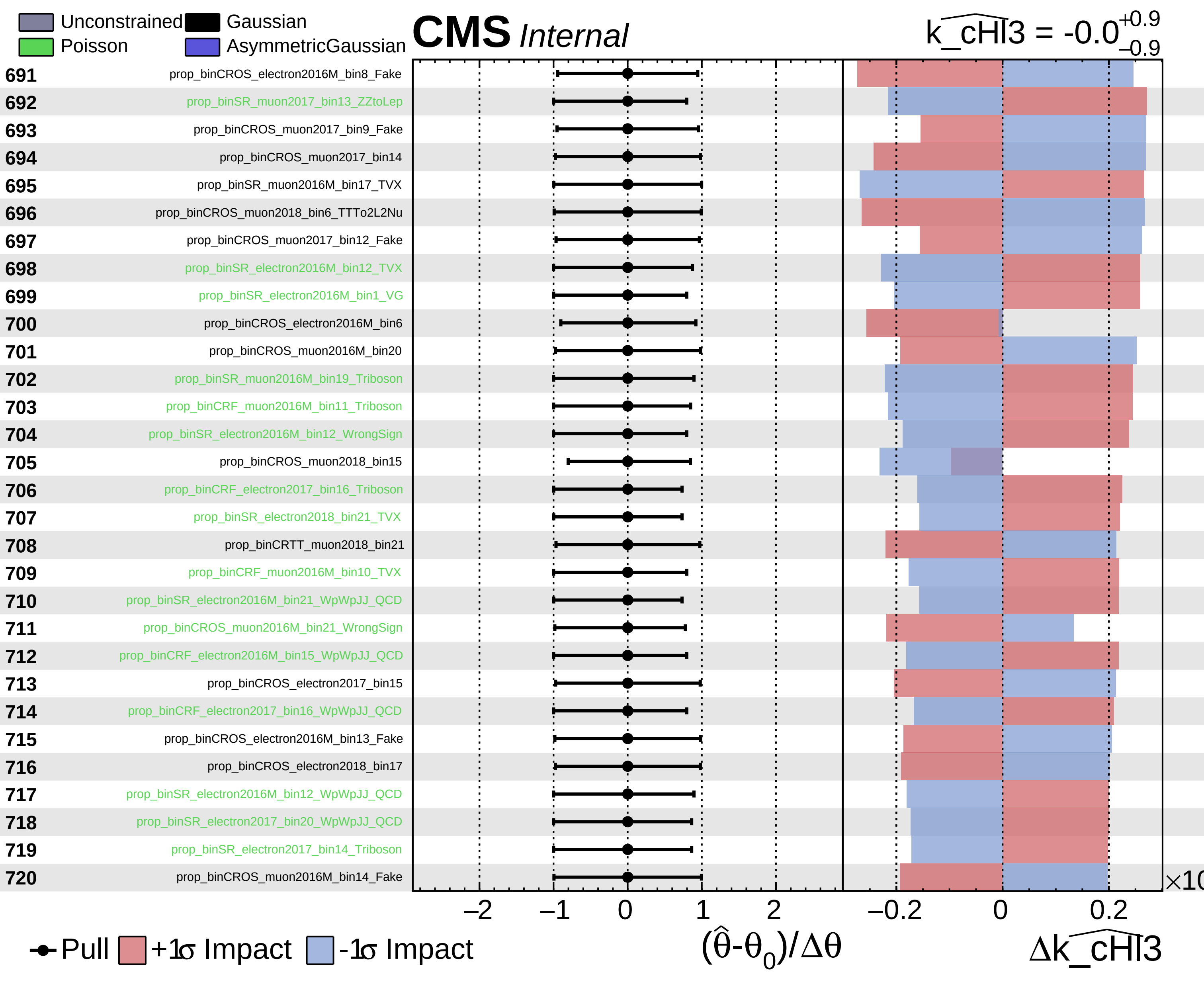
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHl3} = -0.0^{+0.9}_{-0.9}$
601 prop_binCRTT_electron2016M_bin21
602 prop_binSR_electron2017_bin16_WpWpJJ_QCD
603 prop_binCROS_electron2018_bin15_Fake
604 prop_binCROS_electron2016M_bin21_WrongSign
605 prop_binCRTT_electron2017_bin20
606 prop_binCRTT_electron2017_bin11
607 prop_binCROS_electron2018_bin16
608 prop_binCRTT_muon2016M_bin20
609 prop_binCROS_muon2017_bin11_Fake
610 prop_binCRF_electron2018_bin15_TVX
611 prop_binSR_electron2017_bin14_TVX
612 prop_binCRTT_muon2017_bin18
613 prop_binCRTT_electron2016M_bin14
614 prop_binCROS_muon2017_bin10_TTTo2L2Nu
615 prop_binCROS_muon2018_bin11_Fake
616 prop_binCROS_electron2017_bin13_Fake
617 prop_binCRTT_electron2017_bin18
618 prop_binSR_muon2016M_bin18_WrongSign
619 prop_binSR_electron2017_bin19_WpWpJJ_QCD
620 prop_binSR_electron2017_bin12_WrongSign
621 prop_binCRF_electron2018_bin5_Fake
622 prop_binCROS_muon2018_bin11_TTTo2L2Nu
623 prop_binCRTT_muon2016M_bin5
624 prop_binSR_muon2017_bin20_TVX
625 prop_binCRF_electron2016M_bin18_sm
626 prop_binCROS_muon2016M_bin4_TTTo2L2Nu
627 prop_binCRTT_electron2017_bin14
628 prop_binCRTT_electron2016M_bin17
629 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
630 prop_binCROS_electron2017_bin3
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHl3}$



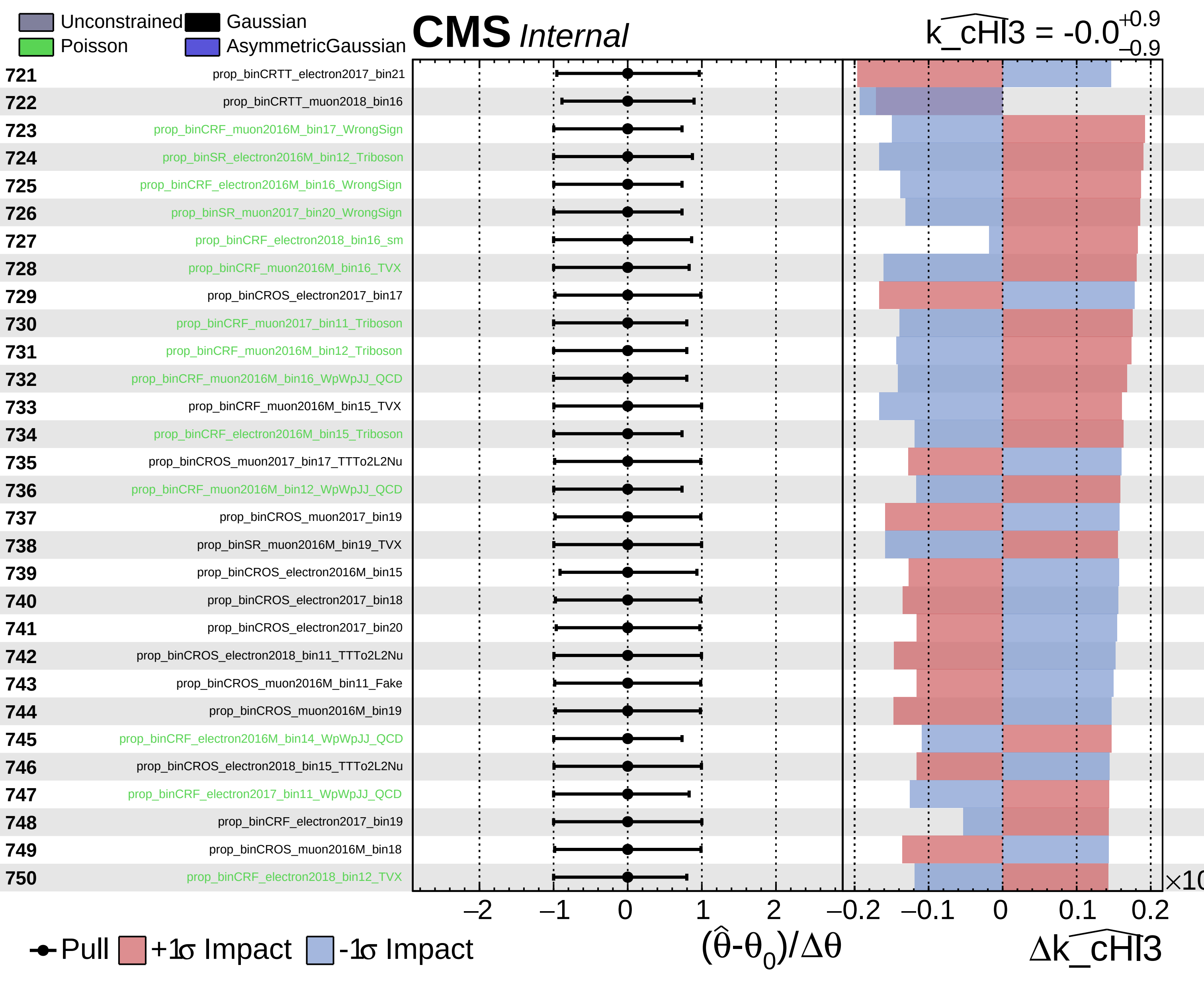
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 -0.9
631 prop_binCRTT_electron2017_bin17
632 prop_binSR_electron2018_bin17_TVX
633 prop_binCRF_muon2018_bin20
634 prop_binCRTT_electron2016M_bin18
635 prop_binCROS_muon2018_bin12_Fake
636 prop_binSR_muon2016M_bin14_Triboson
637 prop_binCRF_muon2018_bin12_TVX
638 prop_binCRF_muon2017_bin8_TVX
639 prop_binCRTT_muon2017_bin21
640 prop_binCROS_electron2017_bin9_Fake
641 prop_binCRF_electron2018_bin15_WrongSign
642 prop_binSR_electron2017_bin16_Triboson
643 prop_binSR_muon2016M_bin16_Triboson
644 prop_binSR_muon2016M_bin18_TVX
645 prop_binCRF_muon2017_bin11_TVX
646 prop_binCRTT_muon2018_bin13
647 prop_binCRF_muon2016M_bin12_WrongSign
648 prop_binSR_electron2018_bin17_WpWpJJ_QCD
649 prop_binCRF_electron2016M_bin15_WrongSign
650 prop_binSR_muon2016M_bin14_WrongSign
651 prop_binCROS_electron2016M_bin17
652 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
653 prop_binCROS_muon2016M_bin12_TTTo2L2Nu
654 prop_binCRF_electron2017_bin18
655 prop_binSR_electron2016M_bin1_WZ
656 prop_binSR_electron2017_bin12_TVX
657 prop_binSR_electron2017_bin12_WpWpJJ_QCD
658 prop_binCROS_electron2017_bin11_Fake
659 prop_binSR_electron2018_bin21_WrongSign
660 prop_binCRF_muon2016M_bin15_WrongSign
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



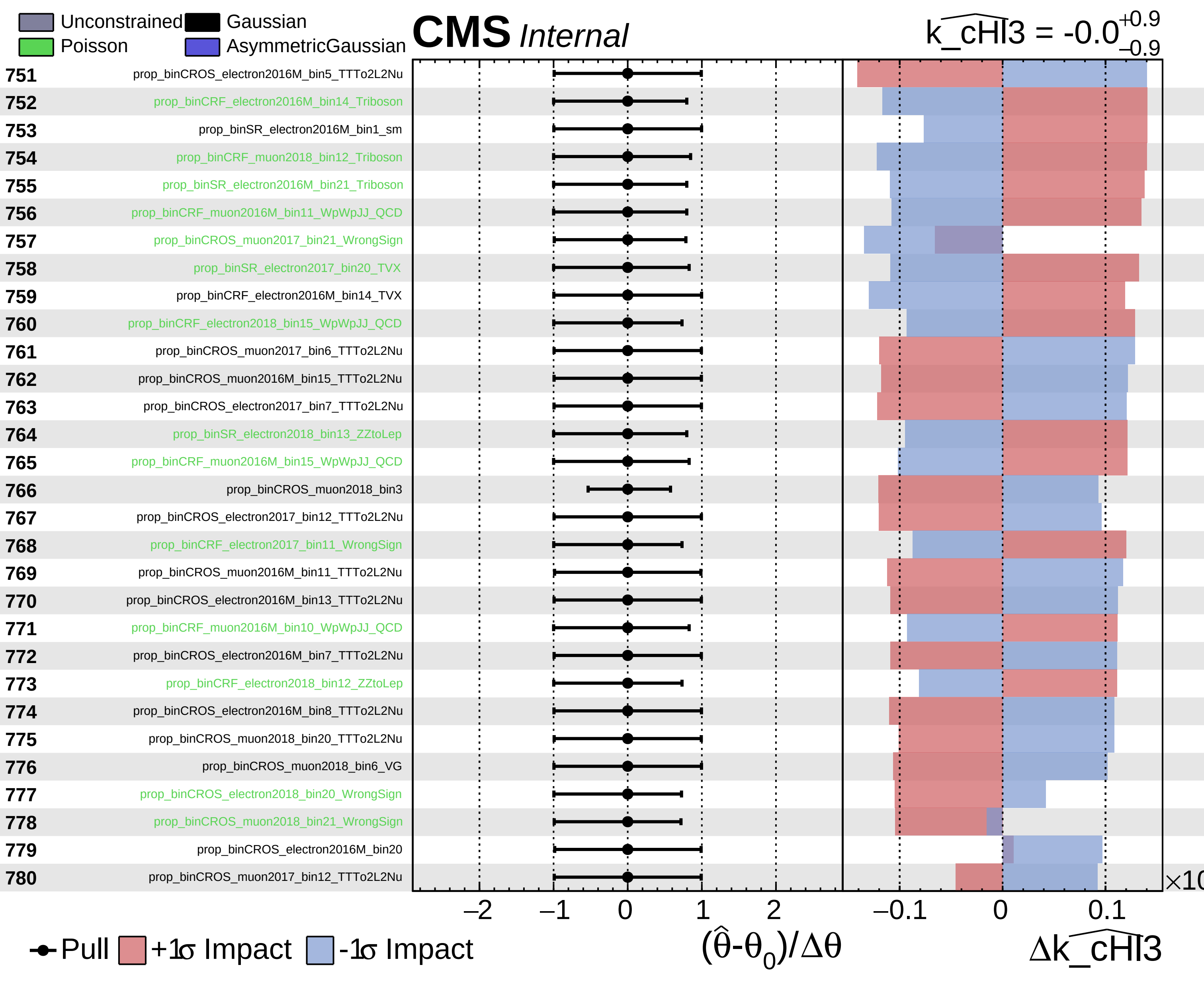
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
661 prop_binCROS_muon2016M_bin6_TTTo2L2Nu
662 prop_binCRF_electron2016M_bin15_TVX
663 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
664 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
665 prop_binCRF_muon2016M_bin17_sm
666 prop_binCRF_muon2016M_bin10_WrongSign
667 prop_binCROS_electron2017_bin7_Fake
668 prop_binCRTT_electron2016M_bin9
669 prop_binSR_electron2017_bin19_Triboson
670 prop_binSR_muon2016M_bin17_Triboson
671 prop_binCRTT_electron2017_bin19
672 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
673 prop_binCRTT_electron2018_bin21
674 prop_binSR_muon2017_bin20_WpWpJJ_QCD
675 prop_binSR_electron2017_bin12_Triboson
676 prop_binSR_electron2017_bin16_TVX
677 prop_binCRTT_electron2018_bin8
678 prop_binCRTT_electron2018_bin11
679 prop_binSR_electron2018_bin21_WpWpJJ_QCD
680 prop_binSR_muon2016M_bin16_TVX
681 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
682 prop_binSR_electron2018_bin17_WrongSign
683 prop_binCRF_electron2016M_bin19_sm
684 prop_binSR_electron2018_bin17_Triboson
685 prop_binSR_muon2017_bin20_Triboson
686 prop_binCROS_electron2016M_bin13_WrongSign
687 prop_binCRF_electron2016M_bin20
688 prop_binSR_electron2016M_bin16_WrongSign
689 prop_binSR_electron2016M_bin16_Triboson
690 prop_binSR_electron2016M_bin1_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3

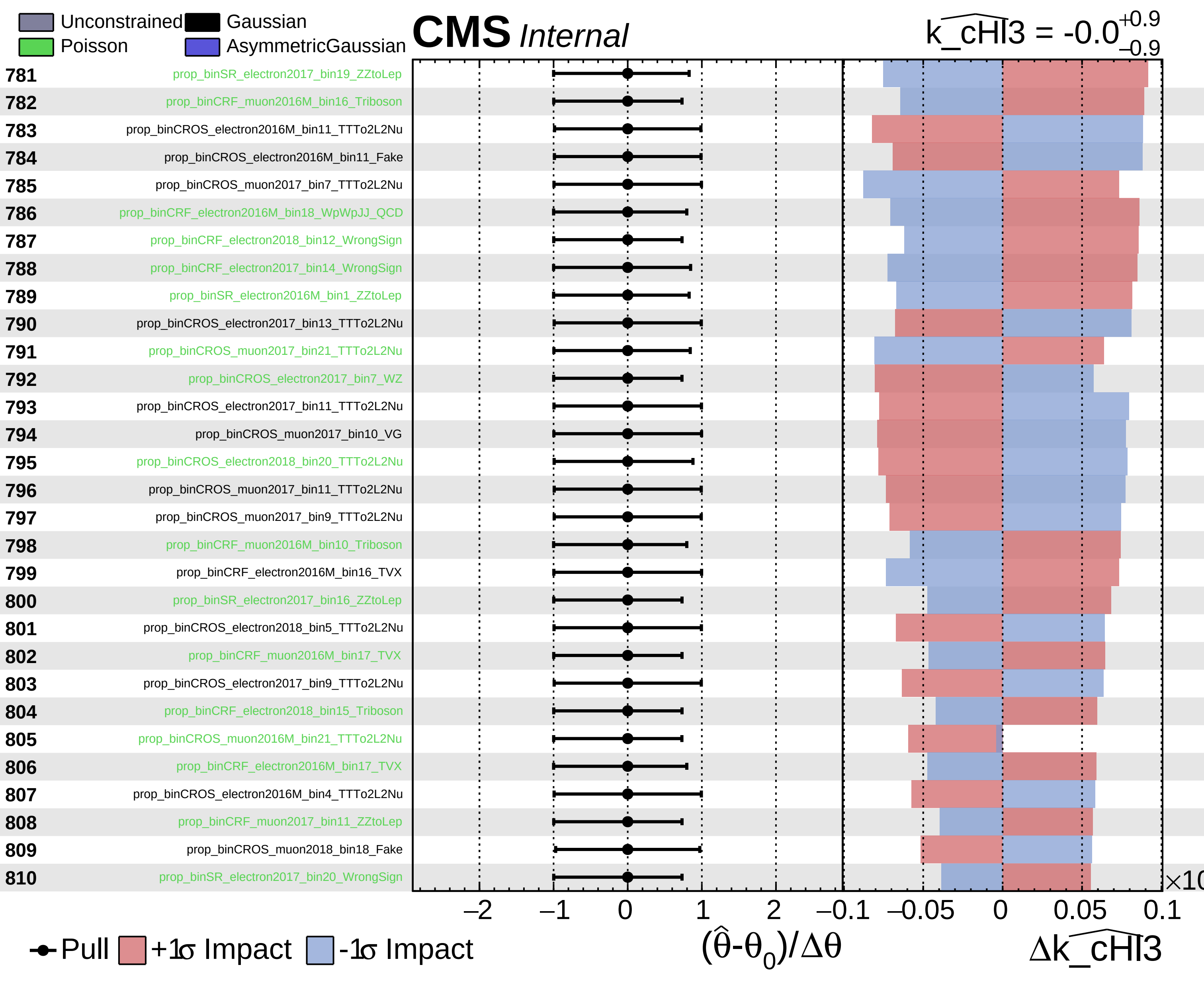


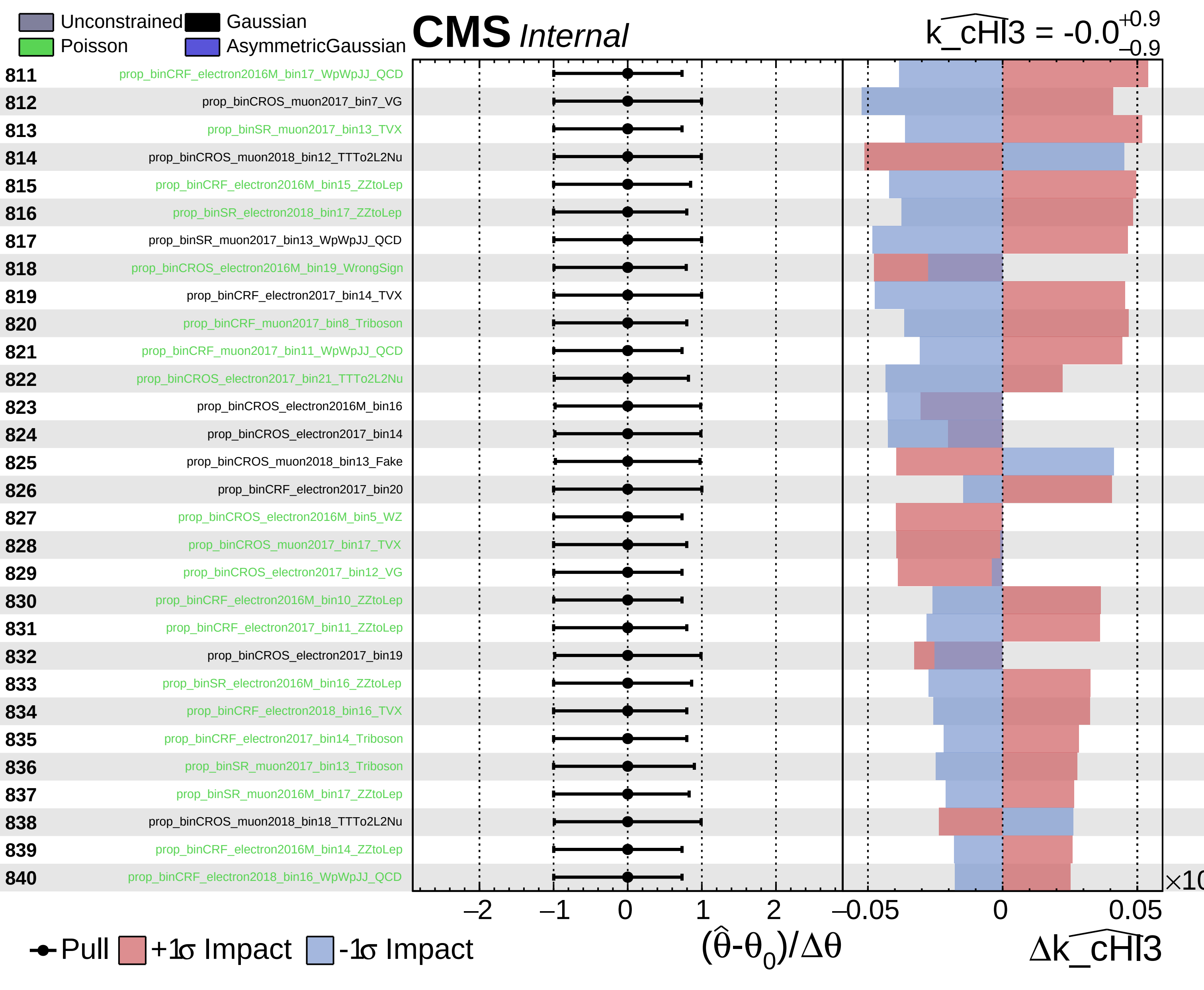
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
691 prop_binCROS_electron2016M_bin8_Fake
692 prop_binSR_muon2017_bin13_ZZtoLep
693 prop_binCROS_muon2017_bin9_Fake
694 prop_binCROS_muon2017_bin14
695 prop_binSR_muon2016M_bin17_TVX
696 prop_binCROS_muon2018_bin6_TTTo2L2Nu
697 prop_binCROS_muon2017_bin12_Fake
698 prop_binSR_electron2016M_bin12_TVX
699 prop_binSR_electron2016M_bin1_VG
700 prop_binCROS_electron2016M_bin6
701 prop_binCROS_muon2016M_bin20
702 prop_binSR_muon2016M_bin19_Triboson
703 prop_binCRF_muon2016M_bin11_Triboson
704 prop_binSR_electron2016M_bin12_WrongSign
705 prop_binCROS_muon2018_bin15
706 prop_binCRF_electron2017_bin16_Triboson
707 prop_binSR_electron2018_bin21_TVX
708 prop_binCRTT_muon2018_bin21
709 prop_binCRF_muon2016M_bin10_TVX
710 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
711 prop_binCROS_muon2016M_bin21_WrongSign
712 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
713 prop_binCROS_electron2017_bin15
714 prop_binCRF_electron2017_bin16_WpWpJJ_QCD
715 prop_binCROS_electron2016M_bin13_Fake
716 prop_binCROS_electron2018_bin17
717 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
718 prop_binSR_electron2017_bin20_WpWpJJ_QCD
719 prop_binSR_electron2017_bin14_Triboson
720 prop_binCROS_muon2016M_bin14_Fake
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



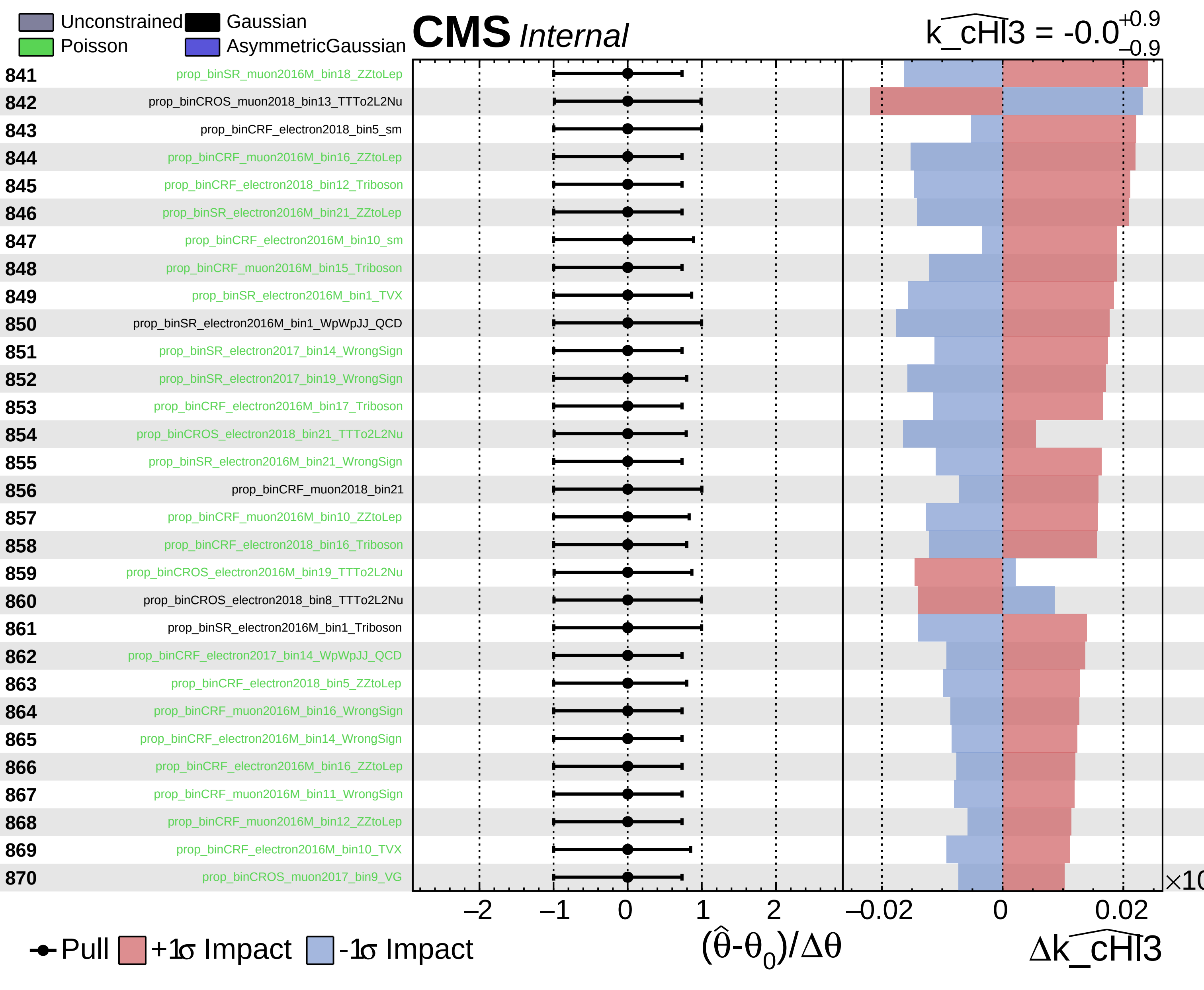
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
721 prop_binCRTT_electron2017_bin21
722 prop_binCRTT_muon2018_bin16
723 prop_binCRF_muon2016M_bin17_WrongSign
724 prop_binSR_electron2016M_bin12_Triboson
725 prop_binCRF_electron2016M_bin16_WrongSign
726 prop_binSR_muon2017_bin20_WrongSign
727 prop_binCRF_electron2018_bin16_sm
728 prop_binCRF_muon2016M_bin16_TVX
729 prop_binCROS_electron2017_bin17
730 prop_binCRF_muon2017_bin11_Triboson
731 prop_binCRF_muon2016M_bin12_Triboson
732 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
733 prop_binCRF_muon2016M_bin15_TVX
734 prop_binCRF_electron2016M_bin15_Triboson
735 prop_binCROS_muon2017_bin17_TTTo2L2Nu
736 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
737 prop_binCROS_muon2017_bin19
738 prop_binSR_muon2016M_bin19_TVX
739 prop_binCROS_electron2016M_bin15
740 prop_binCROS_electron2017_bin18
741 prop_binCROS_electron2017_bin20
742 prop_binCROS_electron2018_bin11_TTTo2L2Nu
743 prop_binCROS_muon2016M_bin11_Fake
744 prop_binCROS_muon2016M_bin19
745 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
746 prop_binCROS_electron2018_bin15_TTTo2L2Nu
747 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
748 prop_binCRF_electron2017_bin19
749 prop_binCROS_muon2016M_bin18
750 prop_binCRF_electron2018_bin12_TVX
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHl3} = -0.0^{+0.9}_{-0.9}$
751 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
752 prop_binCRF_electron2016M_bin14_Triboson
753 prop_binSR_electron2016M_bin1_sm
754 prop_binCRF_muon2018_bin12_Triboson
755 prop_binSR_electron2016M_bin21_Triboson
756 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
757 prop_binCROS_muon2017_bin21_WrongSign
758 prop_binSR_electron2017_bin20_TVX
759 prop_binCRF_electron2016M_bin14_TVX
760 prop_binCRF_electron2018_bin15_WpWpJJ_QCD
761 prop_binCROS_muon2017_bin6_TTTo2L2Nu
762 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
763 prop_binCROS_electron2017_bin7_TTTo2L2Nu
764 prop_binSR_electron2018_bin13_ZZtoLep
765 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
766 prop_binCROS_muon2018_bin3
767 prop_binCROS_electron2017_bin12_TTTo2L2Nu
768 prop_binCRF_electron2017_bin11_WrongSign
769 prop_binCROS_muon2016M_bin11_TTTo2L2Nu
770 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
771 prop_binCRF_muon2016M_bin10_WpWpJJ_QCD
772 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
773 prop_binCRF_electron2018_bin12_ZZtoLep
774 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
775 prop_binCROS_muon2018_bin20_TTTo2L2Nu
776 prop_binCROS_muon2018_bin6_VG
777 prop_binCROS_electron2018_bin20_WrongSign
778 prop_binCROS_muon2018_bin21_WrongSign
779 prop_binCROS_electron2016M_bin20
780 prop_binCROS_muon2017_bin12_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHl3}$

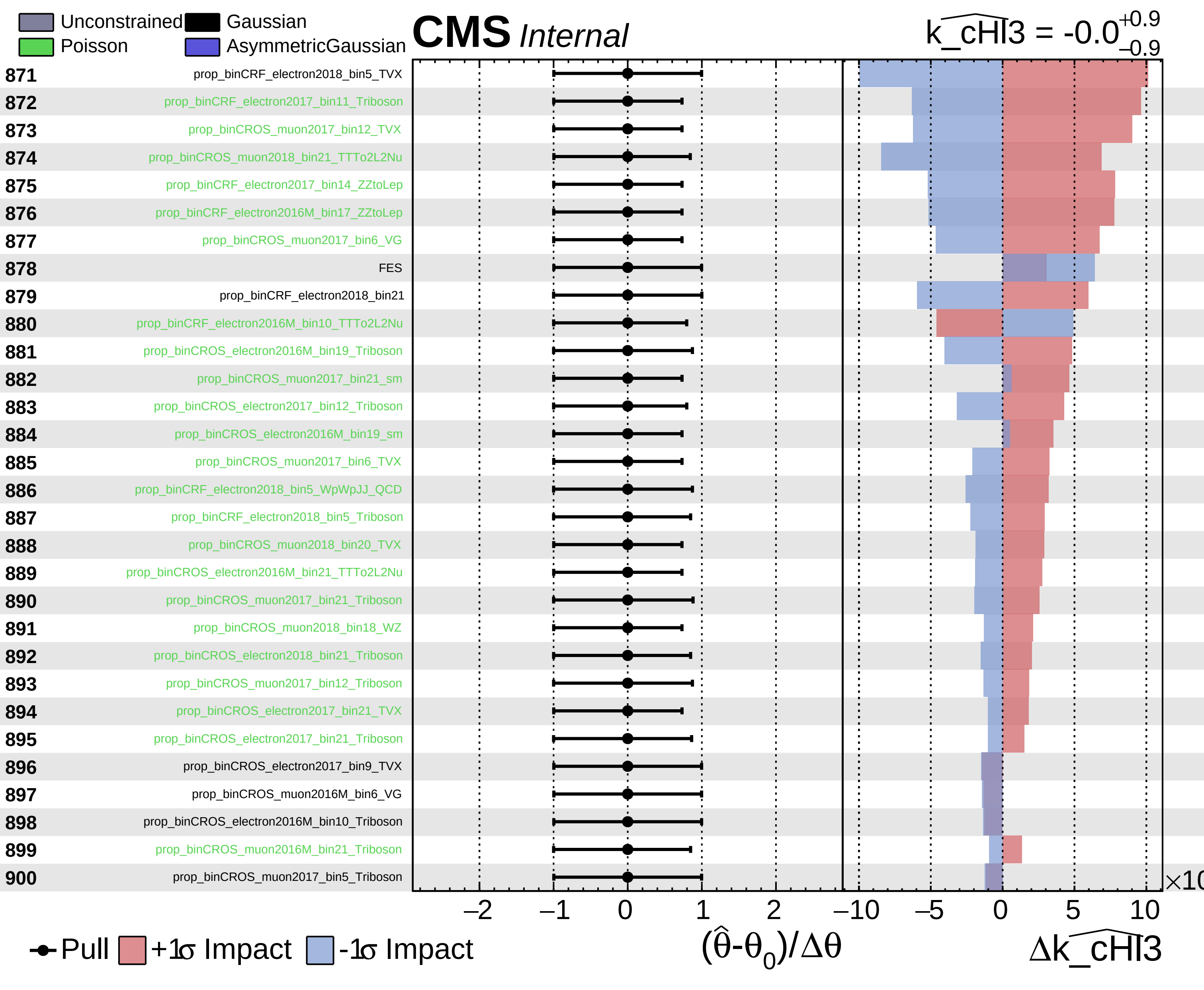

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
781 prop_binSR_electron2017_bin19_ZZtoLep
782 prop_binCRF_muon2016M_bin16_Triboson
783 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
784 prop_binCROS_electron2016M_bin11_Fake
785 prop_binCROS_muon2017_bin7_TTTo2L2Nu
786 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
787 prop_binCRF_electron2018_bin12_WrongSign
788 prop_binCRF_electron2017_bin14_WrongSign
789 prop_binSR_electron2016M_bin1_ZZtoLep
790 prop_binCROS_electron2017_bin13_TTTo2L2Nu
791 prop_binCROS_muon2017_bin21_TTTo2L2Nu
792 prop_binCROS_electron2017_bin7_WZ
793 prop_binCROS_electron2017_bin11_TTTo2L2Nu
794 prop_binCROS_muon2017_bin10_VG
795 prop_binCROS_electron2018_bin20_TTTo2L2Nu
796 prop_binCROS_muon2017_bin11_TTTo2L2Nu
797 prop_binCROS_muon2017_bin9_TTTo2L2Nu
798 prop_binCRF_muon2016M_bin10_Triboson
799 prop_binCRF_electron2016M_bin16_TVX
800 prop_binSR_electron2017_bin16_ZZtoLep
801 prop_binCROS_electron2018_bin5_TTTo2L2Nu
802 prop_binCRF_muon2016M_bin17_TVX
803 prop_binCROS_electron2017_bin9_TTTo2L2Nu
804 prop_binCRF_electron2018_bin15_Triboson
805 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
806 prop_binCRF_electron2016M_bin17_TVX
807 prop_binCROS_electron2016M_bin4_TTTo2L2Nu
808 prop_binCRF_muon2017_bin11_ZZtoLep
809 prop_binCROS_muon2018_bin18_Fake
810 prop_binSR_electron2017_bin20_WrongSign
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



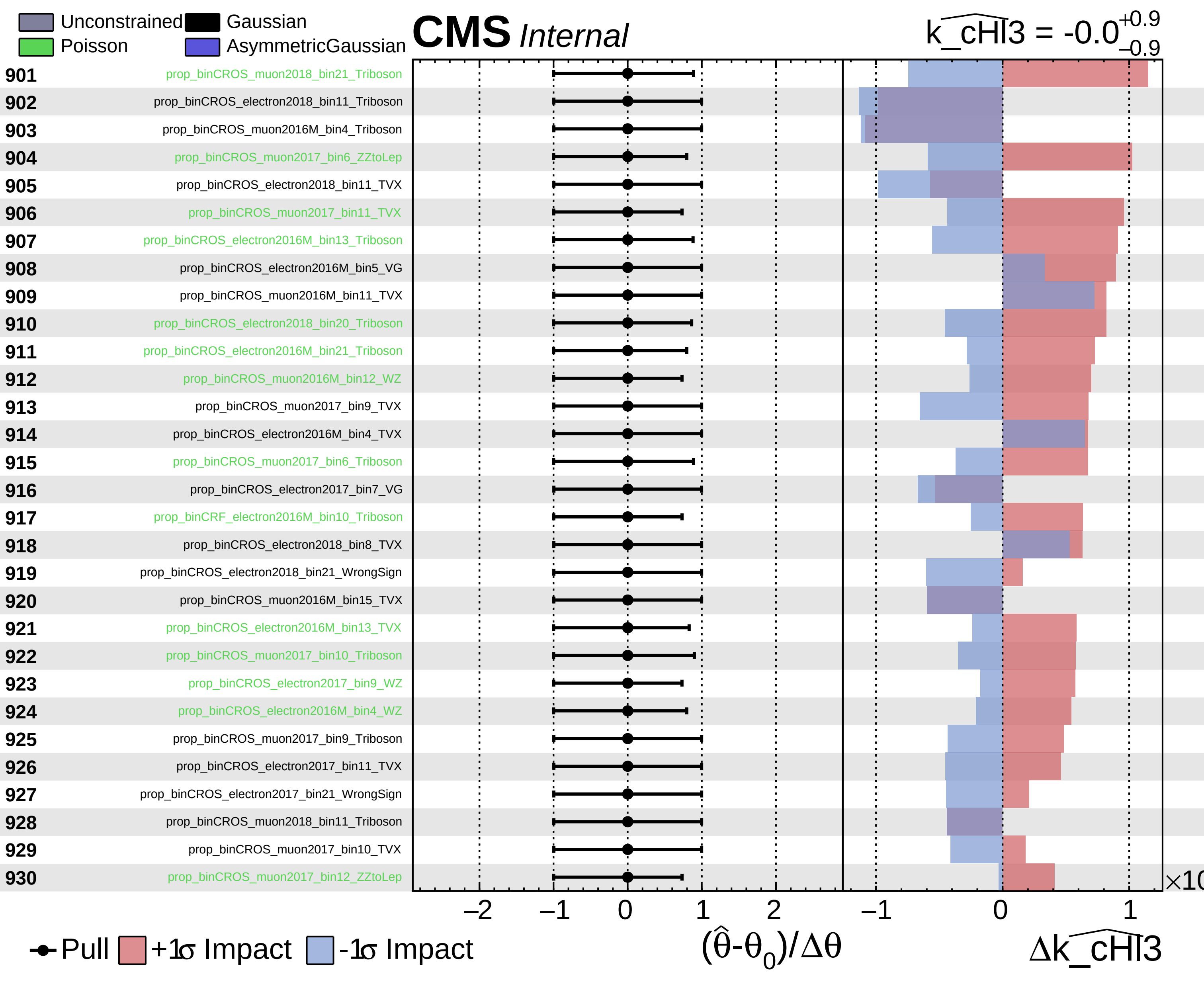
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
811 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
812 prop_binCROS_muon2017_bin7_VG
813 prop_binSR_muon2017_bin13_TVX
814 prop_binCROS_muon2018_bin12_TTTo2L2Nu
815 prop_binCRF_electron2016M_bin15_ZZtoLep
816 prop_binSR_electron2018_bin17_ZZtoLep
817 prop_binSR_muon2017_bin13_WpWpJJ_QCD
818 prop_binCROS_electron2016M_bin19_WrongSign
819 prop_binCRF_electron2017_bin14_TVX
820 prop_binCRF_muon2017_bin8_Triboson
821 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
822 prop_binCROS_electron2017_bin21_TTTo2L2Nu
823 prop_binCROS_electron2016M_bin16
824 prop_binCROS_electron2017_bin14
825 prop_binCROS_muon2018_bin13_Fake
826 prop_binCRF_electron2017_bin20
827 prop_binCROS_electron2016M_bin5_WZ
828 prop_binCROS_muon2017_bin17_TVX
829 prop_binCROS_electron2017_bin12_VG
830 prop_binCRF_electron2016M_bin10_ZZtoLep
831 prop_binCRF_electron2017_bin11_ZZtoLep
832 prop_binCROS_electron2017_bin19
833 prop_binSR_electron2016M_bin16_ZZtoLep
834 prop_binCRF_electron2018_bin16_TVX
835 prop_binCRF_electron2017_bin14_Triboson
836 prop_binSR_muon2017_bin13_Triboson
837 prop_binSR_muon2016M_bin17_ZZtoLep
838 prop_binCROS_muon2018_bin18_TTTo2L2Nu
839 prop_binCRF_electron2016M_bin14_ZZtoLep
840 prop_binCRF_electron2018_bin16_WpWpJJ_QCD
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

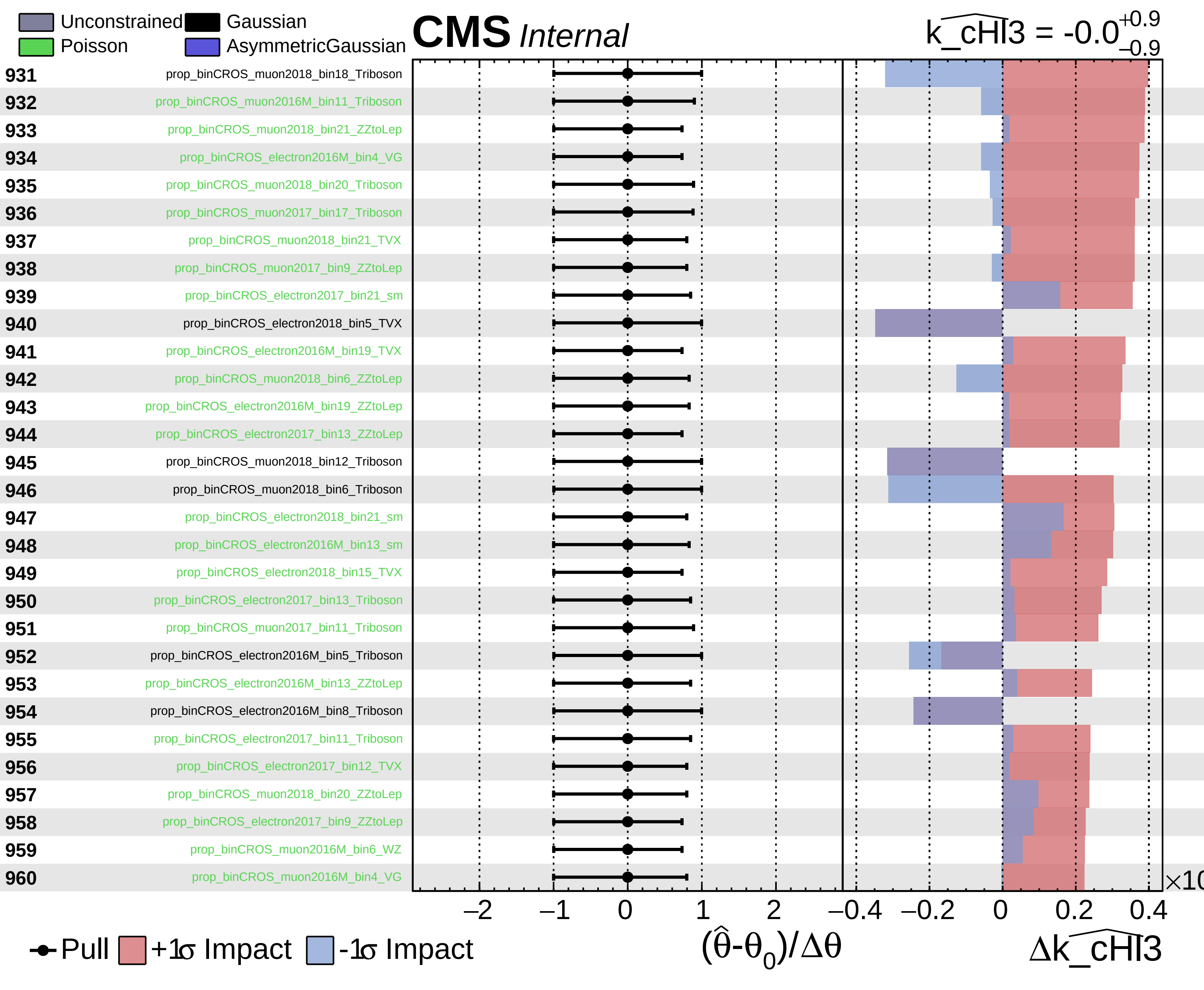


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
841 prop_binSR_muon2016M_bin18_ZZtoLep
842 prop_binCROS_muon2018_bin13_TTTo2L2Nu
843 prop_binCRF_electron2018_bin5_sm
844 prop_binCRF_muon2016M_bin16_ZZtoLep
845 prop_binCRF_electron2018_bin12_Triboson
846 prop_binSR_electron2016M_bin21_ZZtoLep
847 prop_binCRF_electron2016M_bin10_sm
848 prop_binCRF_muon2016M_bin15_Triboson
849 prop_binSR_electron2016M_bin1_TVX
850 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
851 prop_binSR_electron2017_bin14_WrongSign
852 prop_binSR_electron2017_bin19_WrongSign
853 prop_binCRF_electron2016M_bin17_Triboson
854 prop_binCROS_electron2018_bin21_TTTo2L2Nu
855 prop_binSR_electron2016M_bin21_WrongSign
856 prop_binCRF_muon2018_bin21
857 prop_binCRF_muon2016M_bin10_ZZtoLep
858 prop_binCRF_electron2018_bin16_Triboson
859 prop_binCROS_electron2016M_bin19_TTTo2L2Nu
860 prop_binCROS_electron2018_bin8_TTTo2L2Nu
861 prop_binSR_electron2016M_bin1_Triboson
862 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
863 prop_binCRF_electron2018_bin5_ZZtoLep
864 prop_binCRF_muon2016M_bin16_WrongSign
865 prop_binCRF_electron2016M_bin14_WrongSign
866 prop_binCRF_electron2016M_bin16_ZZtoLep
867 prop_binCRF_muon2016M_bin11_WrongSign
868 prop_binCRF_muon2016M_bin12_ZZtoLep
869 prop_binCRF_electron2016M_bin10_TVX
870 prop_binCROS_muon2017_bin9_VG
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3

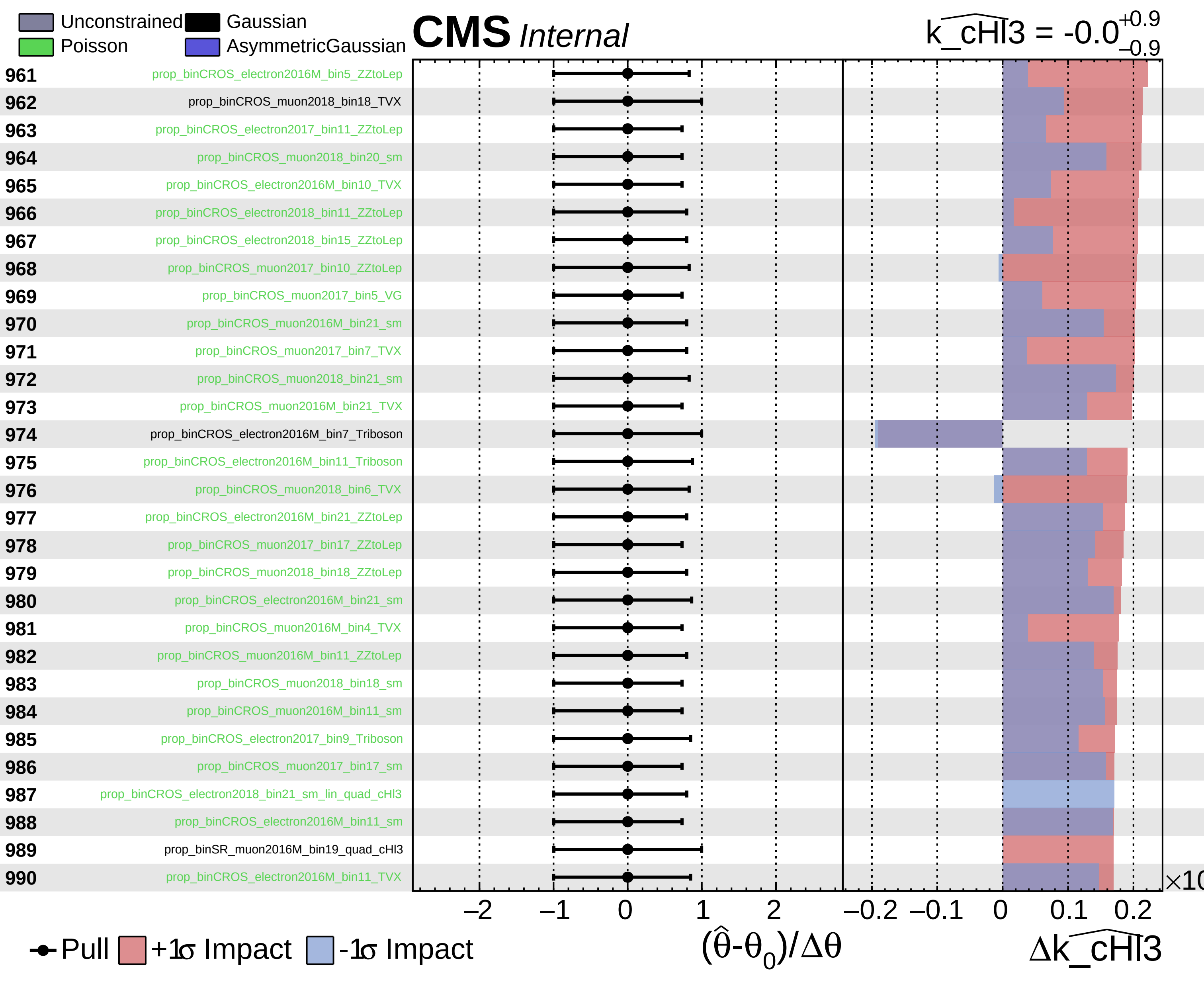


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHl3}$ = $-0.0^{+0.9}_{-0.9}$
871 prop_binCRF_electron2018_bin5_TVX
872 prop_binCRF_electron2017_bin11_Triboson
873 prop_binCROS_muon2017_bin12_TVX
874 prop_binCROS_muon2018_bin21_TTTo2L2Nu
875 prop_binCRF_electron2017_bin14_ZZtoLep
876 prop_binCRF_electron2016M_bin17_ZZtoLep
877 prop_binCROS_muon2017_bin6_VG
878 FES
879 prop_binCRF_electron2018_bin21
880 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
881 prop_binCROS_electron2016M_bin19_Triboson
882 prop_binCROS_muon2017_bin21_sm
883 prop_binCROS_electron2017_bin12_Triboson
884 prop_binCROS_electron2016M_bin19_sm
885 prop_binCROS_muon2017_bin6_TVX
886 prop_binCRF_electron2018_bin5_WpWpJJ_QCD
887 prop_binCRF_electron2018_bin5_Triboson
888 prop_binCROS_muon2018_bin20_TVX
889 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
890 prop_binCROS_muon2017_bin21_Triboson
891 prop_binCROS_muon2018_bin18_WZ
892 prop_binCROS_electron2018_bin21_Triboson
893 prop_binCROS_muon2017_bin12_Triboson
894 prop_binCROS_electron2017_bin21_TVX
895 prop_binCROS_electron2017_bin21_Triboson
896 prop_binCROS_electron2017_bin9_TVX
897 prop_binCROS_muon2016M_bin6_VG
898 prop_binCROS_electron2016M_bin10_Triboson
899 prop_binCROS_muon2016M_bin21_Triboson
900 prop_binCROS_muon2017_bin5_Triboson
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHl3}$

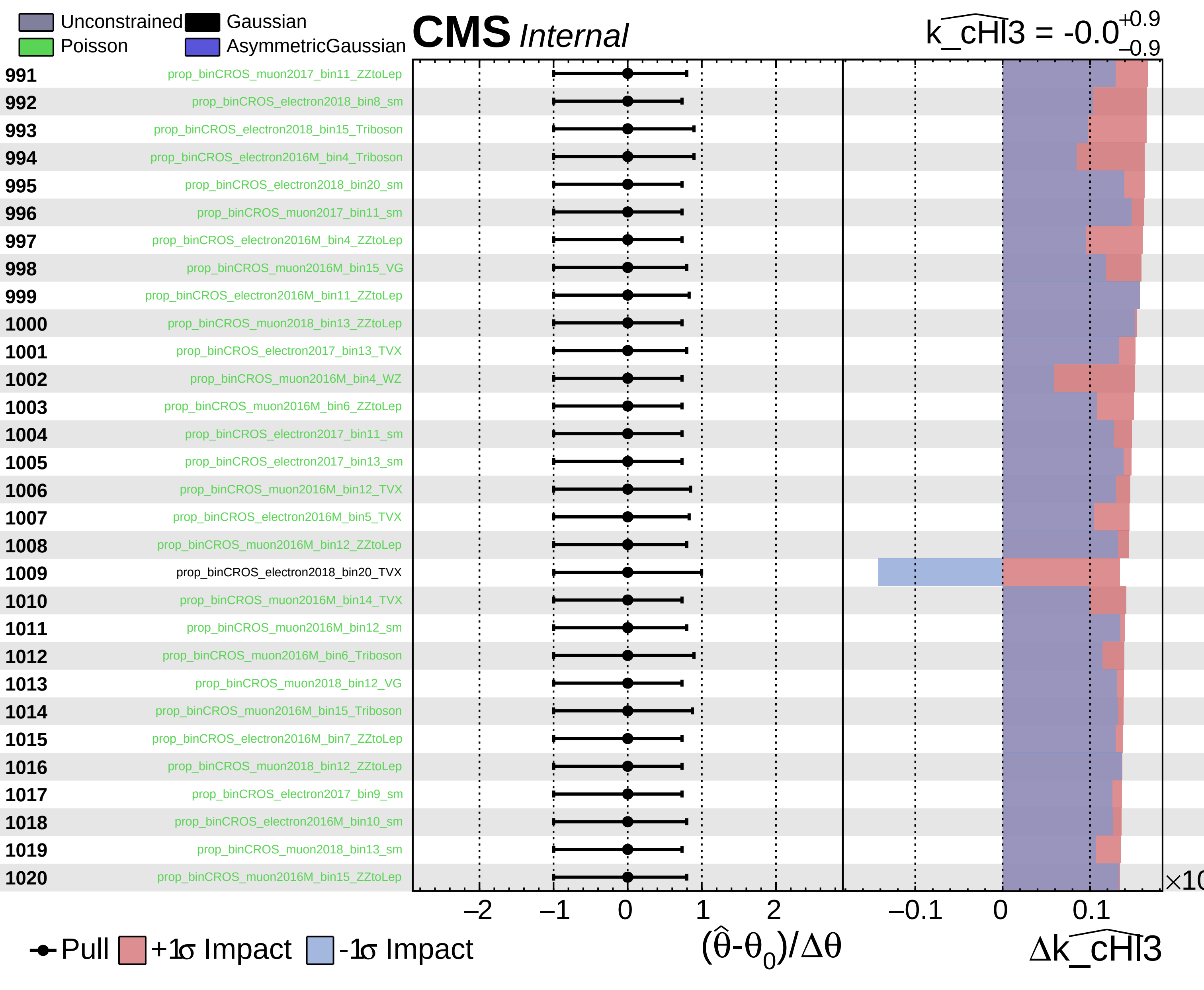

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
901 prop_binCROS_muon2018_bin21_Triboson
902 prop_binCROS_electron2018_bin11_Triboson
903 prop_binCROS_muon2016M_bin4_Triboson
904 prop_binCROS_muon2017_bin6_ZZtoLep
905 prop_binCROS_electron2018_bin11_TVX
906 prop_binCROS_muon2017_bin11_TVX
907 prop_binCROS_electron2016M_bin13_Triboson
908 prop_binCROS_electron2016M_bin5_VG
909 prop_binCROS_muon2016M_bin11_TVX
910 prop_binCROS_electron2018_bin20_Triboson
911 prop_binCROS_electron2016M_bin21_Triboson
912 prop_binCROS_muon2016M_bin12_WZ
913 prop_binCROS_muon2017_bin9_TVX
914 prop_binCROS_electron2016M_bin4_TVX
915 prop_binCROS_muon2017_bin6_Triboson
916 prop_binCROS_electron2017_bin7_VG
917 prop_binCRF_electron2016M_bin10_Triboson
918 prop_binCROS_electron2018_bin8_TVX
919 prop_binCROS_electron2018_bin21_WrongSign
920 prop_binCROS_muon2016M_bin15_TVX
921 prop_binCROS_electron2016M_bin13_TVX
922 prop_binCROS_muon2017_bin10_Triboson
923 prop_binCROS_electron2017_bin9_WZ
924 prop_binCROS_electron2016M_bin4_WZ
925 prop_binCROS_muon2017_bin9_Triboson
926 prop_binCROS_electron2017_bin11_TVX
927 prop_binCROS_electron2017_bin21_WrongSign
928 prop_binCROS_muon2018_bin11_Triboson
929 prop_binCROS_muon2017_bin10_TVX
930 prop_binCROS_muon2017_bin12_ZZtoLep
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



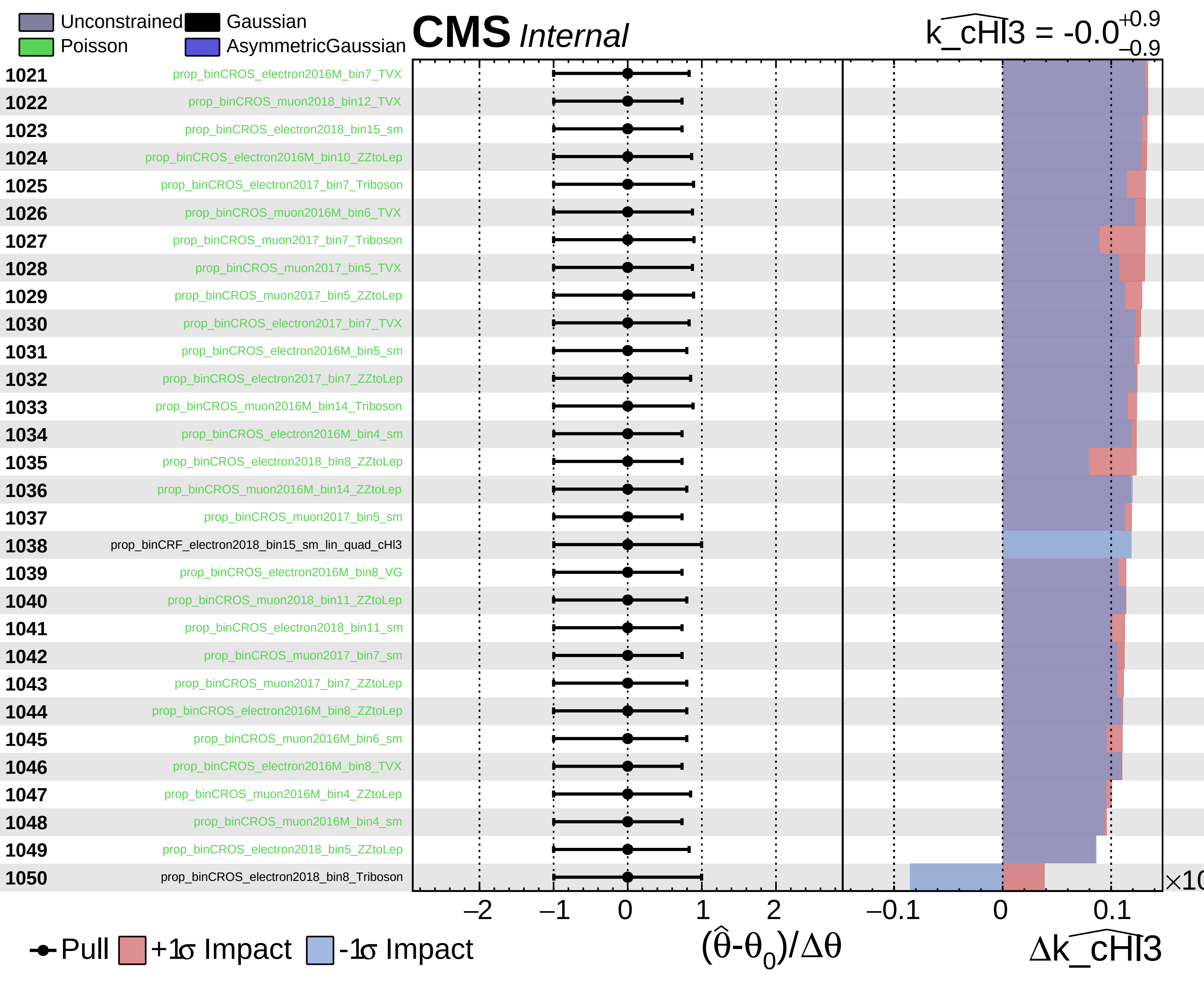
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 -0.9
931 prop_binCROS_muon2018_bin18_Triboson
932 prop_binCROS_muon2016M_bin11_Triboson
933 prop_binCROS_muon2018_bin21_ZZtoLep
934 prop_binCROS_electron2016M_bin4_VG
935 prop_binCROS_muon2018_bin20_Triboson
936 prop_binCROS_muon2017_bin17_Triboson
937 prop_binCROS_muon2018_bin21_TVX
938 prop_binCROS_muon2017_bin9_ZZtoLep
939 prop_binCROS_electron2017_bin21_sm
940 prop_binCROS_electron2018_bin5_TVX
941 prop_binCROS_electron2016M_bin19_TVX
942 prop_binCROS_muon2018_bin6_ZZtoLep
943 prop_binCROS_electron2016M_bin19_ZZtoLep
944 prop_binCROS_electron2017_bin13_ZZtoLep
945 prop_binCROS_muon2018_bin12_Triboson
946 prop_binCROS_muon2018_bin6_Triboson
947 prop_binCROS_electron2018_bin21_sm
948 prop_binCROS_electron2016M_bin13_sm
949 prop_binCROS_electron2018_bin15_TVX
950 prop_binCROS_electron2017_bin13_Triboson
951 prop_binCROS_muon2017_bin11_Triboson
952 prop_binCROS_electron2016M_bin5_Triboson
953 prop_binCROS_electron2016M_bin13_ZZtoLep
954 prop_binCROS_electron2016M_bin8_Triboson
955 prop_binCROS_electron2017_bin11_Triboson
956 prop_binCROS_electron2017_bin12_TVX
957 prop_binCROS_muon2018_bin20_ZZtoLep
958 prop_binCROS_electron2017_bin9_ZZtoLep
959 prop_binCROS_muon2016M_bin6_WZ
960 prop_binCROS_muon2016M_bin4_VG
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



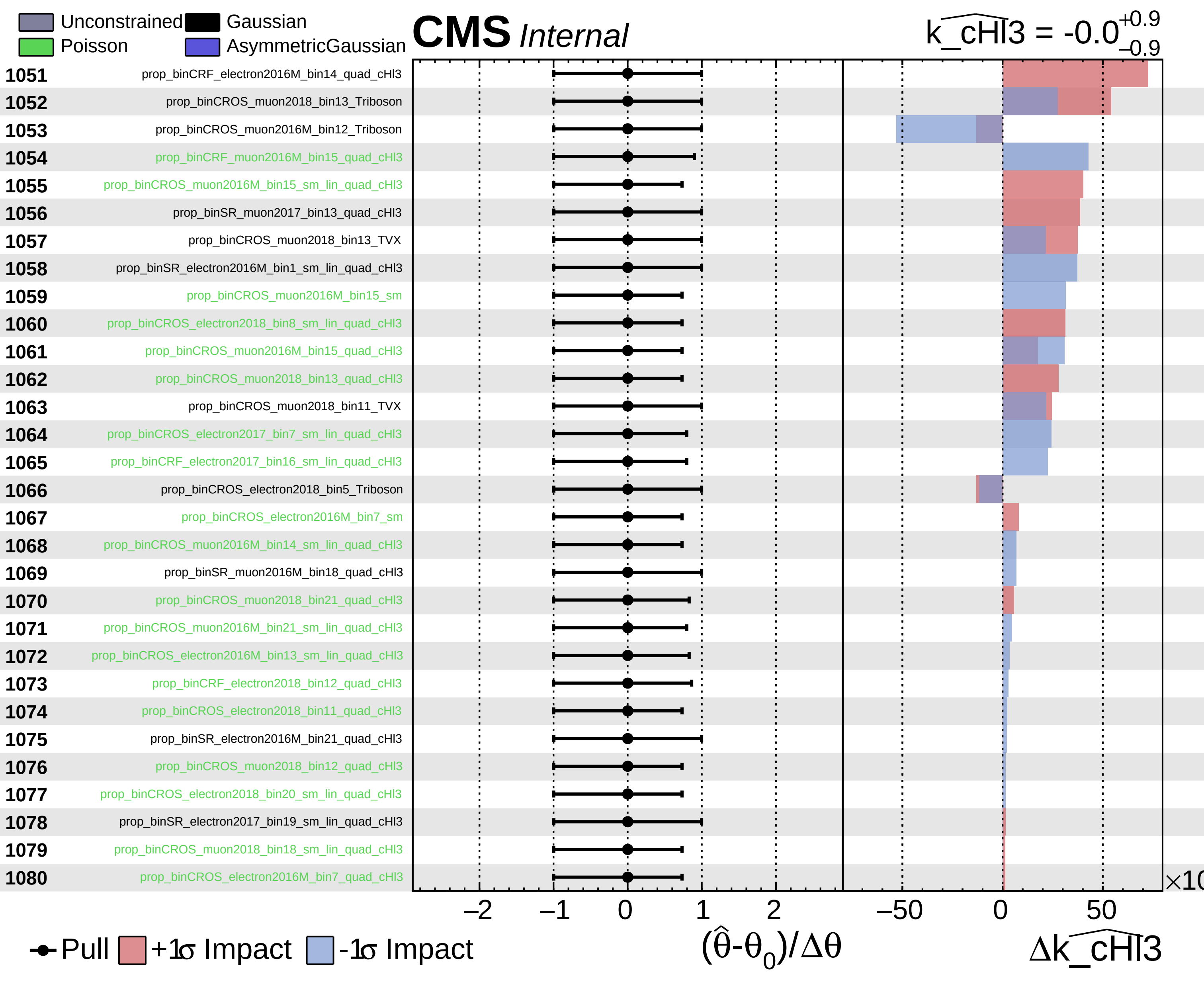
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 -0.9
961 prop_binCROS_electron2016M_bin5_ZZtoLep
962 prop_binCROS_muon2018_bin18_TVX
963 prop_binCROS_electron2017_bin11_ZZtoLep
964 prop_binCROS_muon2018_bin20_sm
965 prop_binCROS_electron2016M_bin10_TVX
966 prop_binCROS_electron2018_bin11_ZZtoLep
967 prop_binCROS_electron2018_bin15_ZZtoLep
968 prop_binCROS_muon2017_bin10_ZZtoLep
969 prop_binCROS_muon2017_bin5_VG
970 prop_binCROS_muon2016M_bin21_sm
971 prop_binCROS_muon2017_bin7_TVX
972 prop_binCROS_muon2018_bin21_sm
973 prop_binCROS_muon2016M_bin21_TVX
974 prop_binCROS_electron2016M_bin7_Triboson
975 prop_binCROS_electron2016M_bin11_Triboson
976 prop_binCROS_muon2018_bin6_TVX
977 prop_binCROS_electron2016M_bin21_ZZtoLep
978 prop_binCROS_muon2017_bin17_ZZtoLep
979 prop_binCROS_muon2018_bin18_ZZtoLep
980 prop_binCROS_electron2016M_bin21_sm
981 prop_binCROS_muon2016M_bin4_TVX
982 prop_binCROS_muon2016M_bin11_ZZtoLep
983 prop_binCROS_muon2018_bin18_sm
984 prop_binCROS_muon2016M_bin11_sm
985 prop_binCROS_electron2017_bin9_Triboson
986 prop_binCROS_muon2017_bin17_sm
987 prop_binCROS_electron2018_bin21_sm_lin_quad_cHl3
988 prop_binCROS_electron2016M_bin11_sm
989 prop_binSR_muon2016M_bin19_quad_cHl3
990 prop_binCROS_electron2016M_bin11_TVX
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



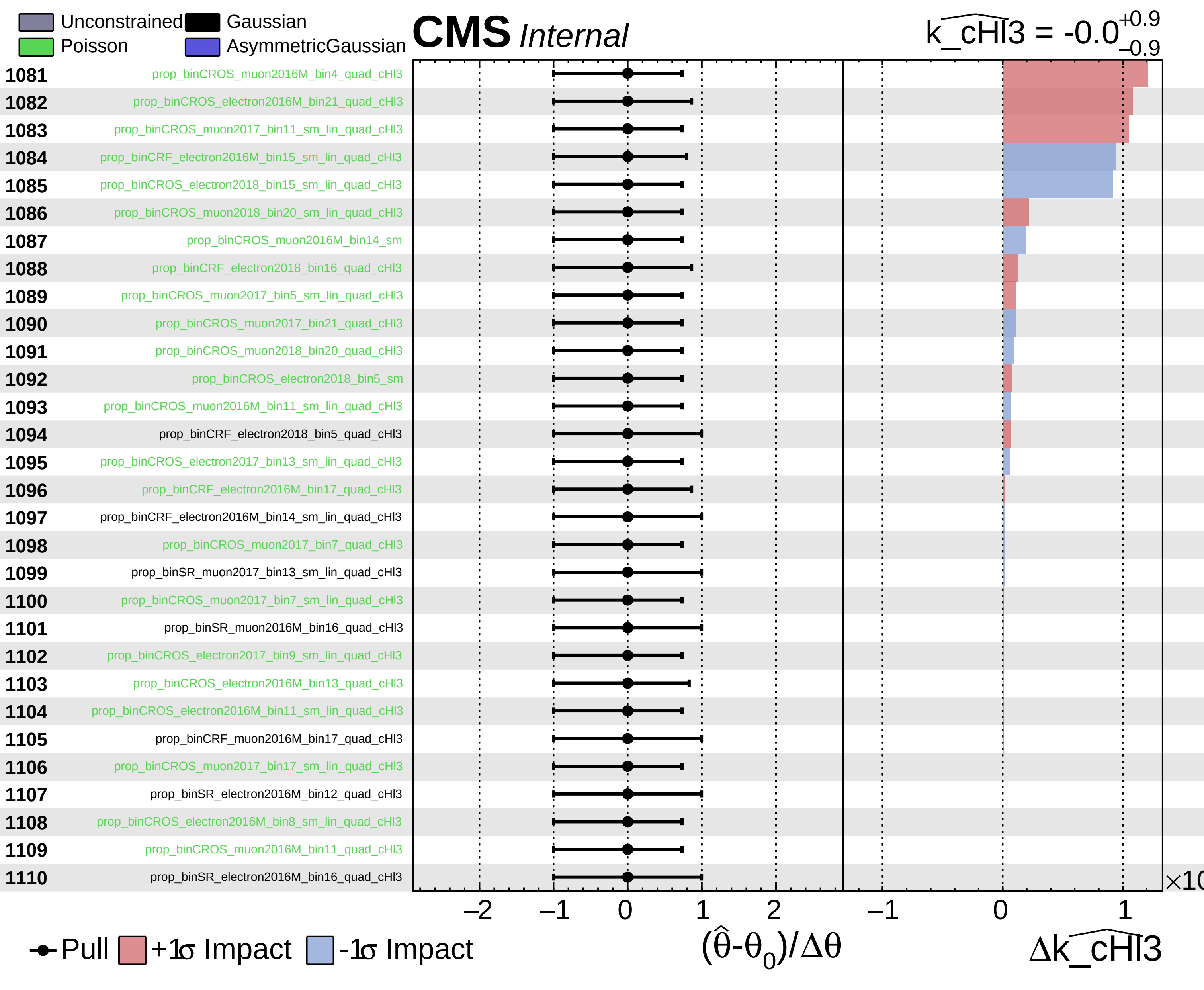
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHl3} = -0.0^{+0.9}_{-0.9}$
991 prop_binCROS_muon2017_bin11_ZZtoLep
992 prop_binCROS_electron2018_bin8_sm
993 prop_binCROS_electron2018_bin15_Triboson
994 prop_binCROS_electron2016M_bin4_Triboson
995 prop_binCROS_electron2018_bin20_sm
996 prop_binCROS_muon2017_bin11_sm
997 prop_binCROS_electron2016M_bin4_ZZtoLep
998 prop_binCROS_muon2016M_bin15_VG
999 prop_binCROS_electron2016M_bin11_ZZtoLep
1000 prop_binCROS_muon2018_bin13_ZZtoLep
1001 prop_binCROS_electron2017_bin13_TVX
1002 prop_binCROS_muon2016M_bin4_WZ
1003 prop_binCROS_muon2016M_bin6_ZZtoLep
1004 prop_binCROS_electron2017_bin11_sm
1005 prop_binCROS_electron2017_bin13_sm
1006 prop_binCROS_muon2016M_bin12_TVX
1007 prop_binCROS_electron2016M_bin5_TVX
1008 prop_binCROS_muon2016M_bin12_ZZtoLep
1009 prop_binCROS_electron2018_bin20_TVX
1010 prop_binCROS_muon2016M_bin14_TVX
1011 prop_binCROS_muon2016M_bin12_sm
1012 prop_binCROS_muon2016M_bin6_Triboson
1013 prop_binCROS_muon2018_bin12_VG
1014 prop_binCROS_muon2016M_bin15_Triboson
1015 prop_binCROS_electron2016M_bin7_ZZtoLep
1016 prop_binCROS_muon2018_bin12_ZZtoLep
1017 prop_binCROS_electron2017_bin9_sm
1018 prop_binCROS_electron2016M_bin10_sm
1019 prop_binCROS_muon2018_bin13_sm
1020 prop_binCROS_muon2016M_bin15_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHl3}$



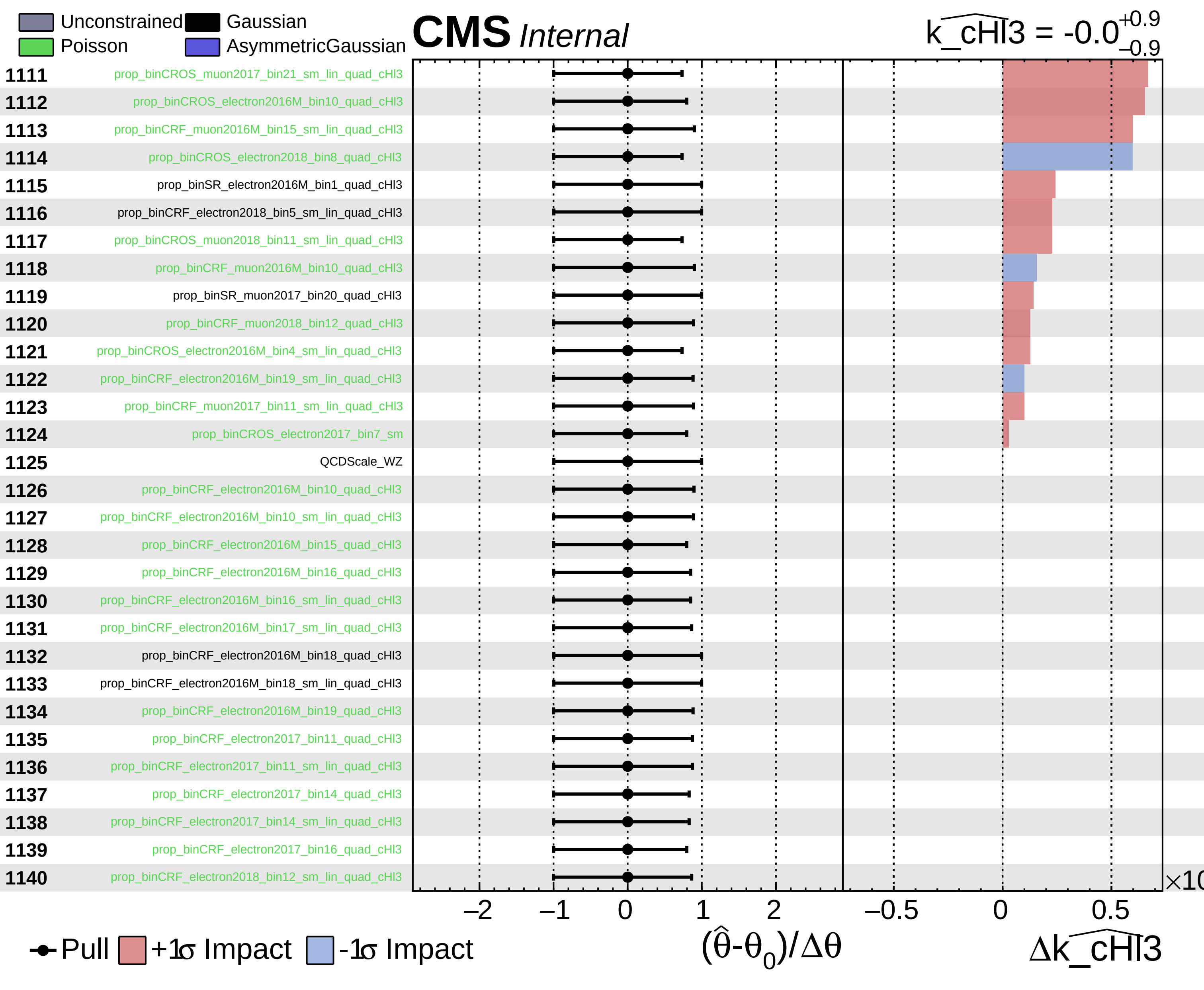
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1021 prop_binCROS_electron2016M_bin7_TVX
1022 prop_binCROS_muon2018_bin12_TVX
1023 prop_binCROS_electron2018_bin15_sm
1024 prop_binCROS_electron2016M_bin10_ZZtoLep
1025 prop_binCROS_electron2017_bin7_Triboson
1026 prop_binCROS_muon2016M_bin6_TVX
1027 prop_binCROS_muon2017_bin7_Triboson
1028 prop_binCROS_muon2017_bin5_TVX
1029 prop_binCROS_muon2017_bin5_ZZtoLep
1030 prop_binCROS_electron2017_bin7_TVX
1031 prop_binCROS_electron2016M_bin5_sm
1032 prop_binCROS_electron2017_bin7_ZZtoLep
1033 prop_binCROS_muon2016M_bin14_Triboson
1034 prop_binCROS_electron2016M_bin4_sm
1035 prop_binCROS_electron2018_bin8_ZZtoLep
1036 prop_binCROS_muon2016M_bin14_ZZtoLep
1037 prop_binCROS_muon2017_bin5_sm
1038 prop_binCRF_electron2018_bin15_sm_lin_quad_cHl3
1039 prop_binCROS_electron2016M_bin8_VG
1040 prop_binCROS_muon2018_bin11_ZZtoLep
1041 prop_binCROS_electron2018_bin11_sm
1042 prop_binCROS_muon2017_bin7_sm
1043 prop_binCROS_muon2017_bin7_ZZtoLep
1044 prop_binCROS_electron2016M_bin8_ZZtoLep
1045 prop_binCROS_muon2016M_bin6_sm
1046 prop_binCROS_electron2016M_bin8_TVX
1047 prop_binCROS_muon2016M_bin4_ZZtoLep
1048 prop_binCROS_muon2016M_bin4_sm
1049 prop_binCROS_electron2018_bin5_ZZtoLep
1050 prop_binCROS_electron2018_bin8_Triboson
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHl3



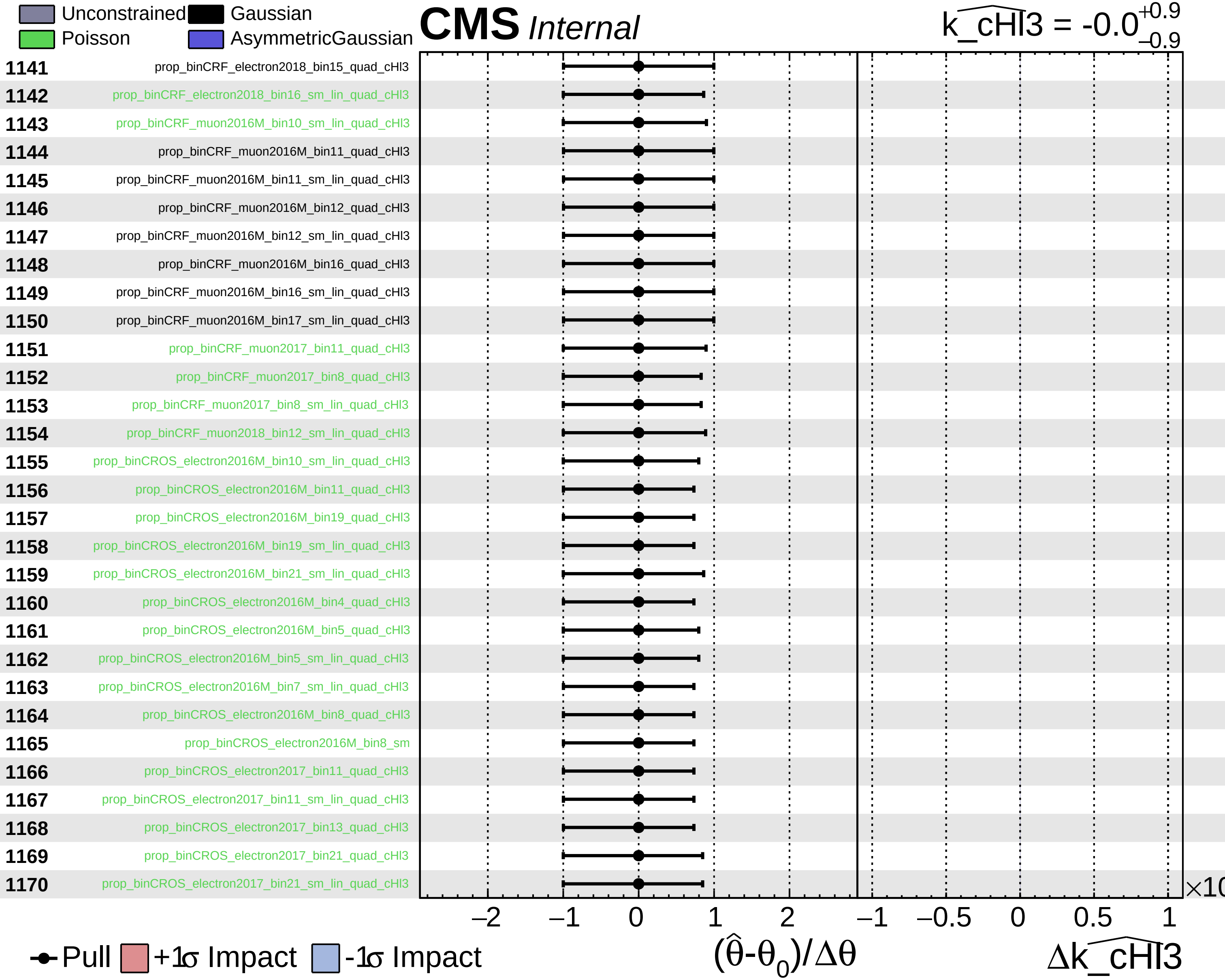
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1051 prop_binCRF_electron2016M_bin14_quad_cHl3
1052 prop_binCROS_muon2018_bin13_Triboson
1053 prop_binCROS_muon2016M_bin12_Triboson
1054 prop_binCRF_muon2016M_bin15_quad_cHl3
1055 prop_binCROS_muon2016M_bin15_sm_lin_quad_cHl3
1056 prop_binSR_muon2017_bin13_quad_cHl3
1057 prop_binCROS_muon2018_bin13_TVX
1058 prop_binSR_electron2016M_bin1_sm_lin_quad_cHl3
1059 prop_binCROS_muon2016M_bin15_sm
1060 prop_binCROS_electron2018_bin8_sm_lin_quad_cHl3
1061 prop_binCROS_muon2016M_bin15_quad_cHl3
1062 prop_binCROS_muon2018_bin13_quad_cHl3
1063 prop_binCROS_muon2018_bin11_TVX
1064 prop_binCROS_electron2017_bin7_sm_lin_quad_cHl3
1065 prop_binCRF_electron2017_bin16_sm_lin_quad_cHl3
1066 prop_binCROS_electron2018_bin5_Triboson
1067 prop_binCROS_electron2016M_bin7_sm
1068 prop_binCROS_muon2016M_bin14_sm_lin_quad_cHl3
1069 prop_binSR_muon2016M_bin18_quad_cHl3
1070 prop_binCROS_muon2018_bin21_quad_cHl3
1071 prop_binCROS_muon2016M_bin21_sm_lin_quad_cHl3
1072 prop_binCROS_electron2016M_bin13_sm_lin_quad_cHl3
1073 prop_binCRF_electron2018_bin12_quad_cHl3
1074 prop_binCROS_electron2018_bin11_quad_cHl3
1075 prop_binSR_electron2016M_bin21_quad_cHl3
1076 prop_binCROS_muon2018_bin12_quad_cHl3
1077 prop_binCROS_electron2018_bin20_sm_lin_quad_cHl3
1078 prop_binSR_electron2017_bin19_sm_lin_quad_cHl3
1079 prop_binCROS_muon2018_bin18_sm_lin_quad_cHl3
1080 prop_binCROS_electron2016M_bin7_quad_cHl3
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

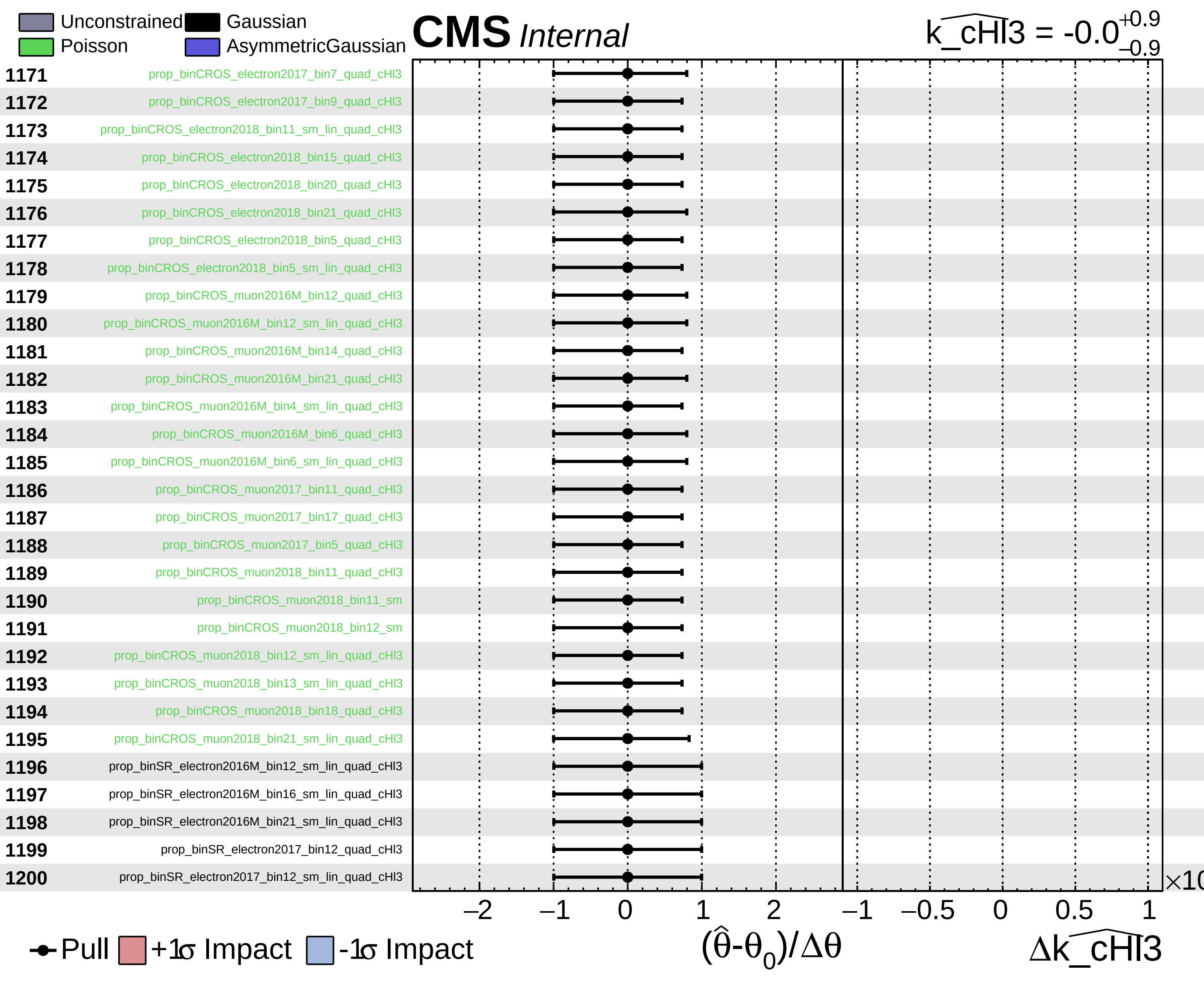


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1081 prop_binCROS_muon2016M_bin4_quad_cHl3
1082 prop_binCROS_electron2016M_bin21_quad_cHl3
1083 prop_binCROS_muon2017_bin11_sm_lin_quad_cHl3
1084 prop_binCRF_electron2016M_bin15_sm_lin_quad_cHl3
1085 prop_binCROS_electron2018_bin15_sm_lin_quad_cHl3
1086 prop_binCROS_muon2018_bin20_sm_lin_quad_cHl3
1087 prop_binCROS_muon2016M_bin14_sm
1088 prop_binCRF_electron2018_bin16_quad_cHl3
1089 prop_binCROS_muon2017_bin5_sm_lin_quad_cHl3
1090 prop_binCROS_muon2017_bin21_quad_cHl3
1091 prop_binCROS_muon2018_bin20_quad_cHl3
1092 prop_binCROS_electron2018_bin5_sm
1093 prop_binCROS_muon2016M_bin11_sm_lin_quad_cHl3
1094 prop_binCRF_electron2018_bin5_quad_cHl3
1095 prop_binCROS_electron2017_bin13_sm_lin_quad_cHl3
1096 prop_binCRF_electron2016M_bin17_quad_cHl3
1097 prop_binCRF_electron2016M_bin14_sm_lin_quad_cHl3
1098 prop_binCROS_muon2017_bin7_quad_cHl3
1099 prop_binSR_muon2017_bin13_sm_lin_quad_cHl3
1100 prop_binCROS_muon2017_bin7_sm_lin_quad_cHl3
1101 prop_binSR_muon2016M_bin16_quad_cHl3
1102 prop_binCROS_electron2017_bin9_sm_lin_quad_cHl3
1103 prop_binCROS_electron2016M_bin13_quad_cHl3
1104 prop_binCROS_electron2016M_bin11_sm_lin_quad_cHl3
1105 prop_binCRF_muon2016M_bin17_quad_cHl3
1106 prop_binCROS_muon2017_bin17_sm_lin_quad_cHl3
1107 prop_binSR_electron2016M_bin12_quad_cHl3
1108 prop_binCROS_electron2016M_bin8_sm_lin_quad_cHl3
1109 prop_binCROS_muon2016M_bin11_quad_cHl3
1110 prop_binSR_electron2016M_bin16_quad_cHl3
−2 −1 0 1 2
−1 0 1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3

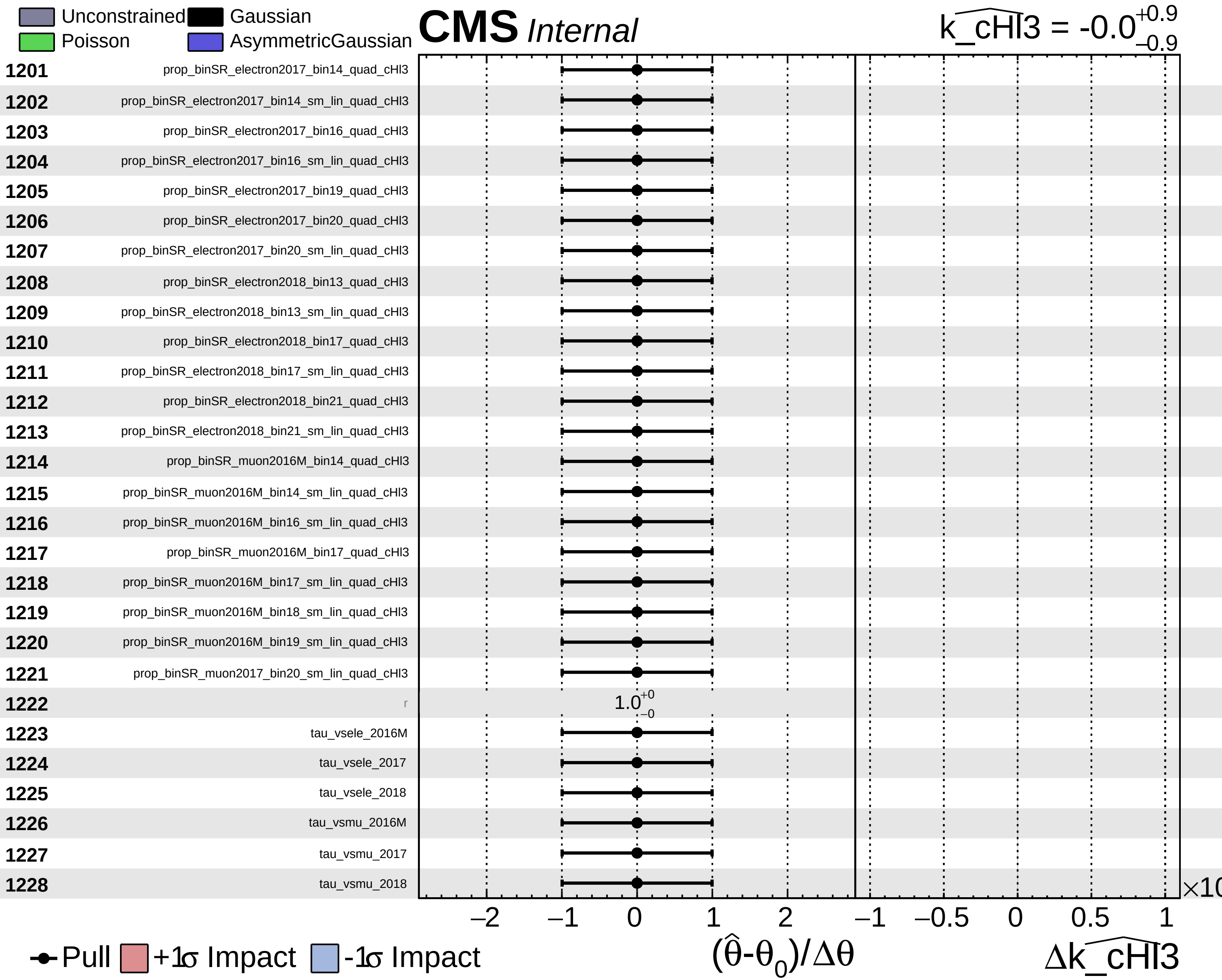


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1111 prop_binCROS_muon2017_bin21_sm_lin_quad_cHl3
1112 prop_binCROS_electron2016M_bin10_quad_cHl3
1113 prop_binCRF_muon2016M_bin15_sm_lin_quad_cHl3
1114 prop_binCROS_electron2018_bin8_quad_cHl3
1115 prop_binSR_electron2016M_bin1_quad_cHl3
1116 prop_binCRF_electron2018_bin5_sm_lin_quad_cHl3
1117 prop_binCROS_muon2018_bin11_sm_lin_quad_cHl3
1118 prop_binCRF_muon2016M_bin10_quad_cHl3
1119 prop_binSR_muon2017_bin20_quad_cHl3
1120 prop_binCRF_muon2018_bin12_quad_cHl3
1121 prop_binCROS_electron2016M_bin4_sm_lin_quad_cHl3
1122 prop_binCRF_electron2016M_bin19_sm_lin_quad_cHl3
1123 prop_binCRF_muon2017_bin11_sm_lin_quad_cHl3
1124 prop_binCROS_electron2017_bin7_sm
1125 QCDScale_WZ
1126 prop_binCRF_electron2016M_bin10_quad_cHl3
1127 prop_binCRF_electron2016M_bin10_sm_lin_quad_cHl3
1128 prop_binCRF_electron2016M_bin15_quad_cHl3
1129 prop_binCRF_electron2016M_bin16_quad_cHl3
1130 prop_binCRF_electron2016M_bin16_sm_lin_quad_cHl3
1131 prop_binCRF_electron2016M_bin17_sm_lin_quad_cHl3
1132 prop_binCRF_electron2016M_bin18_quad_cHl3
1133 prop_binCRF_electron2016M_bin18_sm_lin_quad_cHl3
1134 prop_binCRF_electron2016M_bin19_quad_cHl3
1135 prop_binCRF_electron2017_bin11_quad_cHl3
1136 prop_binCRF_electron2017_bin11_sm_lin_quad_cHl3
1137 prop_binCRF_electron2017_bin14_quad_cHl3
1138 prop_binCRF_electron2017_bin14_sm_lin_quad_cHl3
1139 prop_binCRF_electron2017_bin16_quad_cHl3
1140 prop_binCRF_electron2018_bin12_sm_lin_quad_cHl3
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1141 prop_binCRF_electron2018_bin15_quad_cHl3
1142 prop_binCRF_electron2018_bin16_sm_lin_quad_cHl3
1143 prop_binCRF_muon2016M_bin10_sm_lin_quad_cHl3
1144 prop_binCRF_muon2016M_bin11_quad_cHl3
1145 prop_binCRF_muon2016M_bin11_sm_lin_quad_cHl3
1146 prop_binCRF_muon2016M_bin12_quad_cHl3
1147 prop_binCRF_muon2016M_bin12_sm_lin_quad_cHl3
1148 prop_binCRF_muon2016M_bin16_quad_cHl3
1149 prop_binCRF_muon2016M_bin16_sm_lin_quad_cHl3
1150 prop_binCRF_muon2016M_bin17_sm_lin_quad_cHl3
1151 prop_binCRF_muon2017_bin11_quad_cHl3
1152 prop_binCRF_muon2017_bin8_quad_cHl3
1153 prop_binCRF_muon2017_bin8_sm_lin_quad_cHl3
1154 prop_binCRF_muon2018_bin12_sm_lin_quad_cHl3
1155 prop_binCROS_electron2016M_bin10_sm_lin_quad_cHl3
1156 prop_binCROS_electron2016M_bin11_quad_cHl3
1157 prop_binCROS_electron2016M_bin19_quad_cHl3
1158 prop_binCROS_electron2016M_bin19_sm_lin_quad_cHl3
1159 prop_binCROS_electron2016M_bin21_sm_lin_quad_cHl3
1160 prop_binCROS_electron2016M_bin4_quad_cHl3
1161 prop_binCROS_electron2016M_bin5_quad_cHl3
1162 prop_binCROS_electron2016M_bin5_sm_lin_quad_cHl3
1163 prop_binCROS_electron2016M_bin7_sm_lin_quad_cHl3
1164 prop_binCROS_electron2016M_bin8_quad_cHl3
1165 prop_binCROS_electron2016M_bin8_sm
1166 prop_binCROS_electron2017_bin11_quad_cHl3
1167 prop_binCROS_electron2017_bin11_sm_lin_quad_cHl3
1168 prop_binCROS_electron2017_bin13_quad_cHl3
1169 prop_binCROS_electron2017_bin21_quad_cHl3
1170 prop_binCROS_electron2017_bin21_sm_lin_quad_cHl3
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1171 prop_binCROS_electron2017_bin7_quad_cHl3
1172 prop_binCROS_electron2017_bin9_quad_cHl3
1173 prop_binCROS_electron2018_bin11_sm_lin_quad_cHl3
1174 prop_binCROS_electron2018_bin15_quad_cHl3
1175 prop_binCROS_electron2018_bin20_quad_cHl3
1176 prop_binCROS_electron2018_bin21_quad_cHl3
1177 prop_binCROS_electron2018_bin5_quad_cHl3
1178 prop_binCROS_electron2018_bin5_sm_lin_quad_cHl3
1179 prop_binCROS_muon2016M_bin12_quad_cHl3
1180 prop_binCROS_muon2016M_bin12_sm_lin_quad_cHl3
1181 prop_binCROS_muon2016M_bin14_quad_cHl3
1182 prop_binCROS_muon2016M_bin21_quad_cHl3
1183 prop_binCROS_muon2016M_bin4_sm_lin_quad_cHl3
1184 prop_binCROS_muon2016M_bin6_quad_cHl3
1185 prop_binCROS_muon2016M_bin6_sm_lin_quad_cHl3
1186 prop_binCROS_muon2017_bin11_quad_cHl3
1187 prop_binCROS_muon2017_bin17_quad_cHl3
1188 prop_binCROS_muon2017_bin5_quad_cHl3
1189 prop_binCROS_muon2018_bin11_quad_cHl3
1190 prop_binCROS_muon2018_bin11_sm
1191 prop_binCROS_muon2018_bin12_sm
1192 prop_binCROS_muon2018_bin12_sm_lin_quad_cHl3
1193 prop_binCROS_muon2018_bin13_sm_lin_quad_cHl3
1194 prop_binCROS_muon2018_bin18_quad_cHl3
1195 prop_binCROS_muon2018_bin21_sm_lin_quad_cHl3
1196 prop_binSR_electron2016M_bin12_sm_lin_quad_cHl3
1197 prop_binSR_electron2016M_bin16_sm_lin_quad_cHl3
1198 prop_binSR_electron2016M_bin21_sm_lin_quad_cHl3
1199 prop_binSR_electron2017_bin12_quad_cHl3
1200 prop_binSR_electron2017_bin12_sm_lin_quad_cHl3
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHl3 = -0.0 +0.9 −0.9
1201 prop_binSR_electron2017_bin14_quad_cHl3
1202 prop_binSR_electron2017_bin14_sm_lin_quad_cHl3
1203 prop_binSR_electron2017_bin16_quad_cHl3
1204 prop_binSR_electron2017_bin16_sm_lin_quad_cHl3
1205 prop_binSR_electron2017_bin19_quad_cHl3
1206 prop_binSR_electron2017_bin20_quad_cHl3
1207 prop_binSR_electron2017_bin20_sm_lin_quad_cHl3
1208 prop_binSR_electron2018_bin13_quad_cHl3
1209 prop_binSR_electron2018_bin13_sm_lin_quad_cHl3
1210 prop_binSR_electron2018_bin17_quad_cHl3
1211 prop_binSR_electron2018_bin17_sm_lin_quad_cHl3
1212 prop_binSR_electron2018_bin21_quad_cHl3
1213 prop_binSR_electron2018_bin21_sm_lin_quad_cHl3
1214 prop_binSR_muon2016M_bin14_quad_cHl3
1215 prop_binSR_muon2016M_bin14_sm_lin_quad_cHl3
1216 prop_binSR_muon2016M_bin16_sm_lin_quad_cHl3
1217 prop_binSR_muon2016M_bin17_quad_cHl3
1218 prop_binSR_muon2016M_bin17_sm_lin_quad_cHl3
1219 prop_binSR_muon2016M_bin18_sm_lin_quad_cHl3
1220 prop_binSR_muon2016M_bin19_sm_lin_quad_cHl3
1221 prop_binSR_muon2017_bin20_sm_lin_quad_cHl3
1222 r
1.0 +0 −0
1223 tau_vsele_2016M
1224 tau_vsele_2017
1225 tau_vsele_2018
1226 tau_vsmu_2016M
1227 tau_vsmu_2017
1228 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHl3